

XVA Methodologies

Exotic Derivatives
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1 XVAs In Theory

We present here a theoretical derivation of the valuation equations for the risky derivative, containing both PV and the XVAs. As all forms of modelling, it is inherently a simplification of reality. We use it to inform our understanding and choices of valuation framework, in combination with other considerations such as, for instance, accounting standards and market practices.

We focus on the Replication method. This method assumes that the risky derivative can be replicated by simple instruments including the derivative's underlying and bonds (of our counterparty and possibly of ourselves). As mentioned above this is obviously an idealization: in practice it does not seem realistic to assume that our own default risk can be hedged, and even other risks such as the counterparty risk may not be hedgeable in the perfect way assumed by the Replication. Nevertheless, the Replication method, thanks to the rigorous logic it is based on, gives us valuable information that we can use in conjunction with other forms of reasoning. As insights provided by the method and later described in this section, we can cite:

- avoidance of double-counting: in all its possible versions, the method does not seem to suffer from the double-counting issues that have been the subject of numerous debates in the industry
- justifications to various view-points that have been represented in debates, such as the distinction between the bondholders' and shareholders' point of views, or the DVA2 discussed by Hull and White in [Hull16a]
- it provides clarification on technical issues such as the role of various rates involved in the valuation, and invariance with respect to the reference rate assumed for PV (see the valuation adjustment RVA in this section).

The other method represented in the literature is based on expectations of future cash-flows. We give an introduction to it in the appendix section B.1 but we spend more time deriving our main results using the Replication.

Before looking at the specifics of the Replication method, we start with a few words on margining processes in general (model-independent) in section 1.1. We give the main characteristics of the Variation Margin (VM, a.k.a. collateral), the Initial Margin (IM) and the Independent Amounts (IAs). We also briefly describe the distinctions between segregation and non-rehypothecation in relation to their effects on the XVAs.

We then begin on the Replication by explaining its principles in 1.2. This gives an overview of how the method works and what its main logical steps are. Then we turn to specifics, showing how it allows us to derive a consistent valuation framework for PV and the XVAs (CVA, DVA, FVA, MVA and KVA).

Our formulation is largely based on combining the works of [Kjaer17a], [Burgard12a], [Green16a] and [Antonov14a]. These references all use the Replication but with different assumptions, formalisms and notations. We strive to unify all these ideas starting from more generic formulations and then recovering, as particular cases, the common valuation equations obtained in the above references.

1.1 Types of Margins

1.1.1 Segregation and Rehypothecability

Here we would like to describe the aspects of these features that affect the XVAs. Segregation is a notion related to default payments. As such, it affects the CVA and DVA but not (directly) the other XVAs. Rehypothecability, on the other hand, is related to the ability to reuse a security for borrowing or lending: it thus affects the FVA and potentially the other margin VAs, but not the CVA and DVA. Let us go in more details.

Segregation

When a margin is under segregation, it is given to a custodian who keeps the amount (or security) separated from the accounts of the trading entity. This has two effects as far as the XVAs are concerned:

- the segregated margin is not at risk when an entity defaults, i.e. the custodian simply gives its full amount to the surviving entity, not just the recovery part. This introduces an asymmetry between received and

paid segregated margins. Indeed, if the surviving entity has paid a segregated margin to the defaulting entity, that margin is received in full on default. However, the segregated margin that the surviving entity has received from the defaulting one is usable by the surviving entity to reduce its claim (the custodian is allowed to release it to the surviving entity). This implies that the segregated margin paid by the surviving entity is not part of the default VA (CVA or DVA depending on who is the surviving entity) because it is fully recovered, while the segregated margin received by the surviving entity reduces the claim and therefore is part of the default VA.

- the segregated margin is held by a custodian, and in particular it cannot be used by the receiving entity for lending. As a consequence, it is non-rehypothecable (see below).

Rehypothecability

Rehypothecability is not a strict legal term like segregation but has to do with the ability to reuse a received margin. As such it is dependent on possibly varying market conditions related to liquidity. Rehypothecability means that a received margin, whether cash or security, can be reused by the receiving entity for lending. As we will see later in this chapter, lending can be done either directly by lending to another counterparty, or indirectly by reducing our need for funding. This means that the notion of rehypothecability is connected to the funding VAs (FVA, MVA, IAVA, etc...), but not to the default VAs (CVA and DVA).

Rehypothecability also leads to an asymmetry between received and paid margins. When a received margin is non-rehypothecable, it cannot be reused for lending/reducing funding need. Therefore it carries a cost in the sense that the asset is held with no benefit, while one might have to pay a rate on it to the counterparty that paid it to us (although in most situation that rate will be 0%).

On the other hand, when a paid margin is non-rehypothecable, this has no funding implications on pricing from the point of view of the margin payer. Indeed, the margin payer must fund this margin no matter what happens to it on the receiving side.

As a consequence, non-rehypothecability has no effect on pricing for paid margins, it only affects received margins.

Note that segregation implies non-rehypothecability, but that non-segregation does not imply rehypothecability. The two notions are connected but different.

1.1.2 Variation Margin

The Variation Margin (VM, a.k.a. collateral) is in most situations non-segregated and rehypothecable. It is remargined regularly (often daily, but some CSAs may have other frequencies such as monthly) and is subject to thresholds, MTAs and haircuts applied to the margining value (that may or may not be exactly what we call "PV").

1.1.3 Initial Margin

The Initial Margin (IM) is in most situation segregated (and therefore non-rehypothecable). It is remargined regularly based on a margining model depending on who we trade with. This model may be a near-standard like SIMM for bi-lateral trading, or a proprietary model for trading with a clearing house such as LCH. In any case, the margining amount is varying in time in a complex (stochastic) manner.

1.1.4 Independent Amount

The Independent Amount (IA) is in most situation non-segregated and rehypothecable. It is a fixed constant amount (not subject to re-margining).

1.2 Principles of the Replication Method

The replication method consists in the following steps:

1. Postulate a replication of the risky derivative as a linear combination of underlying variables that enable hedging of the primary sources of risk and funding of the necessary cash. These primary sources of risk are represented by the spot price (of the underlying of the derivative), the bonds of the counterparty and our own bonds (used for funding and potentially hedging of our own default risk)
2. Impose that the portfolio of the derivative, its hedge and the funding position is self-financing
3. Specify the close-out conditions (i.e. how to calculate the close-out claim)
4. Write down the dynamics of the portfolio combining the derivative and the hedge
5. Choose a funding strategy: how many bonds, which ones, to fund which part of the required amount
6. Choose a hedging strategy for our own default risk: fully hedged to not hedged at all, with which bonds
7. Find the hedging coefficients that eliminate the risk for the chosen hedge strategy
8. This leads to a PDE on the risky derivative price
9. Introduce the notion of "Present Value" (PV) in order to split the risky value into PV and the XVAs
10. This leads to a PDE on the XVA components
11. Use the Feynman-Kac theorem to transform this PDE into its equivalent discounted expectation of cash flows, which lets us implement a Monte-Carlo simulation
12. Identify model parameters (e.g. bond yields) with observable data (e.g. OIS and funding rates, default intensities, liquidity spreads, etc...).

In this chapter to avoid lengthy equations that do not help in understanding the method, we apply the Replication to the most typical cases of VM (rehypothecable) and IM (segregated). The case of the IA is described in the next chapter where we take a more hands-on approach.

1.3 Replicating Portfolio

We build a portfolio Π_t at time t that contains the derivative and its hedges. We also include the cash position that is necessary for funding. In other word, any amount of cash that is needed to buy/sell the derivative, its hedges, and to post margins, is borrowed/lent using our bond issuance and this funding position enters the portfolio Π_t .

More precisely, since we include the funding position in the portfolio, the portfolio is entered into at t with 0 value. Fair valuation implies that both parties would only trade if this portfolio is also expected to have 0 value when exiting it at the next point in time $t + dt$. The positions are thus all closed at no cost at $t + dt$. One may or may not decide to continue trading with the counterparty by setting up a new portfolio Π_{t+dt} , again at no cost once including the funding position, and repeating the process indefinitely.

All positions evolve between t and $t + dt$, and since $\Pi_t = 0$ and self-financing imposes to exit at 0 cost at $t + dt$, one must have the condition $\delta\Pi = 0$. This is the self-financing condition that leads to valuation PDEs.

In this analysis, "we", the Bank, are denoted by the letter B while the counterparty is denoted by the letter C. Let us give a few more notations and comments

- the risky derivative (including XVAs) has value $\hat{V}$ from our stand point, it can be positive or negative
- the stock position is δS where δ may be positive or negative
- the position in the counterparty bond is $\alpha_C P_C$ where P_C is the value of the counterparty bonds. α_C is typically negative as we need to short the counterparty bonds to hedge the counterparty default risk

- we take the position $\sum_i \alpha_i P_i$ in our bonds P_i with seniority i . The reason why we use several of our bonds by opposition to only one for the counterparty is that our bonds may be used for two purposes, funding (mandatory) and hedging of our own risk (optional). This means that the Treasury sells/buys our bonds to provide us with the necessary cash for funding, but also potentially adjusts the relative positions on our bonds so as to hedge our default risk. More on this mechanism later in this section.

In the below we explain in more details how this portfolio is constructed and how it evolves.

1.3.1 Underlying

We enter the underlying (stock) position through a repo transaction. We borrow the cash amount $\delta S(t)$ on the repo market, with spot price $S(t)$ at t and repo rate q_S . We use it to buy δ stocks and give those as collateral in the repo transaction at t . We do not need extra cash to fund this component of the portfolio. To summarize, at t we enter the transaction at no cost with the following flows

- $\delta S(t)$ borrowed in the repo market
- $-\delta S(t)$ spent on buying the stock.

The flows at t net to 0. At the close of the transaction at $t + dt$, we must pay the principal plus interest on the repo transaction and receive back the δ amount of stock, which we immediately sell at the spot price. Note also that between t and $t + dt$ the stock will pay dividend, which we simplify as a continuously paid rate of γ_S . In other words we see the following flows at $t + dt$

- $-\delta S(t)(1 + q_S dt)$ as principal and interest on the repo transaction
- $+\delta \gamma_S S(t) dt$ received in dividends
- $+\delta S(t + dt)$ by selling the stock in the spot market after receiving it back from the repo.

The stock position does not require funding and yields the following component in the change of the portfolio

$$\delta \Pi|_S = \delta dS + \delta(\gamma_S - q_S) S dt. \quad (1)$$

1.3.2 Counterparty bond

To hedge the counterparty risk we must short the counterparty bond P_C . We do so by entering a reverse repo position. At t we lend the amount $-\alpha_C P_C$ on the repo market with the counterparty bond as collateral, with repo rate q_C . We receive $-\alpha_C$ of the bond as collateral and sell it. This transaction is entered into at t at no cost with the following flows

- $\alpha_C P_C(t)$ from lending on the repo market
- $-\alpha_C P_C(t)$ received by selling the bond.

The flows at t net to 0. At $t + dt$ we close the transactions, i.e. we receive the principal with interest on the repo transaction, we buy the bond and give it back as collateral on the repo. In other words we see the flows

- $-\alpha_C P_C(t)(1 + q_C dt)$ as principal and interest received from the repo
- $+\alpha_C P_C(t + dt)$ from having to buy the bond as collateral.

The counterparty bond position does not require funding and yields the following component in the change of the portfolio

$$\delta \Pi|_{P_C} = \alpha_C dP_C - \alpha_C q_C P_C dt. \quad (2)$$

1.3.3 Bank bonds

While it may seem unrealistic that we would be able to repo our own bonds (very much like selling CDSs on ourselves), we include this possibility here as it leads to valuable insights into the dynamics of the XVAs. We define our own bond position as the summation

$$\sum_{i=B,1..N} \alpha_i P_i = \alpha_B P_B + \sum_{i=1}^N \alpha_i P_i \quad (3)$$

where P_B is a repo-able bond and P_i for $i = 1..N$ are N non-repo-able bonds.

Repo-able bank bond

The repo-able bond P_B , *if it is used*, is used for hedging our own default risk. The situation is symmetric to that of our counterparty bond. We enter a direct repo position this time. At t we borrow the amount $\alpha_B P_B$ on the repo market with our own bond as collateral, with repo rate q_B . To be able to offer this bond as collateral, we buy α_B of it using the amount borrowed in the repo transaction¹. This transaction is entered into at t at no cost with the following flows

- $\alpha_B P_B(t)$ from borrowing on the repo market
- $-\alpha_B P_B(t)$ paid to buy the bond we then give as collateral.

The flows at t net to 0. At $t + dt$ we close the transactions, i.e. we pay the principal with interest on the repo transaction, we receive the collateral bond back and sell it. In other words we see the flows

- $-\alpha_B P_B(t)(1 + q_B dt)$ as principal and interest are paid at maturity of the repo
- $+\alpha_B P_B(t + dt)$ from selling the collateral bond.

The repo-able own bond position does not require funding and yields the following component in the change of the portfolio

$$\delta \Pi|_{P_B} = \alpha_B dP_B - \alpha_B q_B P_B dt. \quad (4)$$

Note that this position is optional, in that it may or may not be used depending on whether we want to hedge in this manner or not. When this position is not used, we simply force its coefficient to $\alpha_B = 0$ such that the bond P_B can be entirely ignored.

Non-repo-able bank bonds

For these the situation is very different from that of the repo-able bonds. We do not go to the repo market to enter our bond position. Instead, our Treasury buys/sells our bonds in order to fund the derivative trading. The position is thus simply a buy/sell adjusted through time to satisfy the funding requirements.

Margining

We are going to enter the derivative transaction whose risky value is $\hat{V}$ to us. This means that we need to pay the amount $-\hat{V}$ to the counterparty to enter the trade. Once the counterparty received this amount, we value the transaction and through the specified mechanism of the CSA, the counterparty posts to us the collateral amount X (a.k.a. the variation margin), or nothing if there is no CSA. Here we will assume rehypothecation of the collateral, so that we are allowed to use the amount X for something else, typically for reducing the funding requirement.

We must also post the initial margin I_B and this amount must be funded. The counterparty posts the initial margin I_C , but this amount is segregated, and in particular non-rehypothecable. As a consequence, we cannot

¹This is an ideal situation with all positions entered into instantaneously. In practice, finite transaction time will prevent this construction.

use it to reduce our funding requirement and therefore the initial margin of the counterparty I_C does not enter the funding requirements.

Funding through capital

For the sake of generality we will consider that a proportion ϕ of the capital requirement K can be used for reducing the funding requirement. ϕ can be set to 0 if that is representative of the bank's internal policy regarding capital usage.

Funding condition

Gathering the above, we must fund the amount $\hat{V} - X + I_B - \phi K$, because:

1. we must pay the derivative's price $\hat{V}$ in cash to the counterparty
2. the counterparty posts the variation margin X which is rehypothecable, so we use it to reduce our need for funding
3. we must post the initial margin I_B and we must fund this cash amount
4. we cannot use the counterparty's initial margin I_C to reduce our funding
5. we reduce our funding by the proportion ϕ of the capital amount K we borrowed from the shareholders to satisfy regulatory requirements.

We fund this amount by receiving it from the Treasury which raises it by issuing $\sum_i \alpha_i P_i$ of our non-repoed bonds ($i = 1..N$). More precisely, we receive the amount $-\sum_{i=1}^N \alpha_i P_i$ by selling the bonds. This amount is used to fund the cash $\hat{V} - X + I_B - \phi K$ necessary to enter the transaction and settle its margins. This gives us the funding condition

Funding condition: $\sum_{i=1}^N \alpha_i P_i + \hat{V} - X + I_B - \phi K = 0.$

(5)

At t we have the following flows, netting to 0:

- receive $-\sum_{i=1}^N \alpha_i P_i(t) = \hat{V}(t) - X(t) + I_B(t) - \phi K(t)$ from selling our bonds
- give $\hat{V}(t)$ to the counterparty to enter the transaction
- give $I_B(t)$ to the counterparty as initial margin
- receive $X(t)$ from the counterparty as collateral
- receive $K(t)$ by borrowing capital from the shareholders, segregate $(1 - \phi)K(t)$ to satisfy regulatory requirements
- receive $I_C(t)$ from the counterparty as initial margin, and segregate it.

At time $t + dt$, we have the position:

- the derivative's value has become $\hat{V}(t + dt)$
- we give back the collateral plus interest to the counterparty, that is, we see the flow $-X(t)(1 + r_X dt)$ where r_X is the collateral rate

- the counterparty (or its custodian) gives us back our initial margin plus interest, that is we see the flow $I_B(t)(1 + r_{I_B}dt)$ where r_{I_B} is the rate on the initial margin account
- our bond position's value has become $\sum_{i=1}^N \alpha_i P_i(t + dt)$
- we retrieve (receive) the regulatory capital amount $(1 - \phi)K(t)$ that had been segregated away for regulatory purposes
- we return the borrowed capital $K(t)$ to the shareholders and pay them an additional cost of γ_K , that is we see the flow $-K(t)(1 + \gamma_K dt)$
- we retrieve the counterparty's initial margin amount $I_C(t)$ from the custodian, and return it to the counterparty plus interest, that is we see the flow $I_C(t) - I_C(t)(1 + r_{I_C}dt)$.

This means that this side of the portfolio has changed by (remember it was worth 0 at t)

$$\begin{aligned}
 \delta\Pi|_V &= \hat{V}(t + dt) - X(t)(1 + r_X dt) + I_B(t)(1 + r_{I_B}dt) - r_{I_C}I_C(t)dt \\
 &\quad + (1 - \phi)K(t) - K(t)(1 + \gamma_K dt) + \sum_{i=1}^N \alpha_i P_i(t + dt) \\
 &= d\hat{V} - r_X X dt + r_{I_B} I_B dt - r_{I_C} I_C dt - \gamma_K K dt + \sum_{i=1}^N \alpha_i dP_i.
 \end{aligned}$$

where we have used the funding condition (5).

1.3.4 Self-financing condition

The total change in the portfolio is the sum of the above components and is self-financed if

$$\delta\Pi = \delta\Pi|_V + \delta\Pi|_S + \delta\Pi|_{P_C} + \delta\Pi|_{P_B} = 0 \quad (6)$$

which explicitly translates into the self-financing condition

$$\begin{aligned}
 0 &= d\hat{V} - r_X X dt + r_{I_B} I_B dt - r_{I_C} I_C dt - \gamma_K K(t)dt + \delta dS + \delta(\gamma_S - q_S)S dt \\
 &\quad + \sum_{i=1}^N \alpha_i dP_i + \alpha_C dP_C - \alpha_C q_C P_C dt + \alpha_B dP_B - \alpha_B q_B P_B dt.
 \end{aligned} \quad (7)$$

1.4 Dynamics

1.4.1 Underlying

The dynamics of the underlying is not crucial for this derivation as it is ultimately absorbed by Feynman-Kac formula. We take the simplest case of a stock, which follows the dynamics

$$dS = \mu_S S dt + \sigma_S S dW \quad (8)$$

where μ_S is the stock drift and σ_S its diffusion.

1.4.2 Bonds

A generic bond price P_E follows the SDE

$$dP_E = r_E P_E dt - (1 - R_{P_E}) P_E dJ_E \quad (9)$$

where E denotes some as yet unspecified bond issue and

- J_E is a jump event going from 0 to 1, with intensity λ_E
- R_{P_E} is the recovery rate on the bond (not to be confused with the recovery rate on the netting set with a counterparty)
- r_E is yield on the unsecured bond P_E .

1.4.3 Bonds vs CDSs

We will see that it is possible to reach valuation equations with no further assumptions than the above on the bond dynamics. When we do so, we will obtain equations that depend on the rates r_E of generic (unsecured) bonds. These rates are not explicit in the sense that they are not, without further specifications, rates that we know from direct observation of the market. In order to reach practical valuation equations, that is, equations with only observable rates, we will need to assume identifications between the rates r_E and the rates we can measure or estimate, such as the funding rate, the OIS rates, repo rates, survival probabilities, liquidity spreads, etc...

To do so we use the following modelling considerations. We introduce the counterparty risk-free rate (typically OIS) r_0 in the currency of the bond. The yield r_E is then decomposed into

$$r_E = r_0 + \lambda_E LGD_{P_E} + \gamma_E \quad (10)$$

where γ_E is the bond-CDS basis for the bond P_E .

We model the bond offerings in the slightly simplified way by assuming that two types of bonds exist, the repo-able and non-repo-able ones:

- Repo-able bonds: these have a very liquid repo market. For this reason, we are able to lend/borrow them on their respective repo markets and observe little liquidity spreads on them. We model this by assuming that the bond-CDS basis for them only contains the OIS-repo basis and no liquidity spread. In other words, for repo-able bonds,

$$\text{Repo-able bonds} \quad (11)$$

$$\gamma_E = s_{q_E} \quad (12)$$

$$q_E = r_0 + s_{q_E} \quad (13)$$

$$r_E = q_E + \lambda_E LGD_{P_E} \quad (14)$$

where s_{q_E} is the OIS-repo basis and q_E is the repo rate.

- Non-repo-able bonds: these do not have a liquid repo market. As a consequence, we are not able to lend/borrow them on the repo market, and we observe the full bond-CDS basis on them, i.e.

$$\text{Non-repo-able bonds} \quad (15)$$

$$r_E = r_0 + \lambda_E LGD_{P_E} + \gamma_E. \quad (16)$$

Default risk hedging

We assume that default risk can only be hedged if there exists a very liquid repo market for the bonds of the entity whose default we are trying to hedge. In other words, when default risk is hedged, it is hedged with the repo-able type of bonds as described above.

Funding

Funding is viewed as an abstract, or aggregated, representation of our ability to borrow/lend. We will represent it by using non-repoable bonds.

Note that this split in roles between default risk hedging with repo-able bonds and funding with non-repoable bonds is not the only method available. One may use any of these types of bonds for both hedging and funding, which is what is most commonly done in the literature (non-repoable bonds used for funding and hedging). However, we think that the framework presented in this document, strongly inspired by the work of [Antonov14a], is equivalent in results but gives more valuable insights into the meaning and logical connections between various XVA terms and strategies.

Counterparty

We assume that the counterparty default risk can be hedged, and as explained above, we will hedge it using one repo-able bond. For this reason the counterparty bond has the following dynamics

$$dP_C = r_C P_C dt - LGD_{P_C} P_C dJ_C \quad (17)$$

$$r_C = q_C + \lambda_C LGD_{P_C} \quad (18)$$

where q_C is the repo rate, LGD_{P_C} is the loss-given default the bond (not to be confused with the recovery rate on the netting set), and λ_C is the default intensity of the process J_C , measured from the CDS market.

Bank

For ourself we will consider two broad families of strategies that depend on whether we hedge our own default risk or not. Without distinguishing these strategies for now, we model our bond offering in the following way:

- One repo-able bond: we call this bond P_B , we use it for hedging our own default risk (when we hedge it). As discussed above, this bond has yield

$$r_B = q_B + \lambda_B LGD_{P_B} \quad (19)$$

where q_B is the repo rate on this bond, LGD_{P_B} is the loss-given default (not to be confused with the recovery rate on the netting set), and λ_B is the default intensity of the process J_B , measured from the CDS market. Note that the default process J_B is common to all of our bonds.

- Several non-repoable bonds: we call them P_i and use them for funding, with $i = 1..N$ and N is the number of bonds used for funding. As they are non-repoable, they have the dynamics

$$r_i = r_0 + \lambda_B LGD_{P_i} + \gamma_i \quad (20)$$

where γ_i is the bond-CDS basis for the specific bond i . In many situations we will assume that the bond-CDS basis is the same for all non-repoable bonds, but this assumption will not always be made so we keep things more general for now. Funding spreads will be identified in the combinations $\lambda_B LGD_{P_i} + \gamma_i$.

1.4.4 Derivative

With Ito's lemma we obtain the dynamics of the derivatives as

$$d\hat{V} = \frac{\partial \hat{V}}{\partial t} dt + \frac{\partial \hat{V}}{\partial S} dS + \frac{1}{2} \sigma_S^2 S^2 \frac{\partial^2 \hat{V}}{\partial S^2} dt + (g_C - \hat{V}) dJ_C + (g_B - \hat{V}) dJ_B \quad (21)$$

where $g_E - \hat{V}$ is the change in the derivative price upon default of entity E and g_E is the derivative value on-default.

1.5 Close-out

A few words are deserved regarding the calculation of the on-default values g_E for the close-out, in particular with the additional complexity of the margining.

First of all, until this point we have accounted for flows and their evolutions based on the fact that the margins have been settled (i.e. given back with interest). This means that if default were to happen now, the margin accounts would be empty. In reality the margin accounts would contain their pre-default amounts, which we account for here by receiving the balance $X - I_B + I_C$ on-default (i.e. right before default).

On top of this, we receive a claim that depends on who defaulted between the counterparty or us. The claim is calculated as follows:

- we calculate the close-out amount Q following the rules specified in the close-out document (say the CSA, or another form of contract for uncollateralized counterparties).
- the claim subject to default resolution is on the close-out amount Q minus the margins that the surviving party can use to reduce the claim. The collateral is available for this purpose because it is not segregated, and so is the initial margin held by the custodian of the surviving party (i.e. paid by defaulting party). The initial margin that the surviving party had posted is returned to it unconditionally (i.e. not subject to a recovery rate) because it was held safely by the defaulting party's custodian.
- if the claim is positive to the surviving company, it receives only the recovery rate on it, while it must pay it in full if it is negative.

This logic results in the following on-default flows when the counterparty defaults first

$$\begin{aligned} g_C &= X - I_B + I_C + I_B + R_C(Q - X - I_C)^+ + (Q - X - I_C)^- \\ &= Q - LGD_C(Q - X - I_C)^+ \end{aligned} \quad (22)$$

and if we default first

$$\begin{aligned} g_B &= X - I_B + I_C - I_C + (Q - X + I_B)^+ + R_B(Q - X + I_B)^- \\ &= Q - LGD_B(Q - X + I_B)^-. \end{aligned} \quad (23)$$

where Q is the close-out amount and R_E, LGD_E are the recovery rates and losses given default on the netting set for the counterparty and ourselves.

1.6 Hedging Constraints

The sources of randomness related to the stock and the counterparty bond must be removed in the portfolio evolution (7). For the stock, randomness appears only through the term dS that is present both in the evolution of the derivative $d\hat{V}$ in (21) and in the evolution of the stock position in (7). These must cancel out, which implies the condition

$$\delta = -\frac{\partial \hat{V}}{\partial S}. \quad (24)$$

Next we gather the components proportional to the stochastic part of the evolution of the counterparty bond P_C , that is dJ_C . These vanish under the condition

$$\alpha_C P_C = \frac{g_C - \hat{V}}{1 - R_{P_C}}. \quad (25)$$

We are not going to choose the funding and own-default hedging strategies yet, keeping them for a later part of the discussion. We can now use the hedging ratios above to rewrite the self-financing condition (7) as

$$\begin{aligned} 0 = & \left[\frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) + \alpha_B (r_B - q_B) P_B + \sum_{i=1}^N r_i \alpha_i P_i \right] dt \\ & + \epsilon_h dJ_B \end{aligned} \quad (26)$$

where the coefficient of the on-default change $g_C - \hat{V}$ is

$$\frac{r_C - q_C}{LGD_{P_C}} = \lambda_C \quad (27)$$

thanks to the properties of the repo-able bonds used for hedging as discussed in section 1.4.3. The hedge error ϵ_h , i.e. the coefficient of the remaining stochastic component for our bond dJ_B , is

$$\epsilon_h = g_B - \hat{V} - \alpha_B P_B LGD_{P_B} - \sum_{i=1}^N \alpha_i P_i LGD_{P_i} \quad (28)$$

and we have defined the differential operator $\mathcal{A}$

$$\mathcal{A}_t F = \frac{1}{2} \sigma_S^2 S^2 \frac{\partial^2 F}{\partial S^2} - (\gamma_S - q_S) S \frac{\partial F}{\partial S}. \quad (29)$$

The hedge error ϵ_h in (28) may or may not be forced to 0 depending on the hedging strategy.

1.7 Funding Constraints

This document is largely following the methodology of [Antonov14a] with a clean split between hedging and funding through the use of repo-able and non-repo-able bonds as explained in previous sections. This enables us to fully separate the two issues, discuss them in isolation and use them in combination with any choice of hedging. We will consider two families of funding strategies as explained in this section.

1.7.1 Two-Bond Funding

This strategy uses two different non-repoable bonds P_1 and P_2 for funding the positive and negative parts of the funding requirement (5) which reads here

$$\alpha_1 P_1 + \alpha_2 P_2 + \hat{V} - X + I_B - \phi K = 0. \quad (30)$$

In asymmetric funding this is solved by

$$\alpha_1 P_1 = -(\hat{V} - X + I_B - \phi K)^+ \quad (31)$$

$$\alpha_2 P_2 = -(\hat{V} - X + I_B - \phi K)^- \quad (32)$$

i.e. by borrowing with P_1 and lending with P_2 . While this strategy is simple in its meaning, we will see later in this document that when it is truly asymmetric, i.e. when the two bonds are truly different, it tends to lead to non-linear valuation equations that are difficult to implement in standard Monte-Carlo methods.

Note that in the particular case where the two bonds are identical, the funding is effectively symmetric and the resulting PDEs become linear.

1.7.2 Three-Bond Funding

This strategy is inspired by the work of [Kjaer17a] leading to the so-called "No-Shortfall" valuation. In this document we note that while this strategy can, as in [Kjaer17a], be applied when we do not hedge our own default risk, it can be applied as well when we do hedge our own default risk. Furthermore, in its most general form it allows for asymmetric funding while retaining linearity of the valuation equations.

It is based on a split of the funding requirement into the XVA part funded symmetrically by the first bond and the non-XVA part funded, symmetrically or not, by other bonds. This split has been used in its symmetric form in [Kjaer17a] to arrive at the No-Shortfall strategy, but linearity remains even with asymmetric funding.

More precisely, the three-bond funding constraint is

$$\alpha_1 P_1 + \alpha_2 P_2 + \alpha_3 P_3 + \hat{V} - X + I_B - \phi K = 0. \quad (33)$$

and is solved in asymmetric funding by

$$\alpha_1 P_1 = -(\hat{V} - Q) \quad (34)$$

$$\alpha_2 P_2 = -(Q - X + I_B - \phi K)^+ \quad (35)$$

$$\alpha_3 P_3 = -(Q - X + I_B - \phi K)^- \quad (36)$$

where $\hat{V} - Q$ is what we called "the XVA part" and $Q - X + I_B - \phi K$ is "the non-XVA part". The reason why linearity of the PDEs is preserved is that the asymmetry of the funding is not applied to the risky derivative $\hat{V}$ but only to its close-out part Q . This is achieved, of course, only when the close-out Q is calculated risk-free.

Note that similarly to the two-bond strategy in the previous section, symmetric funding is effectively reached as the particular case where the bonds P_2 and P_3 are identical. Furthermore when the three bonds are identical, this strategy reduces to the two-bond strategy with identical bonds.

1.8 From Replication to Expectations

In this section we gather the results obtained so far before choosing specific strategies for funding and hedging our own risk. All explicit valuation equations presented later in this document build upon the results gathered in this section.

We have two constraints that must be satisfied in all cases (self-financing and funding) and one constraint that may or may not be satisfied (own default risk hedging). It is the specification of the hedging and funding strategies, together with identification of observable rates among the model parameters, that fully determines the valuation equations. We will give examples of strategies later in sections 1.9 and 1.10.

The constraints obtained by the Replication method lead to PDEs, and these PDEs are transformed into expectations that are subsequently evaluated using Monte-Carlo simulations. We will give more details about these steps below. In section 1.8.1 we explain how the various constraints (self-financing, funding, hedging) combined into PDEs. We find, for instance, that consistently with [Antonov14a] we can identify two broad types of PDE that correspond to our own default risk being hedged (examples in section 1.9) or not hedged (examples in 1.10). In section 1.8.2 we then explain how these PDEs are transformed into expectations that are suitable for evaluation by Monte-Carlo methods. Finally in section 1.8.3 we briefly mention the issue of how to connect the parameters involved in the Replication (namely bond yields) to observable rates.

1.8.1 From Replication to PDEs

The three types of constraints that are self-financing, funding, and hedging, lead to PDEs on the derivative's risky value $\hat{V}$. We say a few words on these below and then express the two broad families of PDEs corresponding to hedging or not our own default risk.

1. Self-financing: it must be satisfied, and leads to the equation

$$0 = \left[\frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) + \alpha_B (r_B - q_B) P_B + \sum_{i=1}^N r_i \alpha_i P_i \right] dt + \epsilon_h dJ_B. \quad (37)$$

The dt component must be equal to 0 no matter the strategy, while the jump component dJ_B may be forced to 0 (hedging) or not. If it is not forced to 0, the dt component is only equal to 0 before default, and a jump in valuation is observed on default.

2. funding: it must be satisfied, and reads

$$\sum_{i=1}^N \alpha_i P_i + \hat{V} - X + I_B - \phi K = 0. \quad (38)$$

Note that, in our formulation, only non-repoable bonds are used for funding. Fixing the funding strategy means specifying how many such bonds are used, which ones they are (namely their seniority and basis to CDSs), and what part of the funding they account for. Possible funding strategies have been described in section 1.7.

3. own default risk hedging: may or may not be satisfied, i.e. forcing to 0 or not the hedge error

$$\epsilon_h = g_B - \hat{V} - \alpha_B P_B LGD_{P_B} - \sum_{i=1}^N \alpha_i P_i LGD_{P_i} \stackrel{?}{=} 0 \quad (39)$$

Hedged PDEs

By this we mean the PDEs obtained when we hedge our own risk. This is achieved by forcing the hedge error to 0 i.e. by solving equation (39) in terms of the unknowns that are the ratios of own repo-able bond α_B and non-repo-able bonds $\alpha_i, i = 1..N$. This may be done in numerous ways. Most references in the literature do not consider repo-able bonds (that is, choose $\alpha_B = 0$) and hedge (and fund) only with non-repo-able bonds. We prefer to follow [Antonov14a] and hedge solely with the repo-able bond. This means that we impose the equality to 0 in (39) and solve it for the coefficient α_B , which is immediate and leads to

$$\alpha_B P_B = \frac{g_B - \hat{V} - \sum_{i=1}^N \alpha_i P_i LGD_{P_i}}{LGD_{P_B}}. \quad (40)$$

We can now replace the α_B above in the dt component of the self-financing condition (37) which leads to the hedged PDE

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) + \lambda_B (g_B - \hat{V}) \\ + \sum_{i=1}^N \alpha_i P_i (r_i - LGD_{P_i} \lambda_B) = 0 \end{aligned} \quad (41)$$

where the default intensity λ_B arrived in the PDE through the coefficient

$$\frac{r_B - q_B}{LGD_{P_B}} = \lambda_B, \quad (42)$$

a property of the repo-able bonds as discussed in section 1.4.3.

Thanks to the hedging, both the dt and the dJ_B jump components of (37) are 0 such that no jump in value should be observed on default. Note that already at the level of the PDE (41) we can foresee the symmetric bi-lateral valuation arising from the presence of both CVA (coming from $\lambda_C(g_C - \hat{V})$) and DVA (coming from $\lambda_B(g_B - \hat{V})$). The generic PDE (41) will lead to different particular cases based on the funding strategy specifying the remaining unknown ratios $\alpha_i, i = 1..N$.

Non-Hedged PDEs

This category corresponds to when we do not hedge our own default risk. This means that the hedge error (39) is ignored (not vanishing) and in turns we do not need to include the repo-able bond in our portfolio such that $\alpha_B = 0$. Only the dt component of (37) is 0 which leads to the non-hedged PDE

$$\frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C(g_C - \hat{V}) + \sum_{i=1}^N r_i \alpha_i P_i = 0 \quad (43)$$

valid up until the default time, at which point there will typically be a jump in value due to the non-zero hedge error.

Similarly to the hedged case, the generic PDE (43) will lead to different particular cases based on the funding strategy specifying the unknown ratios $\alpha_i, i = 1..N$.

1.8.2 From PDEs to Expectations

Once the funding and hedging strategies have been specified, we obtain PDEs that the risky value $\hat{V}$ must satisfy. To reach practical and familiar valuation equations, in particular in the form of discounted expectations and with a split between PV and XVAs, we need to go through the following actions:

1. write explicitly the on-default payments g_C and g_B defined in (22) and (23)
2. specify the calculation method for the close-out amount Q .

The so-called 'Risky Close-Out', considered by various authors such as [Elouerkaoui16a] and references within [Green16a], is the case where $Q = \hat{V}$. In this situation, the PDEs are non-linear and applying Feynman-Kac to them² leads to recursive valuation equations that are not very practical.

This is typically avoided and the close-out amount Q is most often taken 'risk-free', i.e. modelled as the discounted value of the default-free cash-flows at some rate we denote by r_ϵ . In other words

$$Q = V^{r_\epsilon} \quad (44)$$

with the rate r_ϵ specified in the relevant documents (or assumed). For a generic rate x , we denote this expectation by

$$V_t^x = E_t \left[\int_t^T D_x(t, s) \pi_s ds \right] \quad (45)$$

where $D_x(t, s)$ is the discount factor at time t , for maturity s , with the rate x . π_s is the default risk-free cash-flow received at time s . Note for later reference that V_t^x satisfies the PDE

$$\frac{\partial V^x}{\partial t} + \mathcal{A}_t V^x = x V^x. \quad (46)$$

²If that is even correct theoretically.

3. choose a definition for PV, typically the discounted expectation of the future flows (no default risk) with the reference rate r preferred by the company. r may be the OIS rate in a preferred currency, or OIS with a spread, or any other rate.
4. introduce the split $\hat{V} = PV + XVAs$, that is, in our notations and given the above choice for PV,

$$\hat{V} = V^r + U \quad (47)$$

where U is the sum of all the XVAs. This is in fact a *definition* of U , which is dependent on the choice of rate r made previously for $PV = V^r$. It is important to note here that the choice of r is arbitrary, and that while it affects both PV and the XVAs it does not affect their sum, the original (total) risky value of the derivative $\hat{V}$. This total risky value is therefore identical under any choice of r . We will come back to this issue of reference rate invariance later once we obtained the valuation equations.

5. Finally, this leads to a PDE for U that is transformed into discounted expectations through the use of the Feynman-Kac theorem. The total value $\hat{V}$ is then the sum of $V^r + U$.

1.8.3 Identifying Observable Rates

Once we have obtained valuation equations under the form of expectations at the end of the previous step, these equations are written in terms of the parameters of the Replication method, which are here the various bond yields and default intensities. We have already explained in section 1.4.3 that the default intensities correspond to those derived from the CDS market. As only the yields of our non-repoable bonds remain in the generic PDEs (41) and (43), these are the parameters that need to be fixed to obtain explicit valuation equations.

Our non-repoable bonds have the following generic expressions for their yields

$$r_i = r_0 + \lambda_B LGD_{P_i} + \gamma_i. \quad (48)$$

The hedged PDE (41) only depends on the difference $r_i - \lambda_B LGD_{P_i} = r_0 + \gamma_i$ such that hedged PDEs are fully determined by the specification of the bond-CDS basis γ_i . Non-hedged PDEs under (43), on the other hand, only depend on r_i such that these PDEs are fully determined by the specification of the combination $\lambda_B LGD_{P_i} + \gamma_i$. In some instances this combination may be identified as a whole as a funding spread. In other instances, we will need to specify both the LGD and the bond-CDS basis.

1.9 Hedged Strategies

FT's valuation, the "Firm Value" of [Kjaer17a] and "Zero-Cost" of [Antonov14a] are particular cases of the strategies described in this section, i.e. that hedge our own default risk.

As explained in section 1.8.1, hedging of our own default risk leads to the generic form of PDE (41), which we recall here for easier reading

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \hat{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) + \lambda_B (g_B - \hat{V}) \\ + \sum_{i=1}^N \alpha_i P_i (r_i - LGD_{P_i} \lambda_B) = 0. \end{aligned} \quad (49)$$

Explicit valuation equations are reached after specifying the funding strategy (coefficients α_i above), of which we gave two possible candidates in section 1.7. The two-bond funding of 1.7.1 leads to non-linear PDEs in its generic form (asymmetric funding), see subsection 1.9.1. The three-bond strategy allows to reach linear PDEs even with asymmetric funding³ which is much more amenable to practical implementation. We describe these in more details in subsection 1.9.2.

³Under risk-free close-out. All valuation frameworks are non-linear under Risky Close-Out.

1.9.1 Two-Bond Funding

The strategies in this section lead to non-linear valuation equations in their most generic forms (risky close-out and/or asymmetric funding). As such they are difficult to implement in practice, but we introduce them for the sake of completeness and to show how known strategies identified in the literature fit in the framework presented in this document. In this category, we can find for instance the Zero-Cost strategy of [Antonov14a] and FT's valuation framework. We will also see that they reduce to linear equations with risk-free close-out and symmetric funding.

These strategies are based on the two-bond funding described in section 1.7.1. Plugging (31) and (32) in the hedged PDE (49) we arrive at

$$\begin{aligned} & \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C(g_C - \hat{V}) + \lambda_B(g_B - \hat{V}) \\ & - (r_1 - LGD_{P_1} \lambda_B)(\hat{V} - X + I_B - \phi K)^+ - (r_2 - LGD_{P_2} \lambda_B)(\hat{V} - X + I_B - \phi K)^- = 0. \end{aligned} \quad (50)$$

Unfortunately so long as the funding is truly asymmetric (i.e. that P_1 and P_2 are different bonds), this PDE remains non-linear. Nevertheless we are able to write down the corresponding valuation equations, so long as we assume that the Feynman-Kac applies in this situation as well. The resulting valuation equations are

Hedged-Generic 2 Bonds	
$\hat{V}$	$= PV + CVA + DVA + FCA + FBA + DVA_2 + CVA_2$ $+ ColVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
DVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_B LGD_B (Q - X + I_B)^- \right]$
FCA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_1 - r) (\hat{V} - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_2 - r) (\hat{V} - X + I_B - \phi K)^- \right]$
DVA_2	$= E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_B LGD_{P_1} (\hat{V} - X + I_B - \phi K)^+ \right]$
CVA_2	$= E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_B LGD_{P_2} (\hat{V} - X + I_B - \phi K)^- \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right]$
KVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (\lambda_B + \lambda_C) (V^r - Q) \right]$

(51)

with generic r_1 and r_2 i.e.

$$r_1 = r_0 + \lambda_B LGD_{P_1} + \gamma_1 \quad (52)$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2. \quad (53)$$

DVA_2 (and CVA_2)

Under this generic strategy we can see the appearance of the DVA_2 term as proposed and called by Hull-White in [Hull16a]. In the absence of bond-CDS basis i.e. for $\gamma_1 = 0$ (and for the standard choice $r = r_0$), DVA_2 cancels out against FCA as suggested by Hull-White

$$FCA + DVA_2 = 0. \quad (54)$$

Although less formally stated in [Hull16a], we think that this is very much in line with the opinions mentioned there. [Hull16a] considers that if we have a funding cost, that is because the counterparty that lends to us wants compensation for our default risk. That compensation comes under the form of the credit spread they make us pay, which we see as FCA . On the other hand, [Hull16a] argue that when we default we do not repay the loan, and that this is a benefit. Since our default is, in principle, the source of the credit spread imposed to us for borrowing, then this benefit, DVA_2 , should cancel against FCA .

We can see this happening here and we can go further. If, on the other hand, the lending counterparty has not evaluated the credit spread perfectly, which here would be modelled by a non-zero liquidity spread $\gamma_1 \neq 0$, then FCA and DVA_2 no longer cancel and their sum is proportional to the liquidity spread, that is

$$FCA + DVA_2 \sim \gamma_1. \quad (55)$$

Note that this discussion on FCA and DVA_2 has a symmetric on FBA and what we called CVA_2 here, which cancel out when the *second* bond has no liquidity, and remain otherwise

$$FBA + CVA_2 \sim \gamma_2. \quad (56)$$

RVA

The RVA term does not appear often in the quantitative literature. The reason is that in most references close-out is risk-free and assumed to be calculated with the same risk-free rate as PV. This implies that $Q = V^r$ in our notations and that RVA trivially vanishes. However, we can see here that in general when the close-out is not at PV there should be an additional XVA, RVA. Unfortunately from the practical point of view, RVA is costly to calculate as it requires a second simulation to calculate the forward exposures with the reference rate as discounting.

1.9.1.1 DVA_2 Only

This valuation is the particular case of Generic 2-Bond Hedged where the second bond's LGD is 0. Furthermore, the rates on the two bonds are identified with the borrowing and lending rates. The valuation is

DVA_2 Only	
$\hat{V}$	$= PV + CVA + DVA + FCA + FBA + DVA_2$ $+ ColVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
DVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_B LGD_B (Q - X + I_B)^- \right]$
FCA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_F^b - r) (\hat{V} - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_F^l - r) (\hat{V} - X + I_B - \phi K)^- \right]$
DVA_2	$= E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_B LGD_{P_1} (\hat{V} - X + I_B - \phi K)^+ \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right]$
KVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (\lambda_B + \lambda_C) (V^r - Q) \right]$

(57)

with

$$\begin{aligned}
 r_1 &= r_0 + \lambda_B LGD_{P_1} + \gamma_1 = r_F^b \\
 r_2 &= r_0 + \lambda_B \times 0 + \gamma_2 = r_F^l.
 \end{aligned}
 \tag{58}$$

In particular there is no CVA_2 in this valuation, which is here formalized by the choice $LGD_{P_2} = 0$. Note that this valuation seems to suffer from the DVA-FBA double counting, but this is probably not what it is meant to achieve in the following sense. The absence of CVA_2 is translated here in the fact that the second bond has 100% recovery. As often, the intent of an asymmetric funding strategy is to model true funding costs for borrowing but assume that lending, if any, is done at the risk-free rate, which is consistent with assuming $LGD_{P_2} = 0$. If that is the intent behind this modelling, then the lending rate r_F^l would be r such that FBA would be 0 (assuming a simple situation where $r_F^l = r = OIS$).

1.9.1.2 Antonov's Zero-Cost Strategy

This strategy is derived by Antonov et al. in [Antonov14a], a reference that is the source of our formalism in terms of repo-able and non-repo-able bonds. Antonov's Zero-Cost strategy can be seen as the particular case of FT where the lending rate is indeed chosen as $r_F^l = r$. The valuation equations reduce to

Antonov	
$\hat{V}$	$= PV + CVA + DVA + FCA + DVA_2$ $+ ColVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
DVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_B LGD_B (Q - X + I_B)^- \right]$
FCA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_F^b - r) (\hat{V} - X + I_B - \phi K)^+ \right]$
DVA_2	$= E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_B LGD_{P_1} (\hat{V} - X + I_B - \phi K)^+ \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right]$
KVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} (\lambda_B + \lambda_C) (V^r - Q) \right]$

(59)

with

$$\begin{aligned}
 r_1 &= r_0 + \lambda_B LGD_{P_1} & +\gamma_1 &= r_F^b \\
 r_2 &= r_0 + \lambda_B \times 0 & +\gamma_2 &= r.
 \end{aligned}$$
(60)

1.9.1.3 Symmetric Funding

Symmetric Funding is achieved here by assuming the two funding bonds are the same. In this particular case (and assuming risk-free close-out), the valuation equations become linear. We could simply write the valuation as the particular case of (51) where $r_1 = r_2$ and $LGD_{P_1} = LGD_{P_2}$ (strictly speaking linearity is achieved with the slightly less stringent constraint $\gamma_1 = \gamma_2$).

However, this would hide the fact that another equivalent formulation exists where the valuation equations are linear. We write this equivalent linear formulation here

Hedged-Symmetric Funding	
$\hat{V}$	$= PV + CVA + DVA + FCA + FBA + DVA_2 + CVA_2$ $+ ColVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
DVA	$= -E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} \lambda_B LGD_B (Q - X + I_B)^- \right]$
FCA	$= -E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} (r_F - r) (Q - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} (r_F - r) (Q - X + I_B - \phi K)^- \right]$
DVA_2	$= E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} \lambda_B LGD_F (Q - X + I_B - \phi K)^+ \right]$
CVA_2	$= E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} \lambda_B LGD_F (Q - X + I_B - \phi K)^- \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right]$
KVA	$= -E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r_F + R_F \lambda_B + \lambda_C} (r_F + R_F \lambda_B + \lambda_C - r) (V^r - Q) \right]$

(61)

with $r_1 = r_2$ and the identification to the funding rate r_F as

$$r_1 = r_0 + \lambda_B LGD_F + \gamma = r_F \quad (62)$$

$$r_2 = r_0 + \lambda_B LGD_F + \gamma = r_F. \quad (63)$$

and $R_F = 1 - LGD_F$.

1.9.1.4 Firm Value

In this section our intent is to show where the so-called "Firm Value" valuation sits in our overall framework. As Firm Value is rather well known through the work of [Burgard12a], we find it valuable to see its relation to the other valuation equations developed in this document.

Firm Value is obtained as the particular case of Hedged-Symmetric Funding in (61) where there is no liquidity and the reference rate is chosen to be equal to the base rate of the funding bond r_0 . Under these assumptions, as explained before we have cancellation of FCA and FBA on one side against DVA_2 and CVA_2 on the other. The valuation equations become identical to Firm Value of [Burgard12a], i.e.

Firm Value	
$\hat{V}$	$= PV + CVA + DVA + ColVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_{r_0} \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
DVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \lambda_B LGD_B (Q - X + I_B)^- \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_X - r_0) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_{I_C} I_C - (r_{I_B} - r_0) I_B) \right]$
KVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (\gamma_K - \phi r_0) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (\lambda_B + \lambda_C) (V^{r_0} - Q) \right]$

(64)

where

$$r_1 = r_0 + \lambda_B LGD_F + 0 = r_F \quad (65)$$

$$r_2 = r_0 + \lambda_B LGD_F + 0 = r_F \quad (66)$$

$$r = r_0. \quad (67)$$

1.9.2 Three-Bond Funding

We introduce here a family of strategies that systematically leads to linear valuation equations while preserving the freedom to have asymmetric funding. This is inspired by the strategy behind "No-Shortfall" in [Burgard12a] but is extended to a more general setting.

Introducing the funding bond yields of the three-bond funding (34)~(36) in the hedged PDE (49) leads to the linear PDE

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C(g_C - \hat{V}) + \lambda_B(g_B - \hat{V}) \\ - (r_1 - LGD_{P_1}\lambda_B)(\hat{V} - Q) \\ - (r_2 - LGD_{P_2}\lambda_B)(Q - X + I_B - \phi K)^+ - (r_3 - LGD_{P_3}\lambda_B)(Q - X + I_B - \phi K)^- = 0. \end{aligned} \quad (68)$$

We can then write the valuation equations as

Hedged-Generic 3 Bonds	
$\hat{V}$	$= PV + CVA + DVA + FCA + FBA + DVA_2 + CVA_2$ $+ ColVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
DVA	$= -E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} \lambda_B LGD_B (Q - X + I_B)^- \right]$
FCA	$= -E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} (r_2 - r) (Q - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} (r_3 - r) (Q - X + I_B - \phi K)^- \right]$
DVA_2	$= E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} LGD_{P_2}\lambda_B (Q - X + I_B - \phi K)^+ \right]$
CVA_2	$= E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} LGD_{P_3}\lambda_B (Q - X + I_B - \phi K)^- \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right]$
KVA	$= -E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r_1 + R_{P_1}\lambda_B + \lambda_C} (r_1 + R_{P_1}\lambda_B + \lambda_C - r) (V^r - Q) \right]$

(69)

where the rates have their generic expressions

$$r_1 = r_0 + \lambda_B LGD_{P_1} + \gamma_1 \quad (70)$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2 \quad (71)$$

$$r_3 = r_0 + \lambda_B LGD_{P_3} + \gamma_3. \quad (72)$$

1.9.2.1 "Hedged Asymmetric Ad'hoc"

This strategy is a particular case of three-bond funding with specific identification of rates. It results in valuation equations that may be seen as quite intuitive thanks to the form of the discount in $r_0 + \lambda_B + \lambda_C$ (i.e. loosely speaking "risk-free x survival"), and we present it for this reason. If asked the question "Is there a funding strategy that leads to these equations?", the below is an answer.

Hedged-Asymmetric Ad'hoc	
$\hat{V}$	$= PV + CVA + DVA + FCA + FBA + DVA_2 + CVA_2$ $+ ColVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
DVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \lambda_B LGD_B (Q - X + I_B)^- \right]$
FCA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_F^b - r) (Q - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_F^l - r) (Q - X + I_B - \phi K)^- \right]$
DVA_2	$= E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} LGD_{P_2} \lambda_B (Q - X + I_B - \phi K)^+ \right]$
CVA_2	$= E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} LGD_{P_3} \lambda_B (Q - X + I_B - \phi K)^- \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right]$
KVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_0 + \lambda_B + \lambda_C - r) (V^r - Q) \right]$

(73)

with the rates

$$r_1 = r_0 + \lambda_B LGD_{P_1} + 0 \quad (74)$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2 = r_F^b \quad (75)$$

$$r_3 = r_0 + \lambda_B LGD_{P_3} + \gamma_3 = r_F^l. \quad (76)$$

1.9.2.2 "Hedged Reduced Asymmetric"

The goal of this strategy is to keep the possibility to have asymmetric funding and linear equations, but to be more parsimonious and avoid the rather arbitrary (74) in our bond assumptions. To this end, we take the first bond to be identical to the second one, still identifying it with the borrowing bond. Note that this relies on the hypothesis that the amount it must fund is positive, i.e. that

This strategy is a particular case of three-bond funding with specific identification of rates. It results in valuation equations that may be seen as quite intuitive, and we present it for this reason. If asked the question "Is there a funding strategy that leads to these equations?", the below is an answer.

Hedged-Reduced Asymmetric	
$\hat{V}$	$PV + CVA + DVA + FCA + FBA + DVA_2 + CVA_2$ $+ ColVA + MVA + KVA + RVA$
PV	$E_t \left[\int_t^T D_r \pi \right]$
CVA	$-E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
DVA	$-E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} \lambda_B LGD_B (Q - X + I_B)^- \right]$
FCA	$-E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} (r_F^b - r) (Q - X + I_B - \phi K)^+ \right]$
FBA	$-E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} (r_F^l - r) (Q - X + I_B - \phi K)^- \right]$
DVA_2	$E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} LGD_{P_1} \lambda_B (Q - X + I_B - \phi K)^+ \right]$
CVA_2	$E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} LGD_{P_3} \lambda_B (Q - X + I_B - \phi K)^- \right]$
$ColVA$	$-E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} (r_X - r) X \right]$
MVA	$-E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right]$
KVA	$-E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$-E_t \left[\int_t^T D_{r_F^b + R_{P_1} \lambda_B + \lambda_C} (r_F^b + R_{P_1} \lambda_B + \lambda_C - r) (V^r - Q) \right]$

(77)

with the rates

$$r_1 = r_0 + \lambda_B LGD_{P_1} + \gamma_1 = r_F^b \quad (78)$$

$$r_2 = r_0 + \lambda_B LGD_{P_1} + \gamma_1 = r_F^b \quad (79)$$

$$r_3 = r_0 + \lambda_B LGD_{P_3} + \gamma_3 = r_F^l. \quad (80)$$

1.10 Non-Hedged Strategies

The "No-Shortfall" and "Shareholder" of [Kjaer17a] are particular cases of non-hedged strategies.

As explained in section 1.8.1, not hedging of our own default risk leads to the generic form of PDE (43), which we recall here for easier reading

$$\frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{IB} I_B - r_{IC} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) + \sum_{i=1}^N r_i \alpha_i P_i = 0. \quad (81)$$

Explicit valuation equations are reached after specifying the funding strategy, of which we gave two possible candidates in section 1.7. We will examine a few of these below.

A typical feature of non-hedged strategies is that they do not, unless some very specific assumptions are made, have *DVA*. Instead they have "full" *FVA* in the sense that no *DVA*₂ and *CVA*₂ are present.

1.10.1 Two-Bond Funding

Such strategies tend to lead to non-linear valuation except when they degenerate to one-bond funding, i.e. symmetric funding. Introducing the two funding bond yields (31) and (32) in the non-hedged PDE (81) yields the non-linear PDE

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) \\ - r_1 (\hat{V} - X + I_B - \phi K)^+ - r_2 (\hat{V} - X + I_B - \phi K)^- = 0 \end{aligned} \quad (82)$$

So long as the funding is truly asymmetric i.e. that the two funding bonds are different, the above PDE leads to the non-linear valuation equations

Non-Hedged-Generic 2 Bonds	
$\hat{V}$	$= PV + CVA + FCA + FBA$ $+ ColVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r+\lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
FCA	$= -E_t \left[\int_t^T D_{r+\lambda_C} (r_1 - r) (\hat{V} - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r+\lambda_C} (r_2 - r) (\hat{V} - X + I_B - \phi K)^- \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r+\lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r+\lambda_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right]$
KVA	$= -E_t \left[\int_t^T D_{r+\lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r+\lambda_C} \lambda_C (V^r - Q) \right]$

(83)

with generic r_1 and r_2 i.e.

$$r_1 = r_0 + \lambda_B LGD_{P_1} + \gamma_1 \quad (84)$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2. \quad (85)$$

1.10.1.1 Shareholder

Shareholder is the degenerate case of two-bond funding where the two bonds are identical⁴, their yield being then identified with the funding rate r_F . In this situation the PDE becomes linear and thus leads to explicit (non-recursive) valuation equations:

Shareholder	
$\hat{V}$	$= PV + CVA + FCA + FBA$ $+ CollVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r_F + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
FCA	$= -E_t \left[\int_t^T D_{r_F + \lambda_C} (r_F - r) (Q - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r_F + \lambda_C} (r_F - r) (Q - X + I_B - \phi K)^- \right]$
$CollVA$	$= -E_t \left[\int_t^T D_{r_F + \lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r_F + \lambda_C} \left(r_{IC} I_C - (r_{IB} - r) I_B \right) \right]$
KVA	$= -E_t \left[\int_t^T D_{r_F + \lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r_F + \lambda_C} (r_F + \lambda_C - r) (V^r - Q) \right]$

(86)

with rates

$$r_1 = r_0 + \lambda_B LGD_{P_1} + \gamma_1 = r_F \quad (87)$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2 = r_F. \quad (88)$$

⁴Strictly speaking it is the bond yields that are identical in this strategy, i.e. $r_1 = r_2$, not necessarily the bonds themselves in the sense of their seniority and liquidity spread.

1.10.2 Three-Bond Funding

Similarly to the case of the hedged strategies in section 1.9.2, linear strategies under no own default hedge are obtained using the three-bond funding in section 1.7.2. Introducing the three funding bond yields (34)~(36) in the non-hedged PDE (81) leads to the linear PDE

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) \\ - r_1 (\hat{V} - Q) \\ - r_2 (Q - X + I_B - \phi K)^+ - r_3 (Q - X + I_B - \phi K)^- = 0 \end{aligned} \quad (89)$$

which is the analogue of (68) but in the non-hedged case now. This leads to the valuation equations

Non-Hedged-Generic 3 Bonds	
$\hat{V}$	$= PV + CVA + FCA + FBA$ $+ CollVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r_1 + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
FCA	$= -E_t \left[\int_t^T D_{r_1 + \lambda_C} (r_2 - r) (Q - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r_1 + \lambda_C} (r_3 - r) (Q - X + I_B - \phi K)^- \right]$
$CollVA$	$= -E_t \left[\int_t^T D_{r_1 + \lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r_1 + \lambda_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right]$
KVA	$= -E_t \left[\int_t^T D_{r_1 + \lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r_1 + \lambda_C} (r_1 + \lambda_C - r) (V^r - Q) \right].$

(90)

with the generic rates

$$r_1 = r_0 + \lambda_B LGD_{P_1} + \gamma_1 \quad (91)$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2 \quad (92)$$

$$r_3 = r_0 + \lambda_B LGD_{P_3} + \gamma_3. \quad (93)$$

1.10.2.1 "Non-Hedged Asymmetric Ad'hoc"

This strategy is a particular case of three-bond funding with specific identification of rates. It results in valuation equations that may be seen as quite intuitive thanks to the form of the discount in $r_0 + \lambda_B + \lambda_C$ (i.e. loosely speaking "risk-free x survival"), and we present it for this reason. If asked the question *"Is there a funding strategy that leads to these equations?"*, the below is an answer.

Non-Hedged-Asymmetric Ad'hoc	
$\hat{V}$	$= PV + CVA + FCA + FBA$ $+ CollVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
FCA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_F^b - r) (Q - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_F^l - r) (Q - X + I_B - \phi K)^- \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \left(r_{IC} I_C - (r_{IB} - r) I_B \right) \right]$
KVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_0 + \lambda_B + \lambda_C - r) (V^r - Q) \right]$.

(94)

with rates

$$r_1 = r_0 + \lambda_B \times 100\% + 0 \quad (95)$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2 = r_F^b \quad (96)$$

$$r_3 = r_0 + \lambda_B LGD_{P_3} + \gamma_3 = r_F^l. \quad (97)$$

where the LGD on the first bond is 100% and its liquidity spread is 0. Note that this very specific choice of first bond is the choice made in the No-Shortfall strategy we examine below, see [Green16a] for a more detailed account.

1.10.2.2 No-Shortfall (FBA)

This is the strategy discussed in [Burgard12a]. We can think of it as the particular case of (94) with symmetric funding. The valuation is

No-Shortfall (FBA)	
$\hat{V}$	$= PV + CVA + FCA + FBA$ $+ ColVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
FCA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_F - r) (Q - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_F - r) (Q - X + I_B - \phi K)^- \right]$
$ColVA$	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \left(r_{IC} I_C - (r_{IB} - r) I_B \right) \right]$
KVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_0 + \lambda_B + \lambda_C - r) (V^r - Q) \right]$

(98)

with the rates

$$r_1 = r_0 + \lambda_B \times 100\% + 0 \quad (99)$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2 = r_F \quad (100)$$

$$r_3 = r_0 + \lambda_B LGD_{P_3} + \gamma_3 = r_F. \quad (101)$$

1.10.2.3 No-Shortfall (DVA)

This strategy is described in [Green16a] and is also a particular case of (94). It is designed to have a DVA term, which is obtained by a very specific choice of 3rd funding bond P_3 such that what was called FBA in all previous strategies looks more like a DVA. The valuation is

$$\begin{aligned}
 \hat{V} &= PV + CVA + DVA + FCA \\
 &\quad + ColVA + MVA + KVA + RVA \\
 PV &= E_t \left[\int_t^T D_{r_0} \pi \right] \\
 CVA &= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right] \\
 DVA &= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} \lambda_B LGD_B (Q - X + I_B - \phi K)^- \right] \\
 FCA &= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_F - r_0) (Q - X + I_B - \phi K)^+ \right] \\
 ColVA &= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_X - r_0) X \right] \\
 MVA &= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (r_{I_C} I_C - (r_{I_B} - r_0) I_B) \right] \\
 KVA &= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (\gamma_K - \phi r_0) K \right] \\
 RVA &= -E_t \left[\int_t^T D_{r_0 + \lambda_B + \lambda_C} (\lambda_B + \lambda_C) (V^{r_0} - Q) \right].
 \end{aligned} \tag{102}$$

with rates

$$r_1 = r_0 + \lambda_B \times 100\% + 0 \tag{103}$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2 = r_F \tag{104}$$

$$r_3 = r_0 + \lambda_B LGD_B + 0 \tag{105}$$

$$r = r_0. \tag{106}$$

Note that on top of assuming no liquidity on the 3rd bond, we must also assume that its LGD is that on the derivative's netting set, and also choose the reference rate to the particular value r_0 . Furthermore, the term now called DVA contains a capital component, which may not be what we would expect of a traditional DVA translating on-default payments as all the DVA terms obtained in other strategies.

1.10.2.4 "Non-Hedged Reduced Asymmetric"

This strategy is the non-hedged version of (77). The first two bonds reduce to one⁵ and the same bond with borrowing rate as yield. This leads to the valuation

Non-Hedged-Generic 3 Bonds	
$\hat{V}$	$= PV + CVA + FCA + FBA$ $+ CollVA + MVA + KVA + RVA$
PV	$= E_t \left[\int_t^T D_r \pi \right]$
CVA	$= -E_t \left[\int_t^T D_{r_F^b + \lambda_C} \lambda_C LGD_C (Q - X - I_C)^+ \right]$
FCA	$= -E_t \left[\int_t^T D_{r_F^b + \lambda_C} (r_F^b - r) (Q - X + I_B - \phi K)^+ \right]$
FBA	$= -E_t \left[\int_t^T D_{r_F^b + \lambda_C} (r_F^l - r) (Q - X + I_B - \phi K)^- \right]$
$CollVA$	$= -E_t \left[\int_t^T D_{r_F^b + \lambda_C} (r_X - r) X \right]$
MVA	$= -E_t \left[\int_t^T D_{r_F^b + \lambda_C} \left(r_{IC} I_C - (r_{IB} - r) I_B \right) \right]$
KVA	$= -E_t \left[\int_t^T D_{r_F^b + \lambda_C} (\gamma_K - \phi r) K \right]$
RVA	$= -E_t \left[\int_t^T D_{r_F^b + \lambda_C} (r_1 + \lambda_C - r) (V^r - Q) \right].$

(107)

with the generic rates

$$r_1 = r_0 + \lambda_B LGD_{P_1} + \gamma_1 = r_F^b \quad (108)$$

$$r_2 = r_0 + \lambda_B LGD_{P_2} + \gamma_2 = r_F^b \quad (109)$$

$$r_3 = r_0 + \lambda_B LGD_{P_3} + \gamma_3 = r_F^l. \quad (110)$$

⁵Again strictly speaking the two bonds do not need to be the same, it is their yield that is the same.

1.11 Replication at Company Level

In this section we take a look at the valuation equations obtained from the Replication method when considering the whole company including all its trade sets with all counterparties. Until now we have looked at the case of a single netting set/counterparty and assumed implicitly that both PV and XVAs are simply summed to obtain the total PV and XVAs for the company. This is implicitly assuming that funding is done at the netting set level, which is not necessarily a realistic assumption. We will relax this assumption below.

Let us now define notations to address the question of company-level valuation:

- $\hat{V}$ is now the risky value of the total portfolio including all netting sets
- N_T, N_N, N_F and N_C are the total numbers of trades, netting sets, funding sets and counterparties
- by 'netting set' here we refer more specifically to the netting set used for a close-out on a default event of a given counterparty. These corresponds to the CSA netting sets when there is a CSA, otherwise to any other counterparty-level agreement we may have. In particular, those netting sets are no longer connected to how we fund. Note that there could be several netting sets with one counterparty.
- by 'funding set' we refer to the group of trades whose value is funded separately from the other trades. In particular, there is no relation assumed here between netting sets and funding sets.
- we assume that all trades belong to one netting set and one funding set (i.e. they always belong to one but never more than one).
- $i = 1..N_T$ counts the trades, $\eta = 1..N_N$ counts the netting sets, $\varphi = 1..N_F$ counts the funding sets and $\gamma = 1..N_C$ counts the counterparties.
- $\mathcal{N}(\gamma)$ is the group of netting sets the bank has with counterparty C^γ

We will now go through the Replication arguments again with the goal to value the total company-wide risky derivative $\hat{V}$.

1.11.1 Replicating Portfolio

Underlying

Nothing changes on this part. We still simplify things by assuming a single underlying with the same dynamics as before.

Counterparty bonds

Now we have to consider several counterparties. To generalize the single counterparty case, we assume one jump process per counterparty to drive the C/P's default. This process is denoted by dJ_{C^γ} and we use the repo-able bond P_{C^γ} to hedge C^γ 's default. The corresponding component of the change of portfolio has the same expression as before, the difference being that there is *one component per counterparty*.

Bank bonds

Remember that bank bonds are split into two categories: repo-able ones that we use for hedging our own risk (or not), and non-repo-able ones that we use for funding. In the situation here, there is no change on the hedging part, for which we still use the single repo-able bond P_B with jump process dJ_B . Things may change on the funding side where we may use different bonds for funding different funding sets. This is not an obligation though, as we may use the same bonds as before but in a different way. We will discuss the new form of the funding strategy later.

Margining

Margining is done at the netting set level. This means that we now have one collateral amount X^η per netting set η and similarly for the other margins.

Funding condition

Funding is needed for the amount that remains after the various rehypothecable margins have been subtracted from the derivative's (risky) value. So the funding condition is now written

$$\text{Funding condition: } \sum_{i=1}^N \alpha_i P_i + \hat{V} - \sum_{\eta=1}^{N_N} X^\eta + \sum_{\eta=1}^{N_N} I_B^\eta - \phi K = 0. \quad (111)$$

where $P_i, i = 1..N$ are the funding bonds and X^η (resp. I_B^η) are the variation (resp. initial) margin associated with the netting set η .

Self-financing condition

The funding condition can again be used to rewrite the self-financing condition in a rather straightforward way as

$$\begin{aligned} 0 = & d\hat{V} - \sum_{\eta=1}^{N_N} r_{X^\eta} X^\eta dt + \sum_{\eta=1}^{N_N} r_{I_B^\eta} I_B^\eta dt - \sum_{\eta=1}^{N_N} r_{I_C^\eta} I_C^\eta dt - \gamma_K K(t) dt + \delta dS + \delta(\gamma_S - q_S) S dt \\ & + \sum_{i=1}^N \alpha_i dP_i + \sum_{\gamma=1}^{N_C} \alpha_{C^\gamma} dP_{C^\gamma} - \sum_{\gamma=1}^{N_C} \alpha_{C^\gamma} q_{C^\gamma} P_{C^\gamma} dt + \alpha_B dP_B - \alpha_B q_B P_B dt. \end{aligned} \quad (112)$$

1.11.2 Dynamics

The dynamics of the underlying and the bonds does not change here, the only difference being that we make use of more of the bonds (the C/P's bonds and our funding bonds). What changes more explicitly is the change in the risky derivative as now there are several jump processes, one for each counterparty. The change of the derivative is simply written as

$$d\hat{V} = \frac{\partial \hat{V}}{\partial t} dt + \frac{\partial \hat{V}}{\partial S} dS + \frac{1}{2} \sigma_S^2 S^2 \frac{\partial^2 \hat{V}}{\partial S^2} dt + \sum_{\gamma=1}^{N_C} (g_{C^\gamma} - \hat{V}) dJ_{C^\gamma} + (g_B - \hat{V}) dJ_B \quad (113)$$

where $g_{C^\gamma} - \hat{V}$ is the change in the derivative upon default of counterparty γ . Here we have made again the common assumption that we can neglect double default events.

1.11.3 Close-Out

Determining the on-default flows g_{C^γ} and g_B follows a very similar logic as before, but now we must be careful about which netting set is subject to default resolution.

If one of our counterparties C^γ defaults first, then we must settle all the netting sets with that counterparty. This means settling all the margin accounts, and then receiving the recovery rate on the netting set with positive exposures, or paying the whole exposure on the netting sets with negative exposures. This leads to

$$g_{C^\gamma} = \sum_{\eta \in N(\gamma)} Q^\eta - \sum_{\eta \in N(\gamma)} LGD_C^\eta (Q^\eta - X^\eta - I_C^\eta)^+ \quad (114)$$

where $\eta \in N(\gamma)$ denotes all the netting sets η we have with the counterparty C^γ .

If we default first, on the other hand, *all* the netting sets must be settled which means that the on-default payment is

$$g_B = Q - \sum_{\eta=1}^{N_N} LGD_B^\eta (Q^\eta - X^\eta + I_B^\eta)^-. \quad (115)$$

where by Q we denote the sum of all close-outs

$$Q = \sum_{\eta=1}^{N_N} Q^\eta. \quad (116)$$

1.11.4 Hedging constraints

Each C/P's default risk is hedged separately using that C/P's bonds. As can be expected, we obtain very similar formulas as for the single C/P case, in that for each C/P we find the hedge ratio through the relation

$$\alpha_{C^\gamma} P_{C^\gamma} = \frac{g_{C^\gamma} - \hat{V}}{LGD_{P_{C^\gamma}}}. \quad (117)$$

Explicitly nothing changes on the side of the bank's own default since the bank's default is still hedged (for a hedged strategy) by the bank's repo-able bond. The self-financing condition thus reads

$$\begin{aligned} 0 = & \left[\frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - \sum_{\eta=1}^{N_N} r_{X^\eta} X^\eta + \sum_{\eta=1}^{N_N} r_{I_B^\eta} I_B^\eta - \sum_{\eta=1}^{N_N} r_{I_C^\eta} I_C^\eta - \gamma_K K + \sum_{\gamma=1}^{N_C} \lambda_{C^\gamma} (g_{C^\gamma} - \hat{V}) \right. \\ & \left. + \alpha_B (r_B - q_B) P_B + \sum_{i=1}^N r_i \alpha_i P_i \right] dt \\ & + \epsilon_h dJ_B \end{aligned} \quad (118)$$

where the hedge error has the same expression as before i.e.

$$\epsilon_h = g_B - \hat{V} - \alpha_B P_B LGD_{P_B} - \sum_{i=1}^N \alpha_i P_i LGD_{P_i}. \quad (119)$$

1.11.5 Funding constraints

This is where the crucial difference lies. As the three-bond split funding is the most flexible strategy, we are going to use it here to illustrate how the valuation changes when looking at it company-wide.

As found above the funding constraint now reads

$$\sum_{i=1}^N \alpha_i P_i = -(\hat{V} - \sum_{\eta=1}^{N_N} X^\eta + \sum_{\eta=1}^{N_N} I_B^\eta - \phi K). \quad (120)$$

Our company has now defined *funding sets*. These funding sets cover the whole funding requirement above on the r.h.s. That is, a funding set is defined by what amounts are netted against what other amounts and which bonds are funding the netted results. In the funding requirement above, we have a few margins as well as a capital term. The funding sets will also specify which of these margins is netted against what other component (such as trade values), such that the sum above can be decomposed as a sum over the netting sets.

In principle a margin could be split across several funding sets and we could also have several margins netted in the same funding set. As the funding requirement above is linear, we have the freedom to redistribute all the amounts to different funding sets, so long as the sum is unchanged (associativity of the sum). A funding strategy thus has to define the redistribution

$$\sum_{\varphi=1}^{N_F} \tilde{X}^\varphi = \sum_{\eta=1}^{N_N} X^\eta \quad (121)$$

$$\sum_{\varphi=1}^{N_F} \tilde{I}_B^\varphi = \sum_{\eta=1}^{N_N} I_B^\eta \quad (122)$$

$$\sum_{\varphi=1}^{N_F} \tilde{K}^\varphi = K \quad (123)$$

Tilded quantities on the l.h.s. represent how our margins/capital are redistributed over our funding sets.

For some funding sets, there may be no margin, or there may be a sum of several margins coming from different netting sets. Similarly, the total capital amount may be funded in one funding set or distributed across several funding sets.

Finally, let us also redistribute the close-out amounts Q^η over the funding sets, again following the rule of associativity of the sum

$$\sum_{\varphi=1}^{N_F} \tilde{Q}^\varphi = \sum_{\eta=1}^{N_N} Q^\eta = Q \quad (124)$$

where $\tilde{Q}^\varphi$ is the aggregated close-out amount funded in funding set φ .

With this specification of our funding sets, and a little algebra, we can rewrite the funding condition as

$$\sum_{i=1}^N \alpha_i P_i = -(\hat{V} - \sum_{\eta=1}^{N_N} X^\eta + \sum_{\eta=1}^{N_N} I_B^\eta - \phi K) \quad (125)$$

$$= -(\hat{V} - \sum_{\varphi=1}^{N_F} \tilde{X}^\varphi + \sum_{\varphi=1}^{N_F} \tilde{I}_B^\varphi - \phi \sum_{\varphi=1}^{N_F} \tilde{K}^\varphi) \quad (126)$$

$$= -(\hat{V} - \sum_{\varphi=1}^{N_F} \tilde{Q}^\varphi) - \sum_{\varphi=1}^{N_F} (\tilde{Q}^\varphi - \tilde{X}^\varphi + \tilde{I}_B^\varphi - \phi \tilde{K}^\varphi) \quad (127)$$

$$= -(\hat{V} - Q) - \sum_{\varphi=1}^{N_F} (\tilde{Q}^\varphi - \tilde{X}^\varphi + \tilde{I}_B^\varphi - \phi \tilde{K}^\varphi) \quad (128)$$

where we have added and subtracted the dummy sum of all close-outs and then used the associativity rule in the last line above.

This formulation lets us generalize the 3-bond split funding strategy as follows:

- bond P_1 funds "the risky part", which here is

$$\hat{V} - Q \quad (129)$$

i.e. the difference between the risky value and the sum of all close-out amounts

- in the most general form of the strategy, we use 2 bonds per funding set to fund the positive and negative parts of each funding requirement. Let us call these bonds $P_\varphi^\pm$

Then the funding strategy can be written as

$$\alpha_1 P_1 = -(\hat{V} - Q) \quad (130)$$

$$\alpha_\varphi^+ P_\varphi^+ = -(\tilde{Q}^\varphi - \tilde{X}^\varphi + \tilde{I}_B^\varphi - \phi \tilde{K}^\varphi)^+ \quad (131)$$

$$\alpha_\varphi^- P_\varphi^- = -(\tilde{Q}^\varphi - \tilde{X}^\varphi + \tilde{I}_B^\varphi - \phi \tilde{K}^\varphi)^- \quad (132)$$

where the second and third rows hold for each funding set φ .

1.11.6 PDEs and Valuation

Let us now introduce the notion of PV. PV of the total derivative is denoted by V and is the sum of the PVs of each trade, where the PV of each trade is defined as the expected value of its flows discounted with a certain rate. The difference with the single netting set case we have seen before is that here we define the PV discount rates to be in line with the funding sets. That is, we consider that each funding set identifies a specific rate that is used to discount all the flows of the trades that belong to this funding set. Let us call these rates r_φ . Due to associativity of the sum, the total PV can be obtained by summing the PVs of all the trades, or alternatively, by summing the PVs of the trades within a funding set and then summing the PVs of the funding sets. Let us define the PV of a funding set as

$$V^\varphi = \sum_{i \in N(\varphi)} V_i^{r_\varphi}. \quad (133)$$

This simply means that the PV of a funding set V^φ is the sum of the PVs of the trades in that funding set, and that each trade i in that set has a PV $V_i^{r_\varphi}$ obtained by discounting its flows with the funding set rate r_φ . Then the associativity rule tells us that

$$V = \sum_{i=1}^{N_T} V_i = \sum_{\varphi=1}^{N_F} V^\varphi \quad (134)$$

i.e. that we can equivalently sum all the trade PVs or sum the funding set PVs.

Now the decision that all trades in a funding set φ are discounted the same rate r_φ has the implication that

$$\frac{\partial V^\varphi}{\partial t} + \mathcal{A}V^\varphi = r_\varphi V^\varphi \quad (135)$$

where V^φ is the PV aggregated over the funding set. In turns this means that

$$\frac{\partial V}{\partial t} + \mathcal{A}V = \sum_{\varphi=1}^{N_F} r_\varphi V^\varphi. \quad (136)$$

We can now decompose the risky derivative as usual into PV and the XVAs, this time at the company level, but with the same expression as before

$$\hat{V} = V + U \quad (137)$$

where this time the PV V is obtained by summing over all funding sets and U is the aggregate XVAs at company level.

For illustration purposes we will derive the valuation equations in the hedged case, but the non-hedged case works in a very similar fashion.

Hedged PDE

First, we enforce vanishing of the hedge error (119) and use the resulting constraint in (118) to obtain the PDE

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} &= \sum_{\eta=1}^{N_N} r_{X^\eta} X^\eta - \sum_{\eta=1}^{N_N} r_{I_B^\eta} I_B^\eta + \sum_{\eta=1}^{N_N} r_{I_C^\eta} I_C^\eta + \gamma_K K - \sum_{\gamma=1}^{N_C} \lambda_{C^\gamma} (g_{C^\gamma} - \hat{V}) \\ &\quad - \lambda_B (g_B - \hat{V}) - \sum_{i=1}^N \alpha_i P_i (r_i - \lambda_B LGD_{P_i}). \end{aligned} \quad (138)$$

Then we use the expressions of the on-default flows (114) and (115) to obtain

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} &= \sum_{\eta=1}^{N_N} r_{X^\eta} X^\eta - \sum_{\eta=1}^{N_N} r_{I_B^\eta} I_B^\eta + \sum_{\eta=1}^{N_N} r_{I_C^\eta} I_C^\eta + \gamma_K K + \sum_{\gamma=1}^{N_C} \lambda_{C^\gamma} \sum_{\eta \in N(\gamma)} LGD_C^\eta (Q^\eta - X^\eta - I_C^\eta)^+ \\ &\quad + \lambda_B \sum_{\eta=1}^{N_N} LGD_B^\eta (Q^\eta - X^\eta + I_B^\eta)^- - \sum_{i=1}^N \alpha_i P_i (r_i - \lambda_B LGD_{P_i}) \\ &\quad - \sum_{\gamma=1}^{N_C} \lambda_{C^\gamma} \left(\sum_{\eta \in N(\gamma)} Q^\eta - \hat{V} \right) - \lambda_B (Q - \hat{V}) \end{aligned} \quad (139)$$

The next step is to introduce the funding strategy described earlier, which leads to

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} &= \sum_{\gamma=1}^{N_C} \lambda_{C^\gamma} \sum_{\eta \in N(\gamma)} LGD_C^\eta (Q^\eta - X^\eta - I_C^\eta)^+ + \lambda_B \sum_{\eta=1}^{N_N} LGD_B^\eta (Q^\eta - X^\eta + I_B^\eta)^- \\ &\quad + \sum_{\varphi=1}^{N_F} (r_\varphi^+ - \lambda_B LGD_{P_\varphi^+}) (\tilde{Q}^\varphi - \tilde{X}^\varphi + \tilde{I}_B^\varphi - \phi \tilde{K}^\varphi)^+ + \sum_{\varphi=1}^{N_F} (r_\varphi^- - \lambda_B LGD_{P_\varphi^-}) (\tilde{Q}^\varphi - \tilde{X}^\varphi + \tilde{I}_B^\varphi - \phi \tilde{K}^\varphi)^- \\ &\quad + \sum_{\eta=1}^{N_N} r_{X^\eta} X^\eta - \sum_{\eta=1}^{N_N} r_{I_B^\eta} I_B^\eta + \sum_{\eta=1}^{N_N} r_{I_C^\eta} I_C^\eta + \gamma_K K \\ &\quad + (r_1 - \lambda_B LGD_{P_1}) (\hat{V} - Q) - \sum_{\gamma=1}^{N_C} \lambda_{C^\gamma} \left(\sum_{\eta \in N(\gamma)} Q^\eta - \hat{V} \right) - \lambda_B (Q - \hat{V}) \end{aligned} \quad (140)$$

At this point we can pause, look at the various terms and see that we can identify components on the r.h.s. as

- the first row containing CVA and DVA
- the second row that hints at FCA and FBA, combined with DVA2 and CVA2
- the third row containing margin terms
- RVA and discounting related terms.

Before introducing PV, let us take the case of a single counterparty with a single netting set. The above PDE becomes

$$\begin{aligned}
\frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} &= \lambda_C LGD_C(Q - X - I_C)^+ + \lambda_B LGD_B(Q - X + I_B)^- \\
&+ \sum_{\varphi=1}^{N_F} (r_{\varphi}^+ - \lambda_B LGD_{P_{\varphi}^+})(\tilde{Q}^{\varphi} - \tilde{X}^{\varphi} + \tilde{I}_B^{\varphi} - \phi \tilde{K}^{\varphi})^+ \\
&+ \sum_{\varphi=1}^{N_F} (r_{\varphi}^- - \lambda_B LGD_{P_{\varphi}^-})(\tilde{Q}^{\varphi} - \tilde{X}^{\varphi} + \tilde{I}_B^{\varphi} - \phi \tilde{K}^{\varphi})^- \\
&+ r_X X - r_{I_B} I_B + r_{I_C} I_C + \gamma_K K \\
&+ (r_1 - \lambda_B LGD_{P_1} + \lambda_B + \lambda_C)(\hat{V} - Q)
\end{aligned} \tag{141}$$

The last step we need is to introduce PV, which we have defined previously, and split valuation between PV and XVAs. This leads to

$$\begin{aligned}
\frac{\partial U}{\partial t} + \mathcal{A}\hat{U} - (r_1 - \lambda_B LGD_{P_1} + \lambda_B + \lambda_C)U &= \lambda_C LGD_C(Q - X - I_C)^+ + \lambda_B LGD_B(Q - X + I_B)^- \\
&+ \sum_{\varphi=1}^{N_F} (r_{\varphi}^+ - \lambda_B LGD_{P_{\varphi}^+})(\tilde{Q}^{\varphi} - \tilde{X}^{\varphi} + \tilde{I}_B^{\varphi} - \phi \tilde{K}^{\varphi})^+ \\
&+ \sum_{\varphi=1}^{N_F} (r_{\varphi}^- - \lambda_B LGD_{P_{\varphi}^-})(\tilde{Q}^{\varphi} - \tilde{X}^{\varphi} + \tilde{I}_B^{\varphi} - \phi \tilde{K}^{\varphi})^- \\
&+ r_X X - r_{I_B} I_B + r_{I_C} I_C + \gamma_K K - \sum_{\varphi=1}^{N_F} r_{\varphi} \tilde{Q}^{\varphi} \\
&+ (r_1 - \lambda_B LGD_{P_1} + \lambda_B + \lambda_C)(V - Q) - \sum_{\varphi=1}^{N_F} r_{\varphi} (V^{\varphi} - \tilde{Q}^{\varphi})
\end{aligned} \tag{142}$$

In lines 2, 3, and 4 we recognize the terms which usually combined into what we call FCA, FBA, DVA2, CVA2, ColVA, MVA and KVA. However here it is not apparent how to recover the usual formulas for all these VAs. The closes we can get without further assumptions, with a little algebra, is

$$\begin{aligned}
\frac{\partial U}{\partial t} + \mathcal{A}\hat{U} - (r_1 - \lambda_B LGD_{P_1} + \lambda_B + \lambda_C)U &= \lambda_C LGD_C(Q - X - I_C)^+ + \lambda_B LGD_B(Q - X + I_B)^- \\
&+ \sum_{\varphi=1}^{N_F} (r_{\varphi}^+ - \lambda_B LGD_{P_{\varphi}^+} - r_{\varphi})(\tilde{Q}^{\varphi} - \tilde{X}^{\varphi} + \tilde{I}_B^{\varphi} - \phi \tilde{K}^{\varphi})^+ \\
&+ \sum_{\varphi=1}^{N_F} (r_{\varphi}^- - \lambda_B LGD_{P_{\varphi}^-} - r_{\varphi})(\tilde{Q}^{\varphi} - \tilde{X}^{\varphi} + \tilde{I}_B^{\varphi} - \phi \tilde{K}^{\varphi})^- \\
&+ \sum_{\varphi=1}^{N_F} (r_X - r_{\varphi})\tilde{X}^{\varphi} \\
&+ r_{I_C} I_C - \sum_{\varphi=1}^{N_F} (r_{I_B} - r_{\varphi})\tilde{I}_B^{\varphi} \\
&+ \sum_{\varphi=1}^{N_F} (\gamma_K - r_{\varphi}\phi)\tilde{K}^{\varphi} \\
&+ (r_1 - \lambda_B LGD_{P_1} + \lambda_B + \lambda_C)(V - Q) - \sum_{\varphi=1}^{N_F} r_{\varphi}(V^{\varphi} - \tilde{Q}^{\varphi})
\end{aligned}
\tag{43}$$

The notations are a little cumbersome so let us summarize the results here:

- We are able to obtain typical expressions for FCA and FBA (and their corresponding DVA2 and CVA2). These are written in terms of the defined funding sets. For each funding set, FVA is the exposure aggregated over the trades specified as belonging to the funding set, multiplied by a "funding spread" for this funding set. The funding spreads are the differences between the defined funding rates on each funding set and the PV rates on the same set.
- on the margin side, we obtain the ColVA, MVA and KVA as sums over the funding sets proportional to the pieces of margins that were attributed to each funding set. The spreads on each of these components are then measured by the differences between the rates paid on the margins per the CSA specifications and the PV rates on each funding set.

1.12 Defaultable Underlyings

In this section we consider an extension of the main XVA framework (netting set level) for the case where the underlying S is defaultable. This is essentially a review of the work of [Lichtner15a] adapted to our notations and with only a few minor extensions (namely to more general repo and LGD structures for the new risks).

1.12.1 Dynamics

The underlying S 's evolution is split into a non-defaultable part S_c following a diffusion as before, and the defaultable S following an additional jump process. On default, S jumps from S_d the diffused value to R_S the recovery value. The dynamics of these quantities is defined as

$$dS = dS_d + (R_S - S_d)dJ_S \quad (144)$$

$$dS_d = \mu_{S_d} S_d dt + \sigma_{S_d} S_d dW_{S_d} \quad (145)$$

where dJ_S is a jump process from 0 to 1 on default of S , with intensity λ_S . Denoting the Counterparty by C , three independent default events are considered:

- S defaults but C does not, with intensity $\lambda_{S/C}$
- C defaults but S does not, with intensity $\lambda_{C/S}$. Note that this event is identified with C/P 's default in the previous cases of non-defaultable underlyings. In the below we will use interchangeably C/S and C to denote this event.
- S and C default, with intensity $\lambda_{S \wedge C}$.

As these default events are independent, we have the following expression of S 's default intensity

$$\lambda_S = \lambda_{S \wedge C} + \lambda_{S/C}. \quad (146)$$

CLNs (bonds) are considered for the 3 above events and the unconditional default of S . We denote these with the letter E taking the values $S, S/C, C/S$ and $S \wedge C$ with obvious notations. These bonds are used for hedging and thus have the repo-able dynamics

$$dP_E = r_E P_E dt - LGD_E P_E dJ_E \quad (147)$$

$$r_E = q_E + LGD_E \lambda_E \quad (148)$$

where λ_E is the intensity of dJ_E and q_E is the repo rate.

Ito's formula for the risky derivative is obtained considering it as a function of 4 independent variables: the diffused underlying S_d and the 3 jump processes $J_{S/C}, J_{C/S}$ and $J_{S \wedge C}$

$$\begin{aligned} d\hat{V} = & \frac{\partial \hat{V}}{\partial t} dt + \frac{\partial \hat{V}}{\partial S_d} dS_d + \frac{1}{2} \sigma_{S_d}^2 S_d^2 \frac{\partial^2 \hat{V}}{\partial S_d^2} dt + (g_C - \hat{V}) dJ_C + (g_B - \hat{V}) dJ_B \\ & + (g_{S/C} - \hat{V}) dJ_{S/C} + (g_{S \wedge C} - \hat{V}) dJ_{S \wedge C} \end{aligned} \quad (149)$$

where $g_E - \hat{V}$ is the change in the derivative price upon default of entity E and g_E is the derivative value on-default. As noted earlier in this section, the event denoted by C here is C/P 's default when the underlying has not defaulted, a.k.a. C/S . g_C is the same on-default value as before i.e. (22). The other on-default values $g_{S/C}$ and $g_{S \wedge C}$ will be defined in specific cases later.

1.12.2 Replicating Portfolio

On top of the same bonds as before for hedging C/P and O/S risk and for funding, and the underlying S itself, we use the 4 CLNs for hedging⁶.

For the same reasons as the C/P and O/S hedging bonds, these bonds lead to the portfolio changes

$$\delta\Pi|_{P_E} = \alpha_E dP_E - \alpha_E q_E P_E dt \quad (150)$$

where α_E is the hedge ratio. Note that the component $E = C = C/S$ is the same as the C/P component we had until now, while the new components are for $E = S/C, S \wedge C$ and $E = S$.

All other components of the portfolio change remain unchanged, and in particular we still have

$$\delta\Pi|_S = \delta dS + \delta(\gamma_S - q_S) S_d dt \quad (151)$$

where we consider that the underlying hasn't defaulted at the beginning of the period such that repo and/or dividend are earned on the diffused underlying.

1.12.3 Hedging

First we look at the hedge of the diffusion part of the underlying. The relevant terms come from the change in the derivative, in the underlying and in the underlying CLN positions. These are collected as

$$\frac{\partial \hat{V}}{\partial S_d} dS_d + \delta dS + \delta(\gamma_S - q_S) S_d dt + \alpha_S dP_D - \alpha_S q_{P_S} P_S dt \quad (152)$$

where q_{P_S} is the repo rate on the CLN P_S , to be distinguished from q_S the repo rate on the underlying. The diffusion component S_d is hedged by setting the hedge ratio as

$$\delta = -\frac{\partial \hat{V}}{\partial S_d}. \quad (153)$$

Note that due to the jump component in dS in (144), we are then left with

$$-\frac{\partial \hat{V}}{\partial S_d} [(R_S - S_d) dJ_S + (\gamma_S - q_S) S_d dt] + \alpha_S P_S (r_S dt - LGD_S dJ_S) - \alpha_S q_{P_S} P_S dt \quad (154)$$

where the jump piece is hedged by setting

$$\alpha_S P_S = -\frac{\partial \hat{V}}{\partial S_d} \frac{R_S - S_d}{LGD_S} \quad (155)$$

resulting in a left-over drift-like term of

$$\frac{\partial \hat{V}}{\partial S_d} [(q_S - \gamma_S + \lambda_S) S_d - R_S \lambda_S] dt. \quad (156)$$

For the default events $C/S, S/C$ and $S \wedge C$ we obtain in a straightforward manner

⁶[SG: check if we could consider one of these bonds redundant together with the corresponding jump and proceed to hedging with only 3 CLNs.]

$$\alpha_C P_C = \frac{g_C - \hat{V}}{LGD_C} \quad (157)$$

$$\alpha_{S/C} P_{S/C} = \frac{g_{S/C} - \hat{V}}{LGD_{S/C}} \quad (158)$$

$$\alpha_{S \wedge C} P_{S \wedge C} = \frac{g_{S \wedge C} - \hat{V}}{LGD_{S \wedge C}}. \quad (159)$$

1.12.4 Expected Bond Price

The process defined in (144-145) is suitable for describing a zero-coupon bond (ZCB), but adjustments are needed to capture the effect of the coupons. No additional source of randomness is added for the coupons such that the framework above does not need to change, but when specifying the derivative and the underlying we will have to take care of the specific cash-flows.

So let us consider that the process in (144-145) describes a ZCB with recovery R_S . The pricing measure in which we applied Feynman-Kac's theorem is the one for which the process has the form

$$dS = dS_d + (R_S - S_d)dJ_S \quad (160)$$

$$dS_d = [(q_S + \lambda_S)S_d - R_S \lambda_S]dt + \sigma_{S_d} S_d dW_{S_d}. \quad (161)$$

This process is integrated as follows

$$S_d(T) = e^{\int_t^T q_S + \lambda_S} \left(S_t - R_S N \int_t^T e^{-\int_t^u q_S + \lambda_S} \lambda_S + \int_t^T e^{-\int_t^u q_S + \lambda_S} \sigma_{S_d} S_d dW_{S_d} \right). \quad (162)$$

In particular we can calculate the discounted expectation of it, which is

$$E_t[e^{-\int_t^T q_S + \lambda_S} S_T] = S_t - R_S N \int_t^T E_t[e^{-\int_t^u q_S + \lambda_S} \lambda_S]. \quad (163)$$

We have dropped the dividend rate as we're specifically talking about a bond here.

1.12.5 Coupon Bond

Now let us denote by c_i the coupon paid at time T_i with year fraction δ_i . Then we consider the coupon-bearing bond S as a series of ZCBs, one for each coupon with 0 recovery, and one for the final nominal payment with recovery R_S . Each of these ZCB components follows the above process and expectation up until its payment date at which it falls to 0. We thus model the coupon-bearing bond process as

$$S_t = N \sum_{T_i \geq t}^{T_N} c_i \delta_i + N_t \quad (164)$$

where the c_i are the coupon ZCBs processes with 0 recovery and N_t is the nominal ZCB process with recovery R_S . Now we want to calculate the same discounted expectation as before:

$$\begin{aligned} E_t[e^{-\int_t^T q_S + \lambda_S} S_T] &= N \sum_{T_i \geq T}^{T_N} \delta_i E_t[e^{-\int_t^T q_S + \lambda_S} c_i(T)] + E_t[e^{-\int_t^T q_S + \lambda_S} N_T] \\ &= N \sum_{T_i \geq T}^{T_N} \delta_i c_i(t) + N_t - R_S N \int_t^T E_t[e^{-\int_t^u q_S + \lambda_S} \lambda_S] \end{aligned} \quad (165)$$

where $c_i(t)$ and N_t are the current market values of the coupon and nominal components. For ease of notations let us define the quantity S_t^T as

$$S_t^T = N \sum_{T_i \geq T}^{T_N} \delta_i c_i(t) + N_t \quad (166)$$

i.e. S_t^T is the market value at t of the bond with all coupons before T set to 0. Then we can rewrite the expectation

$$E_t[e^{-\int_t^T q_S + \lambda_S} S_T] = S_t^T - R_S N \int_t^T E_t[e^{-\int_t^u q_S + \lambda_S} \lambda_S]. \quad (167)$$

1.12.6 PDEs and Valuation

Let us define the new operator $\mathcal{A}_t$ as

$$\mathcal{A}_t F = \frac{1}{2} \sigma_{S_d}^2 S_d^2 \frac{\partial^2 F}{\partial S_d^2} + \left[(q_S - \gamma_S + \lambda_S) S_d - R_S \lambda_S \right] S_d \frac{\partial F}{\partial S_d}. \quad (168)$$

Then we may write the self-financing condition as

$$\begin{aligned} 0 = & \left[\frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) + \alpha_B (r_B - q_B) P_B + \sum_{i=1}^N r_i \alpha_i P_i \right. \\ & \left. + \lambda_{S/C} (g_{S/C} - \hat{V}) + \lambda_{S \wedge C} (g_{S \wedge C} - \hat{V}) \right] dt \\ & + \epsilon_h dJ_B. \end{aligned} \quad (169)$$

Again we have to choose whether to hedge our own risk and what the funding conditions are. The differences with the more standard case that has been the main subject of this document are that there are two additional CVA-like terms for the new default events S/C and $S \wedge C$, and that the drift of the spot process is modified.

Let us take the example of the (own default) hedged case with three-bond funding, as it is a case that displays most generality. Hedging our own default risk leads to

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) + \lambda_B (g_B - \hat{V}) \\ + \lambda_{S/C} (g_{S/C} - \hat{V}) + \lambda_{S \wedge C} (g_{S \wedge C} - \hat{V}) + \sum_{i=1}^N \alpha_i P_i (r_i - LGD_{P_i} \lambda_B) = 0. \end{aligned} \quad (170)$$

and applying the three-bond funding strategy to

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - r_X X + r_{I_B} I_B - r_{I_C} I_C - \gamma_K K + \lambda_C (g_C - \hat{V}) + \lambda_B (g_B - \hat{V}) \\ + \lambda_{S/C} (g_{S/C} - \hat{V}) + \lambda_{S \wedge C} (g_{S \wedge C} - \hat{V}) - (r_1 - LGD_{P_1} \lambda_B) (\hat{V} - Q) \\ - (r_2 - LGD_{P_2} \lambda_B) (Q - X + I_B - \phi K)^+ - (r_3 - LGD_{P_3} \lambda_B) (Q - X + I_B - \phi K)^- = 0. \end{aligned} \quad (171)$$

In this section we want to obtain valuation equations that allow for the inclusion of the security's default payment in PV, margining and close-out. This is slightly different from the work of [Lichtner15a] where the security's default is viewed as a CVA-like adjustment. To obtain such a valuation, we propose to solve (171) by a decomposition of the derivative into PV and XVAs through a two-step process: the total derivative is the sum

of the PV and XVAs as before, except that PV is not fully "risk-free". Our PV will now incorporate the risk of the security defaulting, such that PV can itself be viewed as the value of a risky derivative. We decompose

$$\hat{V} = \bar{V}^r + U \quad (172)$$

where $\bar{V}^r$ is our PV and is defined to be free of C/P and O/S default risk, but containing the security's default risk. This PV satisfies a PDE that is a special case of (171), reducing to

$$\frac{\partial \bar{V}^r}{\partial t} + \mathcal{A}\bar{V}^r - (r + \lambda_{S/C})\bar{V}^r = -\lambda_{S/C}g_{S/C}. \quad (173)$$

Now in a very familiar fashion we can solve this by decomposing $\bar{V}^r$ into a security-risk-free value V^r and a security-VA part W as

$$\bar{V}^r = V^r + W, \quad (174)$$

solve the PDE for W and find the PV $\bar{V}^r$ to be

$$\bar{V}^r = V^{r+\lambda_{S/C}} + E_t \left[\int_t^T D_{r+\lambda_{S/C}} \lambda_{S/C} g_{S/C} \right]. \quad (175)$$

Next we decompose the total derivative into the counterparty-risk-free part $\bar{V}^r$ and the XVA part U , i.e. $\hat{V} = \bar{V}^r + U$, plug the decomposition into (171), and, thanks to (173), find

$$\begin{aligned} \frac{\partial U}{\partial t} + \mathcal{A}U - \gamma U &= \lambda_C(Q - X - I_C)^+ + \lambda_B(Q - X + I_B)^- - \lambda_{S \wedge C} g_{S \wedge C} \\ &\quad + (r_2 - LGD_{P_2} \lambda_B - r)(Q - X + I_B - \phi K)^+ \\ &\quad + (r_3 - LGD_{P_3} \lambda_B - r)(Q - X + I_B - \phi K)^- \\ &\quad + (r_X - r)X - (r_{I_B} - r)I_B + r_{I_C}I_C + (\gamma_K - r\phi)K \\ &\quad + (r_1 + R_{P_1} \lambda_B + \lambda_C - r)(\bar{V}^r - Q) + \lambda_{S \wedge C} \bar{V}^r. \end{aligned} \quad (176)$$

Now we can apply Feynman-Kac's theorem to find

Component	Hedged
XVA Discount	$r_1 + R_{P_1} \lambda_B + \lambda_C + \lambda_{S/C} + \lambda_{S \wedge C}$
CVA	$-\lambda_C LGD_C (Q - X - I_C)^+$
CVA($S \wedge C$)	$\lambda_{S \wedge C} g_{S \wedge C}$
DVA	$-\lambda_B LGD_B (Q - X + I_B)^-$
FCA	$-(r_F^b - r)(Q - X + I_B - \phi K)^+$
FBA	$-(r_F^l - r)(Q - X + I_B - \phi K)^-$
DVA2	$\lambda_B LGD_{P_2} (Q - X + I_B - \phi K)^+$
CVA2	$\lambda_B LGD_{P_3} (Q - X + I_B - \phi K)^-$
ColVA	$-(r_X - r)X$
MVA	$-\left[r_{I_C} I_C - (r_{I_B} - r)I_B\right]$
KVA	$-(\gamma_K - \phi r)K$
GVA	$-(r_1 + R_{P_1} \lambda_B + \lambda_C - r)(\bar{V}^r - Q) - \lambda_{S \wedge C} \bar{V}^r$

Table 1: Valuation under 3-bond split funding, defaultable underlying

1.12.7 Applications

In this section we consider a few particular cases of derivatives on defaultable assets. Unless specified otherwise, for illustration purposes we will assume the following configuration

- no own default ($\lambda_B = 0$)
- no simultaneous default of C/P and underlying ($\lambda_{S \wedge C} = 0$, $\lambda_S = \lambda_{S/C}$)
- symmetric funding
- no IMs, no capital cost
- C/P is posting the Independent Amount (IA) I_C , on which we have to pay the rate r_{I_C} (rehypothecable).

The valuation equations in (1) reduce to

Component	Hedged
XVA Discount	$r_F + \lambda_C + \lambda_S$
PV	$\bar{V}^r = V^{r+\lambda_S} + E_t \left[\int_t^T D_{r+\lambda_S} \lambda_S g_{S/C} \right]$
CVA	$-\lambda_C LGD_C (Q - X - I_C)^+$
FVA	$-(r_F - r)(\bar{V}^r - X - I_C)$
ColVA	$-(r_X - r)X$
IAVA	$-(r_{I_C} - r)I_C$
GVA	$-\lambda_C(\bar{V}^r - Q)$

Table 2: Illustrative Valuation for defaultable underlyings

1.12.7.1 Bond Price

To model a simple bond issuance, we further take the following conditions:

- no collateralization: $X = 0$
- no C/P default (facing issuer directly)
- no IA: $I_C = 0$
- the cash-flows in V^r are all the bond's cash-flows (coupon and final notional) discounted at the reference rate r .

This leads to the value

$$\begin{aligned}
\hat{V} &= PV + FVA \\
&= \bar{V}^r - E_t \left[\int_t^T D_{r_F + \lambda_S} (r_F - r) \bar{V}^r \right] \\
&= \bar{V}^{r_F}
\end{aligned} \tag{177}$$

$$= V^{r_F + \lambda_S} + R_S N E_t \left[\int_t^T D_{r_F + \lambda_S} \lambda_S \right]. \tag{178}$$

where we have used proposition 2 in the third line and the expression of PV in the fourth.

This formula has the following meaning: the bond's total value is the sum of the its future cash-flows discounted with the funding rate r_F and the default intensity λ_S , plus the recovery term on the nominal in case default happens before maturity. This recovery term is matched with what we would expect, i.e. the sum of default probabilities over each intervals until maturity discounted by the funding rate.

Now let us discuss this funding rate further. There is no repo markets for derivatives in general such that the derivative assumed behind generic replication arguments would in general not be repo-able. This means that it would have to be funded at the internal funding rate. However, here we are taking the very specific example of a bond, which may have a repo market. If it does, then the bond can be pledged as collateral on the loan, and thus the borrowing rate would be reduced to secured, i.e. the repo rate.

In other words, if there is a repo market then we should choose

$$r_F = q_S \tag{179}$$

where q_S is the repo rate for the bond. If there is no repo market, we should choose

$$r_F = r_f \tag{180}$$

where r_f is some other source of funding. However, in this situation the dynamics we wrote for the bond, which employs a repo rate q_S , becomes ill-defined since this rate does not exist. We argue that if a bond has no repo market, then effectively it means that we must fund its cost through other means, expressed by r_f . It all appears as if there was a repo market for the bond and the repo rate was r_f , which is what we would put in place of the bond's repo rate q_S . If such a situation happened, different participants with different funding rates would view the bond has having different values. Conversely, one may consider that if there is a liquid market for this bond in spite of there being no repo market, one may imply a funding rate from the bond price in a similar fashion as an FVA spread can be implied from the Totem consensus on a set of XVA prices.

With the considerations above, it is apparent that this particular derivative, which happens to be just a bond, will be funded at a rate that has the same meaning as a repo rate for the bond even though a repo market may not exist. Let us then fix the funding rate to the repo rate, $r_F = q_S$.

The first term in the value of the bond in (178) is the sum of the expectation of its cash-flows discounted with the rate $q_S + \lambda_S$ from the time they are paid. Since we are talking about a coupon-bearing bond, this term has the expression

$$\begin{aligned}
V^{q_S + \lambda_S} &= N \sum_{T_i \geq t} E_t \left[e^{-\int_t^{T_i} q_S + \lambda_S} c_i(T_i) \delta_i \right] + E_t \left[e^{-\int_t^{T_T} q_S + \lambda_S} N_T \right] \\
&= N \sum_{T_i \geq t} c_i(t) \delta_i + N_t - R_S N \int_t^T E_t [e^{-\int_t^u q_S + \lambda_S} \lambda_S] \\
&= S_t - R_S N \int_t^T E_t [e^{-\int_t^u q_S + \lambda_S} \lambda_S]
\end{aligned} \tag{181}$$

where we have used the expectation formula (163) for the coupons and final nominal. Going back to the total price of the derivative, this gives us

$$\hat{V} = S_t \quad (182)$$

i.e. that the derivative's price is the current bond price. This is what we should have expected, although we should note that this happens because we chose as funding the repo rate. If we had chosen a different rate, as discussed above, our price would not have coincided with the market price.

1.12.7.2 Bond forward (put type) with no C/P default

Bond forward types differ about how they treat bond default before the forward maturity. The put type, subject of this section, is the case where on default of the bond the (discounted) forward price is paid against the bond. The modelling assumptions are

- no collateralization: $X = 0$
- no C/P default
- no IA: $I_C = 0$
- default payment: $g_{S/C}(\tau) = P_{q_S}(\tau, T)F - R_S N$, because on default of the bond we deliver the discounted forward against the bond, which is worth its recovery at that point. F is the forward price and $P_{q_S}(t, u)$ is the repo curve
- funding at the repo curve.

This leads to the value

$$\begin{aligned} \hat{V} &= PV + FVA \\ &= \bar{V}^{q_S} \\ &= V^{q_S + \lambda_S} + E_t \left[\int_t^T D_{q_S + \lambda_S} \lambda_S (P_{q_S}(u, T)F - R_S N) \right] \\ &= V^{q_S + \lambda_S} + F E_t \left[\int_t^T e^{-\int_t^u q_S} e^{-\int_t^u \lambda_S} \lambda_S \right] - R_S N E_t \left[\int_t^T e^{-\int_t^u q_S + \lambda_S} \lambda_S \right] \\ &= V^{q_S + \lambda_S} + F E_t \left[e^{-\int_t^T q_S} (1 - e^{-\int_t^T \lambda_S}) \right] - R_S N E_t \left[\int_t^T e^{-\int_t^u q_S + \lambda_S} \lambda_S \right] \end{aligned}$$

Now let us calculate the first term $V^{q_S + \lambda_S}$ in the valuation above. This term is the 'risk-free' value which here means no default of the bond. There is therefore only one cash-flow in it, which is the exchange of the forward price against the (non-defaulted) bond at maturity. Its expression is thus

$$V^{q_S + \lambda_S} = E_t \left[e^{-\int_t^T q_S + \lambda_S} (F - S_T) \right]. \quad (183)$$

Back to the derivative's price, this gives us the simplified formula

$$\hat{V} = P_{q_S}(t, T)F - E_t \left[e^{-\int_t^T q_S + \lambda_S} S_T \right] - R_S N E_t \left[\int_t^T e^{-\int_t^u q_S + \lambda_S} \lambda_S \right] \quad (184)$$

Finally, using the expectation formula (167), we find

$$\hat{V} = P_{q_S}(t, T)F - S_t^T \quad (185)$$

where, as a reminder, S_t^T is the market value at t of the bond with all coupons paying before T set to 0. If we denote by $F_{put}(t, T)$ the fair forward price traded at t for maturity T with the put-bond feature, we thus find

$$F_{put}(t, T) = \frac{S_t^T}{P_{qs}(t, T)} \quad (186)$$

where $P_{qs}(t, T)$ is the funding curve implied by the bond.

1.12.7.3 Bond forward (extinguisher type) with no C/P default

We repeat the previous exercise for the extinguisher type of forwards. When the bond defaults, the trade is simply cancelled at no cost. The valuation is similar to the previous case but now we have $g_{S/C} = 0$. This results simply in

$$\begin{aligned} \hat{V} &= PV + FVA = \bar{V}^{q_S} = V^{q_S + \lambda_S} \\ &= E_t \left[e^{-\int_t^T q_S + \lambda_S} (F - S_T) \right] \\ &= FE_t \left[e^{-\int_t^T q_S + \lambda_S} \right] - S_t^T + R_S N \int_t^T E_t [e^{-\int_t^u q_S + \lambda_S} \lambda_S]. \end{aligned} \quad (187)$$

This gives us the expression of the fair forward price for the extinguisher type as

$$F_{ext}(t, T) = \frac{S_t^T - R_S N \int_t^T E_t [e^{-\int_t^u q_S + \lambda_S} \lambda_S]}{E_t \left[e^{-\int_t^T q_S + \lambda_S} \right]}. \quad (188)$$

A more familiar formula can be reached when assuming deterministic intensity

$$F_{ext}(t, T) = \frac{S_t^T - R_S N \int_t^T P_{qs}(t, u) Q(t, u) \lambda_S}{P_{qs}(t, T) Q(t, T)} \quad (189)$$

where $Q(t, u)$ is the survival probability curve.

1.12.7.4 TRS

Now we turn our attention to the valuation of a TRS, with C/P risk and collateralization. The valuation equations, displayed in table 2, satisfy as usual invariance to the reference rate (PV discount rate). However, we have to specify the default claim in case of C/P default (Q) and the margining value X , and for these it is important to define which discounting rate is used.

Let us first take a look at the PV obtained with a generic rate r and with the funding rate r_F corresponding to the funding of the security. The return leg of the TRS is composed of the following flows:

- coupons of the bond until maturity and so long as the bond is alive
- performance of the bond at maturity if the bond is alive, i.e. difference between bond price at maturity and initial bond price at trade date, denoted by S_0
- in case of default, the default performance is the recovery on the bond minus the initial price, i.e.

$$g_{S/C} = R_S N - S_0 \quad (190)$$

The funding leg is composed of the following flows

- Libor plus trade spread s^* bond until maturity and so long as the bond is alive
- in case of default, cancellation of remaining flows.

Return leg PV with generic reference rate

Let us calculate PV assuming a generic reference rate r . We find

$$\begin{aligned}\bar{V}_{TR}^r &= V_{TR}^{r+\lambda_S} + E_t \left[\int_t^T D_{r+\lambda_S} \lambda_S g_{S/C} \right] \\ &= N \sum_{T_i \geq t}^T \delta_i E_t \left[e^{-\int_t^{T_i} r + \lambda_S} c_i(T_i) \right] + E_t \left[e^{-\int_t^T r + \lambda_S} (S_T - S_0) \right] + (R_S N - S_0) E_t \left[\int_t^T e^{-\int_t^u r + \lambda_S} \lambda_S \right]\end{aligned}$$

We cannot in principle push this calculation further analytically because the expectations only reduce to current market values under the repo rate discount, see (167). To understand the connection between generic and funding discount more clearly, let us make the reasonable assumption that the basis $q_S - r$ is deterministic, where as before q_S is the repo rate. Then we find

$$\begin{aligned}\bar{V}_{TR}^r &= N \sum_{T_i \geq t}^T \frac{P_r(t, T_i)}{P_{q_S}(t, T_i)} \delta_i E_t \left[e^{-\int_t^{T_i} q_S + \lambda_S} c_i(T_i) \right] + \frac{P_r(t, T)}{P_{q_S}(t, T)} E_t \left[e^{-\int_t^T q_S + \lambda_S} (S_T - S_0) \right] \\ &\quad + (R_S N - S_0) E_t \left[\int_t^T \frac{P_r(t, u)}{P_{q_S}(t, u)} e^{-\int_t^u q_S + \lambda_S} \lambda_S \right] \\ &= N \sum_{T_i \geq t}^T \frac{P_r(t, T_i)}{P_{q_S}(t, T_i)} \delta_i c_i(t) + \frac{P_r(t, T)}{P_{q_S}(t, T)} (S_t^T - R_S N \int_t^T E_t [e^{-\int_t^u q_S + \lambda_S} \lambda_S]) \\ &\quad - S_0 \frac{P_r(t, T)}{P_{q_S}(t, T)} E_t \left[e^{-\int_t^T q_S + \lambda_S} \right] + (R_S N - S_0) E_t \left[\int_t^T \frac{P_r(t, u)}{P_{q_S}(t, u)} e^{-\int_t^u q_S + \lambda_S} \lambda_S \right].\end{aligned}\quad (191)$$

While this form of valuation is achievable in principle, it is not very easy to reconcile it with intuitive valuation formulas. We will see below that this formula becomes much simpler when discounting with the funding rate.

Return leg PV with repo as reference

To obtain the PV of the return leg under funding discount, we can simply take the generic formula (191) and replace r with q_S . We find

$$\begin{aligned}\bar{V}_{TR}^{q_S} &= N \sum_{T_i \geq t}^T \delta_i c_i(t) + S_t^T - S_0 E_t \left[e^{-\int_t^T q_S + \lambda_S} \right] - S_0 E_t \left[\int_t^T e^{-\int_t^u q_S + \lambda_S} \lambda_S \right] \\ &= S_t - S_0 E_t \left[e^{-\int_t^T q_S + \lambda_S} \right] - S_0 E_t \left[\int_t^T e^{-\int_t^u q_S + \lambda_S} \lambda_S \right]\end{aligned}\quad (192)$$

where now we can identify the spot bond price S_t . If we further assume independence of the repo rate q_S and the default probability λ_S , this expression simplifies to

$$\bar{V}_{TR}^{q_S} = S_t - S_0 \int_t^T P_{q_S}(t, u) Q(t, u) \lambda_S - P_{q_S}(t, T) Q(t, T) S_0. \quad (193)$$

Finally thanks to proposition 4 one may rewrite this as

$$\bar{V}_{TR}^{q_S} = S_t - S_0 + S_0 \int_t^T P_{q_S}(t, u) Q(t, u) f_{q_S}. \quad (194)$$

where $f_{q_S}(u)$ is the instantaneous forward rate of the repo curve $P_{q_S}(t, u)$.

TRS valuation with repo as reference

We have seen above that valuation with the repo rate as reference leads to a much simpler expression for PV. It is also more in line with market practice to estimate the close-out amount Q and the collateral X with the repo discount.

Let us look at the whole valuation including XVAs using the repo rate as reference rate. The valuation equations (2) reduce to

Component	Hedged
XVA Discount	$r_F + \lambda_C + \lambda_S$
PV	$\bar{V}^{qs} = \bar{V}_{TR}^{qs} + \bar{V}_{FL}^{qs}$
Return Leg	$\bar{V}_{TR}^{qs} = S_t - S_0 + S_0 \int_t^T P_{qs}(t, u) Q(t, u) f_{qs}$
Funding Leg	$\bar{V}_{FL}^{qs} = -S_0 \sum_{i=L}^M P_{qs}(t, T_i) Q(t, T_i) \tau_i (L_i + s^*)$
CVA	$-\lambda_C LGD_C (\bar{V}^{qs} - X - I_C)^+$
FVA	$-(r_F - q_S) (\bar{V}^{qs} - X - I_C)$
ColVA	$-(r_X - q_S) X$
IAVA	$-(r_{I_C} - q_S) I_C$

Table 3: Valuation with funding as reference (no IM, KVA)

where r_F is the funding rate for the derivative and L is the next payment date and $T = T_M$ is the maturity.

2 XVAs In Practice

2.1 Segregation and Rehypothecability

We have briefly explained the mechanisms of segregation and rehypothecability and their relation to XVAs in section 1.1. Then we derived the XVA valuation equations for the specific cases of rehypothecable VM and segregated IM. We did not work with a generic margin for shortness of the equations and in order to avoid obscuring more fundamental aspects of the XVAs.

In this section we review and expand on how margins appear in the XVAs depending on whether they are segregated and/or rehypothecable or not. We do not want to go again through all the theoretical derivations of the previous chapter as most arguments and equations are modified in a way that can be more easily understood by simply explaining where the equations are modified, how and for what reason.

Our goal is to derive a set of rules to quickly write the valuation equations for all types of margins, even non-standard ones that are used less commonly in practice (such as non-segregated non-rehypothecable).

For this purpose it is sufficient to take a simplified situation where the derivative's value is V and we consider the cases of margins M_C paid by the C/P to us and M_B paid by us to the C/P. Both M_C and M_B are positive numbers.

These margin amounts are placed on accounts that earn the rates r_{M_C} and r_{M_B} . These rates may be 0, OIS, or anything else to be specified by inputs.

On top of this we consider that the margins may be rehypothecable (rehypo) or not (non-rehypo) and segregated or not. This leaves 4 possible combinations but only 3 make sense:

- non-segregated and rehypo: the margin is rehypothecable so it can be used freely by the side that receives it, in particular for funding. In case of default, it is at risk because it is not segregated (when we paid it) and we are able to use it to reduce the claim (when we receive it).
- non-segregated and non-rehypo: the margin is non-rehypothecable, so it cannot be reused for funding. However it is non-segregated, so it is at risk on default.
- segregated and rehypo: this combination cannot exist, as if a margin is segregated then it is non-rehypo.
- segregated and non-rehypo: cannot be used for funding and is not at risk on default (when we paid it) because it is held by a custodian. On default when we received it, we are allowed to use it to reduce the claim.

The question of how the margins themselves are calculated is not addressed at this point. For our purpose here it is sufficient to view these margins as given amounts. We look at the effects of the margins on CVA, DVA, FVA and MaVA where MaVA is the VA associated with the margin. MaVA will be CollVA, MVA, or IAVA when the margin is the Variation Margin (VM), the Initial Margin (IM) or the Independent Amount (IA).

2.1.1 Reviewing the Replication

Here we go through the Replication arguments again although much more briefly, benefiting from the understanding of the XVAs gained in the previous chapter. Here we focus on where the margins affect the calculation. Let us assume that 6 margins are included, all defined as positive numbers

- R_C is a non-segregated rehypo margin paid by the C/P to us
- R_B is a non-segregated rehypo margin paid by us to the C/P
- N_C is a non-segregated non-rehypo margin paid by the C/P to us
- N_B is a non-segregated non-rehypo margin paid by us to the C/P
- S_C is a segregated (so non-rehypo) margin paid by the C/P to us

- S_B is a segregated (so non-rehypo) margin paid by us to the C/P.

The cases of R_C and R_B (resp. S_C and S_B) have been handled in the previous chapter through the example of the VM (resp. the IM) which is a rehypo (resp. segregated) margin. The VM was denoted by X (positive when received, negative when paid). The IM was denoted by I_C (always positive) segregated and paid by the C/P to us, and I_B (always positive) segregated and paid by us to the C/P. To derive how R_C and R_B (resp. S_C and S_B) appear we simply repeat the previous results for the VM and IM. The non-segregated non-rehypo case of N_C and N_B has not been considered before so we will handle it here.

Funding Constraint

The core funding constraint (5) becomes

$$\text{Funding condition: } \sum_{i=1}^N \alpha_i P_i + \hat{V} - R_C + R_B + N_B + S_B + \dots = 0. \quad (195)$$

where $\dots$ stands for contributions that do not need repeating here (capital). The reason is the following:

- rehypo margins can be used for funding
- received non-rehypo margins (including segregated) cannot be used for funding
- paid non-rehypo margins (including segregated) must be funded.

In fact, all paid margins must be funded independently of whether they are rehypo or not. As a consequence, the non-rehypo (including segregated) received margins are the only ones that do not appear in the funding condition.

Rate Costs/Benefits

In section 1.3.3 we found that margins lead to analogous rate contributions to the self-financing in (7) in terms rate cost/benefit terms whether they are rehypo or not, and segregated or not. This fact is due to a combination of the different evolution of the non-rehypo/segregated margins in the portfolio replication and of the differences with which they appear in the funding constraint. Without entering into unnecessary details, the self-financing condition (7) can be written as

$$0 = d\hat{V} + r_{R_B} R_B dt - r_{R_C} R_C dt + r_{N_B} N_B dt - r_{N_C} N_C dt + r_{S_B} S_B dt - r_{S_C} S_C dt + \dots \quad (196)$$

This result can be understood as saying that rates are paid/received on the margin accounts independently of whether they are rehypo/segregated or not (although these rates might be 0 depending on the type of margin and instruments in it).

Close-out (CVA and DVA)

Generalizing the discussion in section 1.5, we can think of the margin impacts on default from the point of view of the surviving entity. For the surviving entity,

- any margin that is received can be kept on default of the counterparty, even when it is segregated. This is because the custodian can give it to the surviving entity. So a received margin can be used to reduce the default claim whether it is segregated or not. Therefore all received margins are part of the default claim (received by surviving entity).
- for paid margins the situation is different. If the paid margin is segregated, the defaulting entity will not have been able to use it (held by custodian). On default the custodian will give it back to the surviving entity in full, not only the recovery on it, and therefore it is not part of the default claim. However, if the paid margin is not segregated, it is at risk on default and is part of the claim: this is valid for rehypo and for non-rehypo but non-segregated.

The conclusion is that all margins are in CVA and DVA except the segregated margins paid by the surviving entity. These considerations can be transformed into rules for the CVA and DVA by switching who is the surviving entity. For CVA, the Bank is surviving so S_B is not in CVA. For DVA, the Counterparty is surviving so S_C is not in DVA. Therefore we conclude that

$$CVA \sim -\lambda_C LGD_C (V - R_C + R_B - N_C + N_B - S_C)^+ \quad (197)$$

$$DVA \sim -\lambda_B LGD_B (V - R_C + R_B - N_C + N_B + S_B)^-. \quad (198)$$

Funding strategy (FVA)

Here we keep things simple but retain the core idea. We assume a generic asymmetric funding with two a priori different rates for borrowing and lending, such that symmetric funding is the particular case where both rates are the same. Without entering into details, we can consider that two bonds are used, one funding the positive (borrowing) and negative (lending) parts of the funding requirement in (195).

The borrowing bond leads to an FCA-like term while the lending bond leads to an FBA-like term. After introducing of the reference rate r , this leads to the following FVA terms

$$FCA \sim -(r_F^b - r)(V - R_C + R_B + N_B + S_B)^+ \quad (199)$$

$$FBA \sim -(r_F^l - r)(V - R_C + R_B + N_B + S_B)^-. \quad (200)$$

Margin VAs

While all margins, paid or received, rehypos or not, led to a cost/benefit with their own account's rate in the self-financing (196), the introduction of the reference rate in the previous step and the gathering of the funding terms into the FVAs above releases benefit/costs of the reference rate for the margins, but only for the margins that appear in the funding constraint.

All these terms that correspond to margins multiplied by rates lead to what we call "Margin VAs" of which CollVA, MVA and IAVA are examples. Due to the remarks above, we can conclude that all these terms are the corresponding margins multiplied by spreads between the margin account's rate and the reference rate, *except for the received non-rehypo margins* (N_C and S_C) since these are not part of the funding. As a consequence we have the following MaVAs

$$MaVA(R_B) \sim +(r_{R_B} - r)R_B \quad (201)$$

$$MaVA(R_C) \sim -(r_{R_C} - r)R_C \quad (202)$$

$$MaVA(N_B) \sim +(r_{N_B} - r)N_B \quad (203)$$

$$MaVA(N_C) \sim -r_{N_C}N_C \quad (204)$$

$$MaVA(S_B) \sim +(r_{S_B} - r)S_B \quad (205)$$

$$MaVA(S_C) \sim -r_{S_C}S_C. \quad (206)$$

This can be understood in the following way. All margins lead to benefits/costs due to receiving/paying the rates on the margin's account. However, only the margins that are part of the funding have compensating cost/benefits coming from borrowing/lending money to pay/receive them. Since the received non-rehypo are the only margins (segregated implies non-rehypo) that are not part of the funding, they are the only margins that do not have a compensating funding benefit, and they give us raw costs due to us paying their own accounts' rates.

2.1.2 Asymmetric Funding

Here we gather all the valuation equations obtained above, remaining in the most general case of asymmetric funding. We obtain

$$CVA \sim -\lambda_C LGD_C(V - R_C + R_B - N_C + N_B - S_C)^+ \quad (207)$$

$$DVA \sim -\lambda_B LGD_B(V - R_C + R_B - N_C + N_B + S_B)^- \quad (208)$$

$$FCA \sim -(r_F^b - r)(V - R_C + R_B + N_B + S_B)^+ \quad (209)$$

$$FBA \sim -(r_F^l - r)(V - R_C + R_B + N_B + S_B)^- \quad (210)$$

$$MaVA(R_B) \sim +(r_{R_B} - r)R_B \quad (211)$$

$$MaVA(R_C) \sim -(r_{R_C} - r)R_C \quad (212)$$

$$MaVA(N_B) \sim +(r_{N_B} - r)N_B \quad (213)$$

$$MaVA(N_C) \sim -r_{N_C}N_C \quad (214)$$

$$MaVA(S_B) \sim +(r_{S_B} - r)S_B \quad (215)$$

$$MaVA(S_C) \sim -r_{S_C}S_C. \quad (216)$$

2.1.3 Symmetric Funding

Symmetric funding is simply the particular case of the above where $r_F^b = r_F^l = r_F$. While we can simply write the equations as in the above with the same rate in FCA and FBA, we can also take advantage of the symmetry in FCA and FBA to take all (or part of) the margin terms in FVA into the respective MaVAs. This leads to the equivalent formulation (equivalent in total)

$$CVA \sim -\lambda_C LGD_C(V - R_C + R_B - N_C + N_B - S_C)^+ \quad (217)$$

$$DVA \sim -\lambda_B LGD_B(V - R_C + R_B - N_C + N_B + S_B)^- \quad (218)$$

$$FCA \sim -(r_F - r)V^+ \quad (219)$$

$$FBA \sim -(r_F - r)V^- \quad (220)$$

$$MaVA(R_B) \sim +(r_{R_B} - r_F)R_B \quad (221)$$

$$MaVA(R_C) \sim -(r_{R_C} - r_F)R_C \quad (222)$$

$$MaVA(N_B) \sim +(r_{N_B} - r_F)N_B \quad (223)$$

$$MaVA(N_C) \sim -r_{N_C}N_C \quad (224)$$

$$MaVA(S_B) \sim +(r_{S_B} - r_F)S_B \quad (225)$$

$$MaVA(S_C) \sim -r_{S_C}S_C. \quad (226)$$

There are advantages to this formulation, although we should stress that it is equivalent to the formulation with the margins in the FVA as far as the total XVAs are concerned.

One advantage is that the MaVAs become self-contained, so long as we are talking about funding (as the margins to appear in CVA and DVA in an irreducible manner).

The other advantage is that it may be a little easier to interpret the meanings of all the margin VAs as costs and benefits.

For instance, take R_B . This is a margin that we, the bank, pay to the counterparty. It is easy to understand that since we pay it to the C/P, we must pay a cost of the funding rate on it (hence $-r_F$), but the C/P will give us that margin's rate (hence $+r_{R_B}$). If the funding rate is larger than that margin's rate, this margin results in a cost to us. A similar reasoning applies to R_C .

N_B is a non-rehypo margin we pay to the C/P. That it is non-rehypo by the C/P does not change the fact that we must fund it, so incur a term $-r_F$, while again we receive the rate on that margin's account.

Finally, N_C is a non-rehypo margin that the C/P pays to us. This time, the fact that it is non-rehypo means that we cannot use it for funding which would have given us a benefit of r_F . As a consequence, this margin's VA is a raw cost due to the rate we must give to the C/P (not a spread).

2.1.4 Standard Cases

Here we display the result that can be directly inferred from the general rules above when applied to what we consider as the most standard case, i.e. rehypos VM, segregated IM and rehypos IAs. Note that as the VM is usually written as one term that can take both signs (X), we gather the paying and receiving components of the corresponding margins.

Asymmetric Funding

$$CVA \sim -\lambda_C LGD_C(V - X - A_C + A_B - I_C)^+ \quad (227)$$

$$DVA \sim -\lambda_B LGD_B(V - X - A_C + A_B + I_B)^- \quad (228)$$

$$FCA \sim -(r_F^b - r)(V - X - A_C + A_B + I_B)^+ \quad (229)$$

$$FBA \sim -(r_F^l - r)(V - X - A_C + A_B + I_B)^- \quad (230)$$

$$CollVA \sim -(r_X - r)X \quad (231)$$

$$IAVA(A_B) \sim +(r_{A_B} - r)A_B \quad (232)$$

$$IAVA(A_C) \sim -(r_{A_C} - r)A_C \quad (233)$$

$$MVA(I_B) \sim +(r_{I_B} - r)I_B \quad (234)$$

$$MVA(I_C) \sim -r_{I_C}I_C. \quad (235)$$

Symmetric Funding

Again, we can simply take the above with identical funding rates for borrowing and lending, or we can use the equivalent alternative form

$$CVA \sim -\lambda_C LGD_C(V - X - A_C + A_B - I_C)^+ \quad (236)$$

$$DVA \sim -\lambda_B LGD_B(V - X - A_C + A_B + I_B)^- \quad (237)$$

$$FCA \sim -(r_F - r)(V - X)^+ \quad (238)$$

$$FBA \sim -(r_F - r)(V - X)^- \quad (239)$$

$$CollVA \sim -(r_X - r)X \quad (240)$$

$$IAVA(A_B) \sim -(r_F - r_{A_B})A_B \quad (241)$$

$$IAVA(A_C) \sim +(r_F - r_{A_C})A_C \quad (242)$$

$$MVA(I_B) \sim -(r_F - r_{I_B})I_B \quad (243)$$

$$MVA(I_C) \sim -r_{I_C}I_C \quad (244)$$

where we have decided, as is quite common, to keep the VM in the FVA (and therefore CollVA is the spread to the reference rate, not the funding rate as for the other margins).

2.2 Cross-Currency Rate Conversion

In all the previous sections it was assumed that the cash-flows and rates were in the same currency, say the domestic currency. In many practical situations some rates will be originally defined/measured in a different currency. A typical example would be the collateral rate or the funding rate that will often be in a different currency from that of the cash-flows.

The question we are trying to solve is the following: knowing that a cash-flow earns the rate r_f in the foreign currency, what is the rate that this same cash-flow should earn once it is expressed in the domestic currency after conversion by FX? This would be the situation of, for instance, a USD flow earning the funding rate measured in USD, but converted to JPY in a JPY-domestic XVA engine.

The answer is provided by the change of measure formula from foreign to domestic measure in (442). First let us apply this formula to a USD flow $C_{USD}(T)$ to be paid at time T , collateralized in USD. The rate c_f is in fact $c_f = r_{USD}^{OIS}$ and $c_d = r_{JPY}^{OIS}$ while the cross-currency basis is $q_{df} = b_{JPY}^{USD}$. This flow earns OIS in USD, so that its PV in USD is

$$PV_0 = E_0^f[e^{-\int_0^T r_{USD}^{OIS} C_{USD}}] \quad (245)$$

$$= \frac{1}{S_0} E_0^d[e^{-\int_0^T r_{JPY}^{OIS} + b_{JPY}^{USD} S_T C_{USD}}] \quad (246)$$

$$= \frac{1}{S_0} E_0^d[e^{-\int_0^T r_{JPY}^{OIS} + b_{JPY}^{USD} C_{JPY}}] \quad (247)$$

where we denote by $C_{JPY} = S_T C_{USD}$ the JPY-converted value of the flow. The change of formula above tells us that if a flow earns the USD OIS rate r_{USD}^{OIS} when expressed in USD, then we can equivalently calculate its PV in JPY provided we discount the JPY-converted flow at the rate $r_{JPY}^{OIS} + b_{JPY}^{USD}$.

Now we can apply the same logic to a more general USD flow that earns a generic rate r_{USD} in USD, by the following operations

$$PV_0 = E_0^f[e^{-\int_0^T r_{USD} C_{USD}}] \quad (248)$$

$$= E_0^f[e^{-\int_0^T r_{USD}^{OIS} + r_{USD} - r_{USD}^{OIS} C_{USD}}] \quad (249)$$

$$= \frac{1}{S_0} E_0^d[e^{-\int_0^T r_{JPY}^{OIS} + b_{JPY}^{USD} + r_{USD} - r_{USD}^{OIS} C_{JPY}}]. \quad (250)$$

In other words, we can use the following rule: if a flow earns the rate r_{USD} in USD, then when expressed in JPY this flow earns the rate r_{JPY} equal to

$$r_{JPY} = r_{JPY}^{OIS} + b_{JPY}^{USD} + r_{USD} - r_{USD}^{OIS}. \quad (251)$$

All rates appearing in the XVA engine and discounting or multiplying flows expressed in JPY must be converted using the above rule when the original rate is defined in a different currency such as USD. This is typically the case of the funding rate, for the collateral rate in non-JPY CSAs, and for any other non-JPY margin rate.

Cross-currency conversion, Shareholder vs No-Shortfall

At this point we would like to discuss a technical point that can prove to be quite material. The rules above appear rather acceptable, possibly even intuitive, when applied for instance to the funding rate. We will easily accept the fact that if the funding rate is measured in USD, which is typically the case as it is often estimated from the Totem consensus only available in USD, then the funding rate "seen" by the JPY XVA engine will be adjusted according to (251). As in the Shareholder approach (86) the funding rate appears in the discounting of the XVA flows, we will thus accept that the XVA flows in JPY would be discounted with the rate

$$f_{JPY} = r_{JPY}^{OIS} + b_{JPY}^{USD} + f_{USD} - r_{USD}^{OIS} \quad (252)$$

where f_{USD} is the funding rate measured in USD from the Totem consensus. Say that for convenience we measure this funding rate as a spread over USD OIS, i.e. we define the funding spread $s_{USD}^F = f_{USD} - r_{USD}^{OIS}$, then the XVA flows in the JPY engine under Shareholder are discounted according to

$$f_{JPY} = r_{JPY}^{OIS} + b_{JPY}^{USD} + s_{USD}^F. \quad (253)$$

As discussed before, Shareholder and No-Shortfall are only different through this discounting, where Shareholder discounts with the funding rate and No-Shortfall discounts with the funding bond yield, which is often written as "risk-free rate + default probability".

Now let us take a look at the similar cross-currency conversion from the point of view of No-Shortfall. Let us assume that the No-Shortfall funding bond is indeed at risk-free rate + default probability where the risk-free rate is USD OIS and the default probability is measured from the Bank's CDSs in USD. Let us further assume that the resulting probability curve is very close to the funding spread, for instance because Mizuho's credit quality would make it a perfect representative of the Totem consensus.

Then we can conclude that Shareholder and No-Shortfall would lead to exactly the same valuation. At least, this is the conclusion we easily reach when we write the valuation equations in USD. Let us then write the valuation equations in JPY for No-Shortfall. If we interpret the discounting as "risk-free + default probability", we will be tempted to discount at JPY OIS + default probability. However, doing so would result in No-Shortfall being different from Shareholder in JPY, because Shareholder in JPY has the cross-currency basis in discount, on top of the USD funding spread. We arrive at a situation where Shareholder and No-Shortfall are identical in USD but different in JPY, which is breaking the no-arbitrage rule.

We believe that this seeming paradox is resolved by looking at the original meaning of the XVA flow discount in the Replication method. The XVA flow discount is by the yield of the funding bond. This yield is measured in one currency, and then should be cross-currency converted as any other rate for the no-arbitrage rule to hold. When we re-interpret this discount as "risk-free + default probability", we introduce an ambiguity on what "risk-free" means, i.e. in what currency we choose the notion of "risk-freedom".

Once we consider that the XVA flow discount is through the funding bond yield, we also have to cross-currency adjust this rate, which then preserves the equal value of Shareholder and No-Shortfall in any currency, that is, we remain arbitrage-free.

2.3 Adjusted DVA

A common accounting practice consists in reporting the risky value for the derivative

$$\hat{V} = PV + CVA + FCA + FBA + \overline{DVA} \quad (254)$$

where the adjusted DVA is defined as

$$\overline{DVA} = \text{Max}(DVA - FBA, 0). \quad (255)$$

This practice effectively leads to reporting the highest of FBA and DVA, i.e.

$$\hat{V} = PV + CVA + FCA + \text{Max}(FBA, DVA). \quad (256)$$

2.4 Margin Period Of Risk

We start with a review of the definitions of the MPOR that can be found in the literature. We will see in particular that while there is a broad consensus on the definition of the MPOR, there are differences in how different authors approach the issue. As a consequence, different entities may choose to define the MPOR to capture slightly different effects. We then turn to describe a few practices and issues surrounding the actual implementation of the MPOR correction.

2.4.1 Definition

[Brigo11a] has a good definition and remarks on the MPOR that summarize most of what other references also say: the MPOR is *"the time elapsed between the default event and the completion of the close-out procedure... The Basel III Proposal (2010) warns to increase the margin period of risk to capture the illiquidity of collateral and trades, the length of margin call disputes, as well as the costs of trade replacement and operations, in order to avoid exposure underestimates. For instance, they say that, if the trade involves illiquid collateral, or derivative that cannot be easily replaced, the margin period of risk should be equal to the collateral margining update interval plus 20 days"*.

The MPOR is a model parameter, in the sense that it is not a legal or official feature of the CSA. It is a parameter chosen by the modeller to achieve a desired (typically conservative) valuation effect. However, we

can see in the above definition that there are regulatory guidelines surrounding it. In the below we look at various considerations on this parameter put forward by different authors.

2.4.2 Survey of the literature

Pykhtin et al.

[Pykhtin07a] defines the MPOR as *"the time interval from the last exchange of collateral until the defaulting counterparty is closed out and the resulting market risk is re-hedged"*.

In this definition we can see several distinct effects that can potentially lead to increased exposure:

1. default resolution takes time: there is an amount of time where we will receive neither due cash-flows nor default claim. As a consequence, exposure continues to exist beyond the default time. This is true both for uncollateralized and collateralized netting sets.
2. margining stops: when the counterparty defaults, it stops posting collateral. While default is being resolved (closed-out), the collateral remains at its last posted value but PV keeps evolving. This leads to an additional effect specific to collateralized netting sets this time, where additional exposure is created through presumably increased mismatches between PV and collateral.
3. re-hedging takes time: once we have received the default claim, it may take time before we can re-enter an equivalent trade with a different counterparty. In that time interval PV will move, which leads to a P&L impact on the day we re-enter the trade.

Point 1. above suggests that in an ideal implementation, when default is triggered in the simulation engine non-zero exposures should exist beyond the default time (while a standard engine would have set all of them to 0 after default), and that the default claim is paid later.

Point 2. implies that for collateralized sets these exposures should not be the same as those calculated by the engine without default, but instead should be calculated with a "frozen" collateral that is left at the value it had on default. More precisely, the number of units of collateral will remain at the value they had on the margining date preceding default, while the collateral asset price will keep evolving.

Point 3. implies that the exposures should continue further in the simulation and also that the simulation must consider two flows instead of one: first we receive the default claim at the time of default resolution (i.e. at Point 2.), then a residual exposure exists until we can finally re-enter the trade, at which point we pay the current PV that will not match the PV used in the default claim.

Simulating according to the above would be rather complicated, if not impractical⁷. The authors of [Pykhtin07a] hint at a more practical form of implementation where time delays in estimations of the PV, collateral and exposures are introduced to model these effects (rather than full modelling as suggested above by direct translation of the effects). The authors further state that this delay is usually assumed to be the sum of the margining frequency and the "cure period" that is necessary to close-out and re-hedge. Here we note that there is already an additional simplification in the sense that when default happens, the last exchange of collateral might have been the day before even if the margining frequency is say 1M. This practice of adding margining frequency and cure period may be seen as a conservative simplification.

Gregory

As far as we understand [Gregory12a], the author seems to mostly tie the MPOR to an issue of delay between collateral and PV in the default claim estimate. In that sense it is unclear what MPOR would mean to him in the case of a non-collateralized CSA. This might be due to an estimation of the MPOR effect being negligible compared to the effect of lack of collateralization, but we have not seen such a statement in [Gregory12a] so cannot be sure that is what the author is thinking.

⁷One of the reasons why this is not very practical is that default time is rarely modelled explicitly. In practice default is almost always assumed to be independent of the exposures such that default does not need to be simulated. Correlation of default and exposure is usually handled through different methodologies, see section 2.8 on WWR.

In its introduction on the subject [Gregory12a] states that the MPOR is *"the effective time assumed between a collateral call and receiving the appropriate collateral (or in a worse-case scenario putting the counterparty in default, liquidating existing collateral, closing out and re-hedging the trade"*. In this definition we can see the focus of the author on the collateralized situation, although we do see a hint at the fact that some other effects might have to be considered in his statement regarding re-hedging the trade.

Later in [Gregory12a] the author gives further details on the MPOR where again the focus is on the collateralized case. The author lists the reasons why there may be time delays between when a collateral posting request is made and when the amount is actually paid. He also shows that in principle the process should be simulated with different events at different points in time with the example of the undisputed vs disputed amount, noting that applying a fixed MPOR is only a simple representation of the two-step process and that in practice the MPOR is a *rather crude catch-all parameter*.

Andersen et al.

[Andersen16a] gives a very detailed explanation of the default process until close-out and in particular stresses the fact that various events are not instantaneous as if often assumed by XVA engines for simplification.

2.4.3 Modelling considerations

Following the above references, we will first examine a slightly simplified version of the processes described by [Andersen16a] and write down what the cash flows would be under this default resolution process. Then we will explain how these may be simplified (at the cost of some amount of approximation) in order to reach the more familiar and typical valuation equations that appear in most references.

We consider the following events and timing in chronological order:

1. at τ_M the counterparty posts collateral for the last time before defaulting, the collateral amount is X_{τ_M}
2. at τ_D the counterparty effectively defaults. Nothing happens in terms of cash-flows
3. at τ_E the default claim is determined and paid (if it is paid later we would discount it back to τ_E for pricing). We assume that it is calculated based on the risk-free expected values of remaining cash-flows plus the capitalized values of the cash-flows that should have happened between τ_D and τ_E , let us denote these by U_{DE} . The claim is reduced by the amount of collateral we hold, which is not X_{τ_E} but X_{τ_M} as the counterparty was not able to margin since the last margining date τ_M . The default payment is the recovery R_C if the counterparty owes it to us, or the full amount if we owe it to the counterparty
4. at τ_R we re-enter the trade with another counterparty by buying it at its current value V_{τ_R} .

We then discount all these flows to time τ_D for convenience and for analogy with the simplest forms of modelling where default is instantaneous. This leads to the following flows

$$D_r(\tau_M, \tau_D)X_{\tau_M} + D_r(\tau_D, \tau_E) \left[R_C(V_{\tau_E} + U_{DE} - X_{\tau_M})^+ + (V_{\tau_E} + U_{DE} - X_{\tau_M})^- \right] - D_r(\tau_D, \tau_R)V_{\tau_R}. \quad (257)$$

The flows in (257) do not look at first sight like a usual CVA term, but we can check that they do reduce to it when assuming that the four default resolution events above happen at the same time τ . Indeed, in this approximation, the flows (257) become

$$X_\tau + \left[R_C(V_\tau - X_\tau)^+ + (V_\tau - X_\tau)^- \right] - V_\tau = -LGD_C(V_\tau - X_\tau)^+. \quad (258)$$

In principle we could implement the expression (257) in the simulation engine. In practice this expression is not commonly used, and another problem lies in the fact that it does not fully respect the spirit of the Basel purpose behind the definition of the MPOR as reminded to us by [Brigo11a] in section 2.4.1. To see this, let us first make a two approximations that make it easier to understand the source of the issue. Let us assume that

there are no intermediate cash-flows falling between the default time τ_D and the determination time τ_E , and let us neglect the discounting effects (we expect most of the value to come from PV and collateral mismatches rather than discounting over short periods of time).

The default resolution flows (257) then become

$$X_{\tau_M} + \left[R_C(V_{\tau_E} - X_{\tau_M})^+ + (V_{\tau_E} - X_{\tau_M})^- \right] - V_{\tau_R} = -LGD_C(V_{\tau_E} - X_{\tau_M})^+ - (V_{\tau_R} - V_{\tau_E}). \quad (259)$$

Here we can recognize a "stressed CVA" term $-LGD_C(V_{\tau_E} - X_{\tau_M})^+$ that corresponds to the loss due to the inability of the counterparty to update the collateral, plus an additional term that corresponds to the P&L effect of re-entering the trade with some delay after the recovery payment. The issue here is that this second component may not necessarily be a loss as it is possible for the value of the derivative to go down between recovery payment and re-trading. The Basel rule appears to target a conservative estimate, with the goal to avoid underestimating the exposures due to such timing issues. Modelling the default resolution according to the above could, in principle, lead to profit that are not meant to be priced in here. It could even lead to a positive CVA under particular circumstances.

This is one of the motivations, on top of a preference for simplicity, for using the more ad'hoc formula below

$$-LGD_C(V_{\tau_R} - X_{\tau_M})^+. \quad (260)$$

This formula respects the principles of the MPOR in terms of exposure created by non-instantaneous default resolution, while making sure that the pricing formula is conservative and does not generate profits.

The formula above is the base for many practical implementations of the MPOR correction. It leads to an understanding of the MPOR as a delay between the time when the PV is estimated and the time when the last collateral amount was calculated.

Which XVAs should have an MPOR?

The definition of the MPOR explicitly refers to a default event. The cash-flows discussed above are those of the resolution of the default, i.e. valid under the condition that a default occurred. This part of the definition of the MPOR leads to naturally applying it to CVA. The other part of the definition of the MPOR, i.e. its conservative nature, would lead to the interpretation that it should not be part of DVA as the most common effect of the MPOR being to increase the exposures, MPOR on DVA would be an aggressive measure.

We tend to think that the MPOR should not be, or at least needs not be part of FVA either. The reason is this time in its default-related component. The MPOR essentially implies conditioning with default. On the other hand, the FVA models flows that must be funded so long as the trade is alive, i.e. conditioning on survival, the opposite conditioning as the MPOR implies. Furthermore the effect of the MPOR on FVA should be a small number, as the resolution mechanism implies that nothing would happen to the FVA flows until default and only at that point would we see a slightly increased funding requirement due to the exposure existing slightly longer. The danger is that a typical implementation of the MPOR would lead to increased exposures over the whole life of the trade and therefore would overestimate the MPOR effect on FVA, as the MPOR effect on FVA should only be over the small resolution period.

However, as [Gregory12a] implies with his "catch-all" expression, ultimately we may view the MPOR as a rather heuristic parameter that may or may not be included simply with the view to being conservative or not. In that line of thinking, it would make some sense to apply the MPOR to FCA (but less so to FBA).

2.5 XVAs in XVALib

We are going to describe how XVALib implements the valuation equations for the XVAs. We will give the equations in the standard case for the margins (i.e. VM and IA rehypothecable, IM segregated). Non-standard reypo/segregation configurations follow the rules in section 2.1. From the technical point of view XVALib always calculates DVA independently of conclusions from the Replication method. DVA is then aggregated or not, or aggregated through the adjusted DVA presented in section 2.3, depending on valuation policies.

We start by explaining how XVALib chooses the reference rate r in section 2.5.1. Then we give the valuation equations as implemented by the engine. XVALib has two calculation modes for stochastic and deterministic default time modelling. We will present the result in deterministic default time first as it is the most familiar form of the equations used throughout this document.

2.5.1 Reference rate

In its current version XVALib assumes that the reference rate r is equal to the close-out rate such that $r = r_\epsilon$ in all situations and no RVA is needed. The close-out rate is the OIS rate of the collateral currency when there is a CSA, as this appears to be a common market assumption. When there is no CSA the decision becomes more subjective as there does not seem to be a clear market practice. The close-out rate is chosen by a judgement on counterparty: if it is Japanese, JPY OIS, otherwise USD OIS (except for repackages for which we take JPY OIS).

2.5.2 Deterministic default time

XVALib implements valuation in deterministic default time according to

$$\begin{aligned}
 PV &= E_t \left[\int_t^T D_r \pi \right] \\
 CVA &= -LGD_C E_t \left[\int_t^T D_{r+s_B+\lambda_C} \lambda_C (V^r - X - I_C - A_C + A_B)^+ \right] \\
 DVA &= -LGD_B E_t \left[\int_t^T D_{r+s_B+\lambda_C} s_B (V^r - X + I_B - A_C + A_B)^- \right] \\
 FCA &= -E_t \left[\int_t^T D_{r+s_B+\lambda_C} \left(s_F (V^r - X)^+ + s_{A_B} A_B \right) \right] \\
 FBA &= -E_t \left[\int_t^T D_{r+s_B+\lambda_C} \left(s_F (V^r - X)^- - s_{A_C} A_C \right) \right] \\
 CollVA &= -E_t \left[\int_t^T D_{r+s_B+\lambda_C} s_X X \right] \\
 MVA &= -E_t \left[\int_t^T D_{r+s_B+\lambda_C} \left(s_{I_C} I_C + s_{I_B} I_B \right) \right]
 \end{aligned} \tag{261}$$

where $s_B, s_F, s_X, s_{A_B}, s_{A_C}, s_{I_B}, s_{I_C}$ are placeholder spreads to be specified by inputs. The intent is to specify them according to the XVA valuation principles as developed in the previous chapter with possible cross-currency adjustments, together with estimates of the respective margin accounts' rates. The typical definitions for these spreads are

- s_F the funding spread, s_X the collateral spread
- $s_{A_B} = r_F - r_{A_B}, s_{A_C} = r_F - r_{A_C}$
- $s_{I_B} = r_F - r_{I_B}, s_{I_C} = r_{I_C}$

where $r_{A_B}, r_{A_C}, r_{I_B}$ and r_{I_C} are the rates earned on the corresponding margin accounts. The spread s_B is the one that distinguishes between Shareholder and No-Shortfall in XVALib:

- No-Shortfall: $s_B = \lambda_B$ where λ_B is the default probability curve for the Bank
- Shareholder: $s_B = s_F$ where s_F is the funding spread.

2.5.3 Stochastic default time

XVALib implements valuation in stochastic default time according to

$$\begin{aligned}
 PV &= E_t \left[\int_t^T D_r \pi \right] \\
 CVA &= -LGD_C E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau=\tau_C \in ds < \tau_B} (V^r - X - I_C - A_C + A_B)^+ ds \right] \\
 DVA &= -LGD_B E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau=\tau_B \in ds < \tau_C} (V^r - X + I_B - A_C + A_B)^- ds \right] \\
 FCA &= -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau > s} \left(s_F (V^r - X)^+ + s_{A_B} A_B \right) ds \right] \\
 FBA &= -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau > s} \left(s_F (V^r - X)^- - s_{A_C} A_C \right) ds \right] \\
 CollVA &= -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau > s} s_X X ds \right] \\
 MVA &= -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau > s} \left(s_{I_C} I_C + s_{I_B} I_B \right) ds \right]
 \end{aligned} \tag{262}$$

with the following notations relating to default modelling:

- τ_C (resp. τ_B) is the default time of the counterparty (resp. bank)
- $\tau = \min(\tau_B, \tau_C)$ is the time to first default
- $\mathbf{1}_{\tau=\tau_C \in ds < \tau_B}$ is the event where the counterparty defaults first and defaults in the interval $[s, s + ds]$ while both entities had survived until time s
- $\mathbf{1}_{\tau=\tau_B \in ds < \tau_C}$ is the event where the bank defaults first and defaults in the interval $[s, s + ds]$ while both entities had survived until time s .

The hazard rate process Λ_C for the counterparty follows Hull-White model with initial curve equal to the survival curve of the counterparty. We write this symbolically as

$$\Lambda_C = HW(\lambda_C) \tag{263}$$

where λ_C is the survival curve for the counterparty and HW stands for Hull-White.

The hazard rate process Λ_B for the bank is the sum of a Hull-White model with initial curve equal to the survival curve of the bank and a deterministic "liquidity spread". We write this symbolically as

$$\Lambda_B = HW(\lambda_B) + l_B \tag{264}$$

where λ_B is the survival curve for the bank and l_B is a deterministic liquidity spread. Up to possible cross-currency conversions, the liquidity spread is determined as

- No-Shortfall: $l_B = 0$
- Shareholder: $l_B = s_F - \lambda_B$.

Finally, once the default intensities have been calculated with the above processes, default times are drawn from a Gaussian copula.

2.5.4 MPOR

XVALib applies the MPOR by introducing a shift of MPOR days in the evaluation of PV w.r.t. the collateral. This shift is applied on top of the ones relating to the margining frequency, such that we may view XVALib's MPOR modelling as introducing a different of Margin Frequency + MPOR between PV and collateral. The MPOR is applied to CVA and DVA but not FVA.

2.6 XVAs in MHI's Numerix

The XVAs in MHI are calculated using Numerix for the simulation of the underlying stochastic processes and exposure calculations, together with an MHI-specific script for the margining process and discounting choices. In this section we do not describe the stochastic processes and the margining implementations, focusing instead on the final equations for comparison with the valuation framework described in this document.

We start by explaining how MHI chooses the reference rate r in section 2.6.1. Then we give the valuation equations as implemented by the MHI XVA engine. MHI's XVA engine only implements deterministic default time as it considers that this is the market practice (correlation impact and WWR being handled through scenario modelling).

2.6.1 Reference rate

MHI always takes the reference rate to be the close-out rate, and therefore never needs to calculate RVA. The close-out rate is assumed to be the collateral rate when there is a CSA. As a consequence, MHI does not have a CollVA either. When there is no CSA, the close-out rate is USD OIS.

2.6.2 Valuation equations

MHI does not have a need to model IAs. Its XVA engine implements valuation according to

$$\begin{aligned}
 PV &= E_t \left[\int_t^T D_r \pi \right] \\
 CVA &= -LGD_C E_t \left[\int_t^T D_{r+\lambda_C} \lambda_C (V^r - X - I_C)^+ \right] \\
 DVA &= -LGD_B E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \lambda_B (V^r - X + I_B)^- \right] \\
 FCA &= -E_t \left[\int_t^T D_{r+\lambda_C} s_F (V^r - X)^+ \right] \\
 FBA &= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} s_F (V^r - X)^- \right] \\
 MVA &= -E_t \left[\int_t^T D_{r+\lambda_B+\lambda_C} \left(r_{I_C} I_C + (r_F - r_{I_B}) I_B \right) \right].
 \end{aligned} \tag{265}$$

The funding spread s_F is measured on top of USD OIS from the Totem consensus. When the collateral is not USD, MHI adjusts the funding spread with cross-currency basis according to (251) (adapted to other currencies than JPY).

A particular point to notice is that our own default probability λ_B is not in the discount for CVA and FCA. This is for conservative reasons as well as to decrease the incentive to take more risk when our company's credit quality decreases. The C/P's survival probability is to be taken to 100% for a clearing house.

2.6.3 MPOR

MHI's Numerix applies the MPOR by introducing a shift of MPOR days in the evaluation of PV w.r.t. the last margined collateral. This is not exactly the same as a shift of Margin Frequency + MPOR as the last margin

collateral may have been settled less than the margin frequency ago depending on the date of the exposure. The MPOR is applied to CVA only.

2.7 Tail Risk CVA

Tail Risk CVA denotes the CVA due to residual default risk in the situation where the derivative, as opposed to most cases, is secured rather than ranking *pari passu* to other claims. One such situation happens when using an SPV rather than trading directly with the C/P.

When trading with a C/P, if the C/P defaults on an exposure that is positive to us, we make a loss of the LGD on the exposure. When trading with an SPV, we may make no loss at all as we have access to the assets of the SPV that can be seen as a bond with notional N_{cb} . In most situations these assets will be sufficient to compensate us for the exposure on the derivative such that no loss will be made at all, as opposed to direct trading with the C/P where we would have made a loss.

However, there is a residual risk of loss in the case the assets of the SPV, having lost value due to credit worsening of the bond issuer, no longer cover the value of the derivative. Tail Risk CVA is designed to capture this residual risk.

In modelling Tail Risk CVA we take into account the following points:

- credit quality of the C/P is no longer relevant. The risk of default is on the issuer of the bond held by the SPV.
- the C/P does not 'refill' the SPV in case the bond loses quality/value. The nominal amount of bond in the SPV is fixed at N_{cb} .
- the value of this bond, on the other hand, is not fixed and depends on market and issuer credit conditions. We denote the bond PV as V_{cb} .
- if the issuer defaults, we will recover the whole exposure so long as this exposure is lower than the value of the bond V_b . If it is larger, we can only receive V_{cb} . In other words, the recovery in default is capped at V_{cb} at default.
- in case of default, the value of the bond V_{cb} drops to its default value

$$V_{cb}^T = R_{cb}N_{cb} \quad (266)$$

where R_{cb} is the recovery rate on the bond.

There exist two mechanisms to face the SPV:

1. With CSA: we post the standard rehypos VM R_B . The SPV posts its bond as VM (rehypos), with the condition that it cannot post more than what it was set up with, with nominal N_{cb} (fixed) and value V_{cb} (floating). We value this situation according to

$$CVA \sim -\lambda_{cb}(V - \min(V, V_{cb}^T) + R_B)^+ \quad (267)$$

$$DVA \sim -\lambda_B LGD_B(V - \min(V, V_{cb}^T) + R_B)^- \quad (268)$$

$$FCA \sim -(r_F - r)(V - \min(V, V_{cb}^T) + R_B)^+ \quad (269)$$

$$FBA \sim -(r_F - r)(V - \min(V, V_{cb}^T) + R_B)^- \quad (270)$$

$$CollVA \sim +(r_{R_B} - r)R_B - (r_{R_C} - r)\min(V, V_{cb}^T) \quad (271)$$

where λ_{cb} is the default intensity of the bond issuer, r_{R_B} is the rate the SPV pays on our posting and r_{R_C} is the rate we pay on their posting. Note that while it is straightforward to calculate the value of the bond on default V_{cb}^T (assuming we are confident on the value of the recovery fraction), it is not easy to

calculate its value in a non-default situation V_{cb} as is needed in the *DVA*, *FVA* and *CollVA* components. In XVALib we choose to value the bond at

$$V_{cb} = \alpha \times N_{cb} \quad (272)$$

where α is input parameter. In times of good health of the bond issuer, we would typically choose α to match the current bond price. If we have reasons to expect a significant worsening of the issuer credit in the close future, we may adjust α down compared to its market value. In the most extreme situation where we expect an imminent default of the issuer, we may choose $\alpha = R_{cb}$.

2. Without CSA: no collateral posting on either side.

$$CVA \sim -\lambda_{cb}(V - \min(V, V_{cb}^T))^+ \quad (273)$$

$$DVA \sim -\lambda_B LGD_B V^- \quad (274)$$

$$FCA \sim -(r_F - r)V^+ \quad (275)$$

$$FBA \sim -(r_F - r)V^-. \quad (276)$$

2.8 Wrong Way Risk

Until now we have not specified how to calculate the expectations in the various valuation equations obtained in the previous section. In most situations we make the assumption that defaults of the C/P and Bank are independent from the exposure which allows us to calculate the survival and default probabilities independently from the exposures (outside of the exposures).

Wrong Way Risk (WWR) arises when it is not possible or not accurate to do so as there is a material correlation between exposure and default, and in particular when that results in an increased loss due to the fact that the exposure increases in default situations.

Except for the direct modelling of default time, most WWR modelling methodologies are based on rewriting the unconditional expectations at default times into conditional expectations upon default. The methods then differ in how they calculate the conditional expectations. In this section we start by explaining the first step that introduces the conditional expectations, before discussing possible methods for the calculation of the conditional expectations.

2.8.1 From unconditional to conditional expectations

Let us take the example of CVA for illustration purposes. Without going into details that are not necessary for this section, the familiar form of the CVA under reduced form modelling for default is

$$CVA = -LGD_C E_t \left[\int_t^T D_{r+\lambda_C}(t, s) \lambda_C(s) (V_s - X_s)^+ ds \right] \quad (277)$$

where the intensity $\lambda_C(s)$ is in general stochastic and in particular correlated to the exposure. It is easiest to introduce the conditional expectations by going to the more probabilistic (equivalent) expression

$$CVA = -LGD_C E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau \in [s, s+ds]} (V_s - X_s)^+ \right] \quad (278)$$

where $\mathbf{1}_{\tau \in [s, s+ds]}$ is 1 when default happens over the integration interval $[s, s+ds]$. Then we can use the property of conditional expectations

$$E[X|Y] = \frac{E[X \mathbf{1}_Y]}{P[Y]} \quad (279)$$

to rewrite *CVA* as

$$\begin{aligned}
CVA &= -LGD_C \int_t^T E_t \left[D_r(t, s) \mathbf{1}_{\tau \in [s, s+ds]} (V_s - X_s)^+ \right] \\
&= -LGD_C \int_t^T E_t \left[D_r(t, s) (V_s - X_s)^+ | \tau \in [s, s+ds] \right] P_t \left[\tau \in [s, s+ds] \right].
\end{aligned} \tag{280}$$

The probability of default over the interval $[s, s+ds]$ can be calculated from the initial survival probability curve $\lambda_C^0(s)$ as

$$P_t \left[\tau \in [s, s+ds] \right] = D_{\lambda_C^0}(t, s) \lambda_C^0(s) ds \tag{281}$$

such that CVA can be calculated as

$$CVA = -LGD_C \int_t^T D_{\lambda_C^0}(t, s) \lambda_C^0(s) E_t \left[D_r(t, s) (V_s - X_s)^+ | \tau \in [s, s+ds] \right] ds. \tag{282}$$

This last equation shows formally that CVA can be calculated in a familiar manner by multiplying the exposures by the probability of default estimated from the initial survival curve, provided the exposures are calculated conditionally on default happening. If we denote the conditional positive exposure as

$$EPE(t_i)^{WWR} = E_t \left[D_r(t, t_i) (V_{t_i} - X_{t_i})^+ | \tau \in [t_{i-1}, t_i] \right] \tag{283}$$

then CVA can be discretized over N steps to maturity as

$$CVA = -LGD_C \sum_{i=1}^N EPE(t_i)^{WWR} P \left[\tau \in [t_{i-1}, t_i] \right] \tag{284}$$

which is the same formula as the CVA without WWR but with the EPE replaced by the conditional EPE . Modelling can then be based on specifying how to calculate the EPE under the condition of default.

2.8.2 Scenario methods for conditional expectations

We summarize here the framework introduced by [Turlakov13a]. This reference considers the case where the exposure is sensitive to sovereign default for instance through FX devaluation. The WWR arises when the counterparty's default triggers the default of the corresponding sovereign.

[Turlakov13a] proposes to write the conditional EPE EPE^{WWR} as the weighted sum of a stressed EPE $EPE^{stressed}$ and the non-stressed EPE calculated without WWR . The weighted sum is written as

$$EPE^{WWR} = P(sov|cpty) \times EPE^{stressed} + \left(1 - P(sov|cpty) \right) \times EPE. \tag{285}$$

A parameter λ is introduced in

$$P(sov|cpty) = \lambda P(sov) \tag{286}$$

to model the unknown $P(sov|cpty)$, the probability of default of the sovereign conditional on default of the counterparty. λ characterizes the strength of the connection between sovereign and counterparty, with $\lambda = 1$ being close to independence of the two. The intuition behind the parameter λ can be gained by using Bayes' theorem which, applied to this situation, states that

$$P(sov|cpty) = \frac{P(cpty|sov)}{P(cpty)} P(sov) \tag{287}$$

which implies that

$$\lambda = \frac{P(cpty|sov)}{P(cpty)}. \quad (288)$$

λ must be chosen by the modeller, with strong coupling for large values, near independence for $\lambda = 1$, and weak coupling for $\lambda < 1$. With this parameter, the WWR exposure can be written as

$$EPE^{WWR} = EPE + \lambda P(sov)(EPE^{stressed} - EPE). \quad (289)$$

After choosing λ , the model must then specify how to stress the EPE following a default of the sovereign.

2.9 Methodologies for MVA

In this section we gather thoughts on possible methodologies to calculate MVA, the XVA that captures the cost of funding the Initial Margin (IM). As we have seen in chapter 1, both MVA and the XVAs that depend on the IM in a less direct manner (CVA, DVA, and FVA) have straightforward expressions provided that we have been able to calculate the IM on each Monte-Carlo path at each time. Therefore in the present section we focus the problem of simulating the IM.

We start by gathering the most relevant definitions or features of the IM. Then we turn to a description of the various methodologies that can be found in the literature for the simulation of the theoretical IM. Next we discuss the issue of scaling, designed to adjust the simulation so as to recover the actual IM at least on the valuation date. Finally we close this section by a few thoughts and comments regarding practical implementation of such methods.

This section draws a lot of its inspiration and notations from the work of [Caspers17a].

2.9.1 Generic definition of the Initial Margin

A high level definition of the IM is the 99% confidence level of the changes of value of the netting set over a period of 10 days. The precise way in which this is calculate, however, depends on the situation. In bilateral trading, a near standard model is SIMM that corresponds in its main lines to a Variance-Covariance (VcV) and sensitivity based approach. When trading with a clearing house, the IM will be calculated by the clearing house based on its own model whose definitions we may not be able to know in details.

The features of the IM that are most relevant to its simulation, which is our interest here, are:

- it is the 99% Value-at-Risk (VaR) over a 10D period
- it is calculated at netting set level
- it may be estimated with some amount of stress/conservatism (such is the case of SIMM)
- as a VaR model, it is based on the "clean PV change", i.e. the PV changes over the 10D period assuming no ageing of the portfolio over the period
- some additional regulatory rules may apply, such that, for instance, the exclusion of the final notional payment for cross-currency swaps.

2.9.2 SIMM

In broad terms, the SIMM model is based on a VcV approach using 1st order sensitivities together to be aggregated and combined with fixed volatilities and correlations in a way analogous to the Gaussian VaR. The main difference with Gaussian VaR is that the "volatilities" corresponding to the required percentile (99%) are calculated by percentiles of historical time series (rather than assuming individual normal distribution) and that the time window to estimate them is contains stressed events to provide a conservative estimate.

For the present discussion, it is important to highlight the following aspects of SIMM:

- it contains vega terms

- is is a stressed VaR (i.e. cannot be expected to be directly matched systematically by VaR approaches calibrated to current market data)
- it captures tenor basis risk (through different volatilities and correlations per curve within a currency)
- its total IM is the sum of the IMs restricted to separate asset classes (Rates and FX, Equity, Credit, Commodity).

2.9.3 Modelling considerations

Let us denote by V_i^j and I_i^j the PV and IM of the netting set on path j at time t_i . The IM is the 99% confidence level of the PV changes over the horizon period δ (10D) following t_i , given that we are on the Monte-Carlo scenario j (i.e. with all risk factors having the values taken on path j at time t_i). Furthermore, care should be taken about the fact that the PV changes we are looking at here are "clean", i.e. without portfolio ageing, and in particular, do not include the effect of cash-flows being paid on $[t_i, t_i + \delta]$.

Denoting these clean PV changes as ΔV_i^j , the IM is thfore defined as

$$I_i^j = \mathcal{Q}_{99}[\Delta V_i^j | \mathcal{F}_{t_i}]. \quad (290)$$

As a consequence, the question of calculating the simulated IM is that of estimating the quantile of the PV change distribution conditionally on the state of the world at a specific future point in time on a specific Monte-Carlo path.

We can find 5 types of methods that address this problem:

1. Deterministic Method: the stochastic evolution of the IM is ignored and its term structure is approximated linearly, starting at the current value (as calculated by the counterparties' IM models) and linearly amortizing down to 0 at the last maturity of the netting set. This is the method currently employed by MHSC with XVALib.
2. Simple Method⁸: the term structure of the IM is modelled accurately by actual valuation at time t_i , but the conditional nature (i.e. dependency on the simulated path j) is ignored.
3. Regression methods: these are based on estimating the moments of the conditional distribution of the PV changes ΔV_i^j using regression methods in very much the same way as when we calculate the future (conditional) PVs for the other XVAs using LSMC. Instead of applying LSMC to the path-wise discounted flows to calculate the their conditional expectation (the future PVs), we apply LSMC to the path-wise ΔV_i^j , $(\Delta V_i^j)^2$, etc... to calculate the conditional moments (expectation, variance, etc...) of the distribution of the PV changes. These moments are then used to imply the 99% quantile, i.e. the IM. There are various ways of doing the regression, and distributional assumptions are needed in order to calculate the 99% quantile out of the moments.
4. Differential methods: these are based on calculating differentials of the future PVs V_i^j w.r.t. a number of risk factors and applying a VcV (or Cornish-Fisher) approach to calculating the VaR again after distributional assumptions. These methods avoid the need of a regression but require the knowledge of the VcV matrix. This matrix may be implied from the simulation model or may be given from outside. In particular, it may be identified with the one used by SIMM, such that a particular example of these methods is sometimes called "Simulated SIMM".
5. Nested simulation methods: the main (and possibly only) feasible example of such method is provided by [Green16a] under the name of "Augmented Compression". It consists in running a Monte-Carlo simulation to measure the IM on each path and time point of the original simulation. These are extremely costly. [Green16a] is a version of these that does a Historical Simulation on the nodes.

⁸From the Mathematical stand point this method can be seen as the particular case of regression method where we regress on the constant function only.

Scaling adjustments

Except for the Deterministic Method, none of the methods above will, in general, replicate the IM as calculated by the counterparties' IM models on the valuation date. A common mechanism to cope with this drawback is to scale the models so as to recover exactly the IM at least at $t = 0$. The most simple form of scaling is a proportional scaling using the $t = 0$ value multiplying all simulated values, but more advanced scaling can be considered such as time-dependent ones as suggested by [Anfuso17a]. Finally, one may consider a form of "calibration" whereby an existing model parameter is adjusted (for instance a volatility) in order to match the $t = 0$ IM. We present a few of these scaling adjustments in section 2.9.9.

2.9.4 Deterministic method

There is little more to say on this method. It is obviously very simple to implement and very fast in runtime. However it neglects important features of the IM even at the deterministic level as the IM is not, in general, going to be linear in time. While this gives rather accurate results in [Caspers17a] when applied to a single vanilla swap in a netting set, this reference already shows that such an approximation would be significantly off for non-linear products. Moreover, netting sets with several vanilla swaps are also expected to have an IM evolution that could differ significantly from the linearly interpolated one.

2.9.5 Simple method

This method neglects the state-dependent nature of the IM and interprets the ΔV_i^j as a sample of the stochastic IM at t_i as if it did not depend on the particular realizations of the risk factors on path j at t_i . From this vector of ΔV_i^j we can then simply calculate the 99th percentile which gives us the IM. Note that as a consequence of this approximation, at any given time t_i , the IM is the same on all paths.

2.9.6 Regression methods

These methods are based on the idea of estimating the moments of the distribution of ΔV_i^j and deriving the 99th quantile from the moments. The methods in this family differ both in how they perform the regression and in the distributional assumptions that allow them to calculate the quantile from the moments. We start

Distributional Assumptions

The most simple assumption is normality, which allows us to calculate the quantile simply by knowing the conditional variance, i.e.

$$I_i^j = \mathcal{N}^{-1}(99\%) \times \sigma_i^j \quad (291)$$

where $\mathcal{N}^{-1}()$ is the inverse cumulative of the normal distribution and σ_i^j is the conditional standard deviation as estimated by the regression step. Further sophistication can be brought using the Cornish-Fisher expansion requiring more moments.

Regression

This regression step follows the standard approach of the American MC (LSMC) only applied to the powers of ΔV_i^j depending on the moment we wish to calculate. As such it has the same advantages and drawbacks. In particular, it is dependent on choices of nature and number of basis functions.

In principle we would use for this step the same explanatory variables as those used to calculate the PVs. However, note that while for PV we had the choice to do the regression at trade level or at netting set level, here we no longer have such a choice as the IM is only defined at netting set level. As a consequence, the number of explanatory variables can be large in realistic situations, leading to the added complexity of regression in high dimension.

One may consider a simplification of this by using PV itself as the only explanatory variable. The regression becomes much simpler, but [Caspers17a] reports that this idea leads, as expected, to lower accuracy than the multi-dimensional approach, assuming the multi-dimensional regression has completed successfully.

2.9.7 Differential methods

In differential methods distributional assumptions are made as well, while the VcV matrix of the underlying risk factors is provided from outside and VaR is estimated using the differentials of PV w.r.t. these risk factors. The differentials may be calculated analytically in some cases, or numerically in others, or as a combination of both for instance following [Antonov17a].

One of the advantages of these methods is that it allows for an external set of volatilities to be input in the model, and as such may be more consistent with the banks' actual IM models than other methods. For instance, one may consider using the volatilities and correlations as specified by SIMM, which leads to the model often called "Simulated SIMM". Note that even if the VcV matrix is the same as SIMM, this does not guarantee full identity with SIMM. Indeed, SIMM also specifies a particular process for aggregation and a set of risk factors that may not be modelled in the simulation, and therefore whose differential may not be calculable in the model (for instance the basis risk).

2.9.8 Nested simulation methods

Also called "brute force", such methods simply attempt to calculate VaR on the paths using a sub-simulation at each node of the main simulation. A Monte-Carlo sub-simulation will in general not be feasible, but one may consider doing a Historical Simulation instead.

This is the approach taken by [Green16a], which, as far as we can tell, is the only member of this family that has a chance of being applied in practice. The algorithm, which [Green16a] called "Augmented Compression", takes advantage of the LSMC step applied in the main simulation to be able to reprice the netting set under historical scenarios as specified from outside. This idea assumes that historical returns for all the relevant risk factors have been prepared, stored, and fed to the engine. At each point in time the engine calculates the changed value of the netting set on the scenarios by using the result of the LSMC regression and applying it at the historically-shifted risk factors.

At each point of the simulation, this gives a set of simulated returns which can then be ordered and used to calculate the required percentile. [Green16a] makes the additional remark that this idea applied naively will most probably lead to estimating PV changes based on a range of historical scenarios that go beyond what has been simulated in the regression step. As a consequence, it is not safe to use the regression functions as obtained by that step. [Green16a] proposes that the regression should be done on an augmented set of risk factors that contains the historical returns.

2.9.9 Scaling adjustments

In all models above the simulated IM will in general not match the IM as calculated by the counterparties' IM models. Of course this is true of the IM simulated at future times, but this is also true in most models for the IM as simulated on the valuation date. The easiest improvement for this problem is simply to scale multiplicatively the simulated IM so that it matches the $t = 0$ IM exactly.

Let us call I_0 the IM as calculated at $t = 0$ by the bank's IM model, I_0^U the IM calculated by the simulation at $t = 0$ (the same for all paths) and $I_i^{j,U}$ the simulated IM on path j at time i before normalizing (U standing for "Un-normalized"). The most simple scaling strategy consists in defining our final simulated IM I_i^j as

$$I_i^j = \alpha I_i^{j,U} \quad (292)$$

$$\alpha = \frac{I_0}{I_0^U} \quad (293)$$

where we have define the scaling coefficient α . Since all paths j see the same un-normalized IM at $t = 0$, i.e. $I_0^{j,U} = I_0^U$ at all j , this scaling ensures that all simulated paths I_i^j start exactly at the IM as calculated by the bank's official model.

[Anfuso17a] argues that this scaling may be valid at $t = 0$ but that it will not remain so for later times in the simulation, in particular due to differences in the volatilities assumes by the bank's IM model and those

used by the MVA simulation. As a consequence [Anfuso17a] refines the strategy above by using a parametric time-dependent scaling function α_i and suggests possible calibration methods for its parameters.

An alternative idea consists in modifying the volatility input in the MVA simulation to make the simulation match the $t = 0$ IM. This achieves the same result at $t = 0$ but has the advantage of having more financial meaning than the ad-hoc scalings above. Indeed, SIMM is defined for stressed volatilities while the un-normalized simulation model will run based on the volatilities implied by the market on the valuation date, i.e. not particularly stressed. It thus makes sense to stress the input volatilities as this puts the simulation model more in line with the true IM model. However, this process may be cumbersome and does not address the issue of the long-term scaling mentioned in [Anfuso17a].

2.9.10 Conclusions

A very good review of several of the methodologies presented above is featured in [Caspers17a], whose findings we summarize below.

1. all methods perform rather well on the examples of vanilla swaps. The IM patterns shown in [Caspers17a] suggest that even the Deterministic Method would work well in such cases, although we believe that this conclusion is only valid for swaps that are alone in a netting set. In more realistic situations with multiple swaps, we expect a significant loss of accuracy of the Deterministic Method.
2. for non-linear products such as options, the IM patterns show that the Deterministic Method would be significantly off. Furthermore, [Caspers17a] finds that the Simple Method visibly under-performs as well (although less dramatically).
3. [Caspers17a] find that the simplifying idea of using a single-dimensional regression by regressing on PV yields a rather accurate value at low dimensions but a loss of accuracy at higher dimensions is noted
4. [Caspers17a] warn however about the difficulty of multi-dimensional regressions and admits that it is a complex subject and that while several types of regression methods exist, the authors note that the reliability of such methods still needs to be assessed.

Taking the above into account, our conclusions are the following

- Deterministic method: very simple but expected to be off significantly as soon as several vanilla swaps with different maturities are in the same netting or in the case of non-linear payoffs
- Simple method: probably a good compromise between complexity and accuracy. Not very difficult to implement but leading to correct shapes for the IM evolution (as opposed to the Deterministic Method). However, at least in [Caspers17a] we can see it to be more noisy than the other methods and apparently over-estimating the IM in all test examples.
- Regression methods: the next level in terms of complexity and accuracy. These involve additional regression processes. These method do not appear to be complicated in terms of implementation as they only reuse what is already implemented for the other XVAs. However, they force a regression at netting set level which in principle leads to high dimensionality and the reliability of such regressions is an open question.
- Differential methods: more complex, potentially much more complex to implement as they require the calculation of differentials. However, they have the advantage of not necessarily using regressions at netting set level which can avoid the dimensionality issue. On the other hand, calculating the differentials can be a major difficulty (both in implementation and runtime). The Simulated SIMM method has the advantage of being potentially very consistent with the true IM model.
- Nested simulation methods: considering only the Augmented Compression as example of such methods, on top of a probable slowness at runtime, the implementation appears to be rather complex with in particular a significant work at input level (gathering time series for all risk factors, "augmenting" the regression set, etc...).

We are considering implementing at least the Simple method and the PV-regression method for potential production use. The multi-regression method could also be considered for production if multi-dimensional regression proves to be reliable. Differential methods appear to be unrealistic on the short-term and so is the Augmented Compression, but these could be considered at prototyping/benchmarking level.

2.10 Methodologies for KVA

2.10.1 Market Risk

Citing [Green16a] *"The internal model method for market risk capital currently uses VaR/SVaR and will move to the closely related expected shortfall measure under the FRTB."* This means that similar methodologies as the MVA can be applied.

2.10.2 Counterparty Credit Risk

In this case the internal model method is based on the potential exposure calculation. This makes its simulation more complex than for the Market Risk, but we could imagine a simplification by using interpolated add-ons. Since the Current Exposure is already simulated in the XVA engine for the other XVAs, we could calculate the PFE part by interpolated stored add-ons.

3 Documentation Sign Off

Date	Version	Author	Change Details
25/10/2016	0.0	Sebastien Gurrieri	Initial version
11/11/2016	0.1	Sebastien Gurrieri	FX Forwards, MtM swaps, XCcy FVAs
14/12/2016	0.2	Sebastien Gurrieri	XCcy XVAs
17/01/2017	0.3	Sebastien Gurrieri	Change of measure in drift, FX surface inversion
26/01/2017	0.4	Sebastien Gurrieri	Replication method
06/06/2017	1.0	Sebastien Gurrieri	MVA, KVA, funding and hedging strategies, harmonizing notations
13/06/2017	1.1	Sebastien Gurrieri	Equivalent reshuffling in FVA for easier presentation
12/07/2017	2.1	Sebastien Gurrieri	Introducing the Zero-Cost strategy
13/07/2017	2.2	Sebastien Gurrieri	Reformatting Zero-Cost
18/07/2017	2.3	Sebastien Gurrieri	More work on identification in Firm Value
15/08/2017	3.0	Sebastien Gurrieri	Refactoring, No-Shortfall as non-hedged, clarifying bond-CDS basis
17/08/2017	3.1	Sebastien Gurrieri	2 and 3-bond strategies, No-Shortfall funding in hedged strategies.
18/08/2017	3.2	Sebastien Gurrieri	Cleaning, re-arranging sections.
24/10/2017	3.3	Sebastien Gurrieri	Segregation/Rehypo, Tail-Risk CVA, WWR.
27/10/2017	3.4	Sebastien Gurrieri	Survey of MVA methodologies.
03/11/2017	3.5	Sebastien Gurrieri	Survey on the MPOR.
18/06/2018	4.0	Sebastien Gurrieri	Adding FT, reorganizing strategies, reformatting.
03/07/2018	4.1	Sebastien Gurrieri	Calculating PVs with different reference rates.
21/08/2018	4.2	Sebastien Gurrieri	Defaultable assets, bond forwards and TRSs.
22/08/2018	4.21	Sebastien Gurrieri	Adding IA, clarifying distinction between funding rate and repo rate.

Date	Version	Author	Change Details
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A Toolkit

A.1 Pricing Formulas

Domestic Measure

$$\begin{cases} P_{d,d}(t, T) &= E_t^{Q^d} [e^{-\int_t^T c_d(u) du}] \\ P_{f,d}(t, T)X(t) &= E_t^{Q^d} [e^{-\int_t^T c_d(u) du} X(T)] \\ X(t) &= E_t^{Q^d} [e^{-\int_t^T r_{df}(u) du} X(T)] \end{cases} \quad (294)$$

Foreign Measure

$$\begin{cases} \frac{P_{d,f}(t, T)}{X(t)} = E_t^{Q^f} [e^{-\int_t^T c_f(u) du} \frac{1}{X(T)}] \\ P_{f,f}(t, T) = E_t^{Q^f} [e^{-\int_t^T c_f(u) du}] \\ 1/X(t) = E_t^{Q^f} [e^{\int_t^T r_{df}(u) du} 1/X(T)] \end{cases} \quad (295)$$

A.2 Change Of Measure

A.2.1 Generic formulation with two numeraires

Changing between numeraires N_1 and N_2

$$E_t^{N_1} [\frac{N_1(t)}{N_1(T)} R] = E_t^{N_2} [\frac{N_2(t)}{N_2(T)} R] \quad (296)$$

$$E_t^{N_1} [R] = E_t^{N_2} [\frac{N_1(T)N_2(t)}{N_1(t)N_2(T)} R] \quad (297)$$

or equivalently using the likelihood process $M(t)$

$$E_t^{N_1} [R] = E_t^{N_2} [\frac{M(T)}{M(t)} R]. \quad (298)$$

Then the likelihood process is a Martingale in N_2 since

$$E_t^{N_2} [M(T)] = M(t). \quad (299)$$

A.2.2 MMA and change to T -forward

By taking $N_1(t) = P(t, T)$ and $N_2(t) = e^{\int_0^t c(u) du}$ we obtain the change of measure from T -forward to collateral

$$E_t^T [R] = E_t^Q [\frac{e^{-\int_t^T c(u) du}}{P(t, T)} R]. \quad (300)$$

The likelihood process can be defined as

$$M(t) = \frac{P(t, T) e^{-\int_0^t c(u) du}}{P(0, T)} \quad (301)$$

and can be checked to satisfy the change of measure formula as

$$E_t^T[R] = E_t^Q\left[\frac{M(T)}{M(t)}R\right] \quad (302)$$

with the specific form of $M(\cdot)$ defined in (301). We can then invert to get the change from collateral to T-forward measure as

$$E_t^Q[R] = E_t^T[e^{\int_t^T c(u) du} P(t, T) R]. \quad (303)$$

A.2.3 Cross-currency change of measure

From domestic to foreign

$$E_t^f[R] = \frac{1}{X(t)} E_t^d[e^{-\int_t^T r_{df}} X(T) R] \quad (304)$$

and from foreign to domestic

$$E_t^d[R] = X(t) E_t^f\left[\frac{e^{\int_t^T r_{df}}}{X(T)} R\right]. \quad (305)$$

Alternatively it is often convenient to look at the same formulas but including the discounting in the respective measures, i.e.

$$E_t^f[e^{-\int_t^T c_f} R] = \frac{1}{X(t)} E_t^d[e^{-\int_t^T c_d + q_{df}} X(T) R] \quad (306)$$

and from foreign to domestic

$$E_t^d[e^{-\int_t^T c_d} R] = X(t) E_t^f\left[e^{-\int_t^T c_f - q_{df}} \frac{R}{X(T)}\right]. \quad (307)$$

A.2.4 Change of measure and drift

We start by re-deriving the result for generic numeraires and then apply to the specific case of change of measure between domestic and foreign currencies.

Generic formula

Let us take a generic asset X evolving with a Brownian motion process dW^S under numeraire S . We write its SDE as

$$\frac{dX}{X} = \mu_S dt + \sigma dW^S \quad (308)$$

where σ may be a function of X . We write the change of measure property as

$$E^S[X_T] = E^U\left[\frac{U_0 S_T}{U_T S_0} X_T\right] \quad (309)$$

where X has the following dynamics in the U measure

$$\frac{dX}{X} = \mu_U dt + \sigma dW^U. \quad (310)$$

The goal here is to find the expression of μ_U in terms of the other quantities of the problem. We do so by calculating the expectations on both sides of (309). In this process we will need to integrated expressions of X_T , which read

$$X_T = X_0 e^{\int_0^T (\mu_S - \frac{1}{2}\sigma^2)dt + \sigma dW^S} \quad (311)$$

$$X_T = X_0 e^{\int_0^T (\mu_U - \frac{1}{2}\sigma^2)dt + \sigma dW^U}. \quad (312)$$

We also need to write down the dynamics of the ratio S/U which we do as

$$\frac{dS/U}{S/U} = \sigma_{S/U} dW^{S/U} \quad (313)$$

in the U measure. As it is Martingale it has no drift, and its integrated expression is

$$\frac{S_T}{U_T} = \frac{S_0}{U_0} e^{\int_0^T -\frac{1}{2}\sigma_{S/U}^2 dt + \sigma_{S/U} dW^{S/U}}. \quad (314)$$

Putting everything together in (309) leads to

$$e^{\int_0^T \mu_S dt} = E^U \left[e^{\int_0^T (\mu_U - \frac{1}{2}\sigma^2 - \frac{1}{2}\sigma_{S/U}^2)dt + \sigma dW^U + \sigma_{S/U} dW^{S/U}} \right] \quad (315)$$

and, after calculating the expectation and taking the log,

$$\int_0^T \mu_S dt = \int_0^T (\mu_U - \frac{1}{2}\sigma^2 - \frac{1}{2}\sigma_{S/U}^2)dt + \frac{1}{2} \langle \sigma dW^U + \sigma_{S/U} dW^{S/U}, \sigma dW^U + \sigma_{S/U} dW^{S/U} \rangle \quad (316)$$

$$= \int_0^T \mu_U dt + \langle \sigma dW^U, \sigma_{S/U} dW^{S/U} \rangle. \quad (317)$$

Finally we rewrite this infinitesimally as

$$\mu_U = \mu_S + \langle d \ln X, d \ln \frac{U}{S} \rangle. \quad (318)$$

Cross-currency case

The cross-currency case is the situation where S and U are in different currency. More precisely we consider the change of measure between the domestic and foreign collateral measures. The bank accounts are the numerares in each currencies, but the change of measure formula involves an FX conversion and the cross-currency basis. Let us start for instance from (307). This formula can be understood as the change of measure between the domestic money market account

$$S = e^{\int_0^t c_d} \quad (319)$$

and the foreign one after correction with the basis and conversion in domestic currency

$$U = e^{\int_0^t c_f - q_{df}} FX_t. \quad (320)$$

We can now apply the change of numeraire formula (318). Money market accounts do not have diffusion terms so the formula becomes

$$\mu_f = \mu_d + \langle d \ln X, d \ln FX \rangle \quad (321)$$

where FX is the number of domestic currency for 1 unit of foreign currency.

A.3 Integral Transformations

A.3.1 Intensity-discounting transformations

In an intensity-based default model, the default intensity λ_u satisfies the following properties

Property 1.

$$\int_t^T D_r(t, s) \mathbf{1}_{s < \tau} \pi_s ds = \int_t^T D_{r+\lambda}(t, s) \pi_s ds \quad (322)$$

where $D_{r+\lambda}(t, s)$ is the risky discount factor

$$D_{r+\lambda}(t, s) = e^{-\int_t^s (r_u + \lambda_u) du}.$$

Property 2.

$$\int_t^T D_r(t, s) \mathbf{1}_{\tau \in ds} \pi_s = \int_t^T D_{r+\lambda}(t, s) \lambda_s \pi_s ds \quad (323)$$

Proof. Denoting by $Q(s) = e^{-\int_t^s \lambda_u du}$ the survival probability function,

$$\begin{aligned} \int_t^T D_r(t, s) \mathbf{1}_{\tau \in ds} \pi_s &= \int_t^T D_r(t, s) (Q(s) - Q(s + ds)) \pi_s \\ &= - \int_t^T D_r(t, s) \frac{\partial Q}{\partial s} \pi_s ds \\ &= \int_t^T D_r(t, s) Q(s) \lambda_s \pi_s ds \\ &= \int_t^T D_{r+\lambda}(t, s) \lambda_s \pi_s ds. \end{aligned}$$

□

A.3.2 Spread-discounting transformations

We are interested in transforming expressions that often appear in this work, namely a situation like the following. A base value V_t^r is calculated by discounting the payoffs π_s with a discount factor $D_r(t, s)$ at some generic rate r with the notations used throughout this document i.e.

$$\begin{aligned} D_r(t, s) &= e^{-\int_t^s r(u) du} \\ V_t^r &= \int_t^T D_r(t, s) \pi_s ds. \end{aligned}$$

A correction is added to V_t^r that takes the form

$$\int_t^T D_h(t, s) (r_s - f_s) V_s^c ds$$

where h, f, c are other rates to be specified case-by-case. Our goal is to simplify the total expression of base + correction.

We start with some preliminary notations and properties.

Definition 1. Let $R(u, v)$ and $R(u)$ be defined as

$$\begin{aligned} R(u, v) &= \frac{D_r(u, v)}{D_f(u, v)} \\ R(v) &= R(t, v) \end{aligned}$$

where t is a fixed value.

Property 3. The R functions satisfy the following properties

$$\begin{aligned} R(t) &= 1 \\ \frac{dR(v)}{dv} &= -R \times (r - f) \end{aligned} \tag{324}$$

$$R(v)R(v, u) = R(u) \tag{325}$$

Using these definitions and properties we can prove a few useful properties of the base integrals and their corrections. Some of them are exact, others are approximated and we give the approximation term.

Proposition 1. Case where the exposure in the correction term discounts its flows (from payment to exposure date) with a different rate compared to the base exposure. We have the exact result

$$V_t^r + \int_t^T D_r(t, s)(r_s - f_s)V_s^f ds = V_t^f \tag{326}$$

Proof.

$$\begin{aligned} \int_t^T D_r(t, s)\epsilon_s(r_s - f_s)ds &= - \int_t^T D_f(t, s)R'(s)ds \left(\int_s^T D_f(s, u)\pi_u du \right) \\ &= - \int_t^T \int_s^T dsdu D_f(t, u)R'(s)\pi_u \\ &= - \int_t^T H(u)\pi_u du. \end{aligned}$$

The double integral reduces to a sum of instantaneous payoffs π_u which we expressed by introducing the 1-dimensional integral in the 3rd line above, and all we need to do is find the weight function $H(u)$.

We can gather the terms in the double integral and recognize that each π_u appears many times for specific values of s with the infinitesimal weight

$$ds D_f(t, u)R'(s)$$

but that this weight is effective only when $s < u$. In other words we have found the weight function $H(u)$ to be

$$\begin{aligned} H(u) &= \int_t^u D_f(t, u)R'(s)ds \\ &= D_f(t, u) \int_t^u dR \\ &= D_f(t, u)(R(u) - 1) \end{aligned}$$

which leads to a total price of

$$\begin{aligned}
\bar{V}_t &= V_t^0 - \int_t^T H(u) \pi_u du \\
&= \int_t^T D_f(t, u) \pi(u) (R(u) - R(u) + 1) du \\
&= \int_t^T D_f(t, u) \pi(u) du.
\end{aligned}$$

□

Proposition 2. *Case where the exposure in the correction term discounts its flows (from payment to exposure date) with the same rate as the base exposure, but is discounted from exposure date to 0 at a different rate. We have the exact result*

$$V_t^r + \int_t^T D_f(t, s)(r_s - f_s) V_s^r ds = V_t^f \quad (327)$$

Proof. Start with eq. (326) and rewrite it by bring the correction term on the right side. We get

$$\begin{aligned}
V_t^r &= V_t^f - \int_t^T D_r(t, s)(r_s - f_s) V_s^f ds \\
&= V_t^f + \int_t^T D_r(t, s)(f_s - r_s) V_s^f ds.
\end{aligned}$$

Then exchange the letters "r" and "f" and this gives the proposition.

□

Proposition 3. *Case where the exposure in the correction term discounts its flows (from payment to exposure date) with the same rate as the base exposure, and is discounted from exposure date to 0 at that same rate as well. We have the approximate result*

$$\begin{aligned}
V_t^r + \int_t^T D_r(t, s)(r_s - f_s) V_s^r ds &= \int_t^T D_f(t, s) \pi_s ds + \delta_{r-f} \\
\delta_{r-f} &= \int_t^T D_f(t, u) \pi(u) R(u) (1 - \ln R(u) - \frac{1}{R(u)}) du \\
R(u) &= e^{-\int_t^u r-f}
\end{aligned} \quad (328)$$

where δ_{r-f} is an r -dependent correction at order 2 in $r - f$.

Proof.

$$\begin{aligned}
\int_t^T D_r(t, s) \epsilon_s (r_s - f_s) ds &= - \int_t^T D_f(t, s) R'(s) ds \left(\int_s^T D_f(s, u) R(s, u) \pi_u du \right) \\
&= - \int_t^T \int_s^T ds du D_f(t, u) R(u) \frac{R'(s)}{R(s)} \pi_u \\
&= - \int_t^T G(u) \pi_u du.
\end{aligned}$$

The weight $G(u)$ of the instantaneous payoffs is now

$$ds D_f(t, u) R(u) \frac{R'(s)}{R(s)}$$

and is integrated as

$$\begin{aligned} G(u) &= \int_t^u D_f(t, u) R(u) \frac{R'(s)}{R(s)} ds \\ &= D_f(t, u) R(u) \int_t^u \frac{dR}{R} \\ &= D_f(t, u) R(u) \ln R(u) \end{aligned}$$

which leads to a total price of

$$\begin{aligned} \bar{V}_t &= V_t^0 - \int_t^T G(u) \pi_u du \\ &= \int_t^T D_f(t, u) \pi(u) (R(u) - R(u) \ln R(u)) du \\ &= \int_t^T D_f(t, u) \pi(u) du + \delta_{r-f}. \end{aligned}$$

The correction term δ_{r-f} is dependent on r and has the exact expression

$$\delta_{r-f} = \int_t^T D_f(t, u) \pi(u) R(u) \left(1 - \ln R(u) - \frac{1}{R(u)}\right) du.$$

is last approximation sign is in the following sense. Let us take advantage of the specific form of the function $R(u)$ to introduce the rate correction α as

$$\begin{aligned} \alpha &= \int_t^u (r - f) du \\ R(u) &= e^{-\int_t^u (r-f) du} = e^{-\alpha}. \end{aligned}$$

For small α , the Taylor expansion in α leads to

$$\begin{aligned} R - R \ln R - 1 &= e^{-\alpha} (1 + \alpha) - 1 \\ &= (1 - \alpha + \frac{1}{2} \alpha^2 + o(\alpha^3)) (1 + \alpha) - 1 \\ &= -\frac{1}{2} \alpha^2 + o(\alpha^3). \end{aligned}$$

□

A.3.3 Rate-default transformation

Proposition 4. *Let us consider a discount curve $P_r(t, T)$ with rate r and instantaneous forward f_r , and a survival curve $Q_\lambda(t, s)$ with hazard rate λ . Then we have the equality*

$$P_r(t, T) Q_\lambda(t, T) + \int_t^T P_r(t, s) Q_\lambda(t, s) \lambda ds = 1 - \int_t^T P_r(t, s) Q_\lambda(t, s) f_r ds \quad (329)$$

Proof.

$$\begin{aligned}
\int_t^T P_r(t, s) Q_\lambda(t, s) \lambda ds &= - \int_t^T P_r(t, s) \frac{\partial Q_\lambda}{\partial s}(t, s) ds \\
&= -P_r(t, T) Q_\lambda(t, T) + P_r(t, t) Q_\lambda(t, t) + \int_t^T \frac{\partial P_r}{\partial s}(t, s) Q_\lambda(t, s) ds \\
&= -P_r(t, T) Q_\lambda(t, T) + 1 - \int_t^T P_r(t, s) Q_\lambda(t, s) f_r ds
\end{aligned}$$

In the first line we used the definition of the survival probability in terms of the hazard rate, which is the same relation holding between the yield curve and the instantaneous forward rate used in the third line. In the second line we used integration by part and the fact that $P_r(t, t) = Q_\lambda(t, t) = 1$. $\square$

A.4 PDE Algebra

$$\left[(V - C)^+ + RR_I(V - C)^- \right]^+ = (V - C)^+ \quad (330)$$

$$\left[(V - C)^+ + RR_I(V - C)^- \right]^- = RR_I(V - C)^-. \quad (331)$$

We introduce the funding spreads for borrowing and lending

B More on the XVA Derivation

B.1 The Expectation Method

What we call "Expectation Method" is what [Green16a] calls "Expectation-theoretic approach" and is based on constructing the cash-flows generated by the trade together with the collateral and funding strategy and writing the valuation as the discounted expectation of the flows. These "flows" include default events, i.e. may or may not take place depending on survival of the entities. It is the method used in [Brigo14a] and this section borrows most of its content to this reference.

We start by a high-level treatment of the cash-flows and discounting, before going into more details on all components. Trying to stay close to the notations of [Brigo14a], we consider a trade between two counterparties "C" and "B", where we are "B", the bank, i.e. flows that are received by us/B are positive.

First of all let us set up some notations. Default times are denoted by the letter τ with subscript specifying the entity defaulting. The symbol

$$\tau = (\tau_B \wedge \tau_C) \quad (332)$$

where τ_B is our default time and τ_C is the default time of the counterparty, gives the minimum of the 2, that is the time to first default. We also take the following notations

- $\pi(t, T)$ is the sum of the discounted payoffs between t and T , only coming from the derivative itself
- $\gamma(t, T)$ is the sum of the discounted margining costs over the same period
- $C(t)$ is the amount on the collateral account
- $g_\tau(\epsilon)$ is the on-default cash-flow
- Γ is the close-out value
- $\varphi(t, T)$ is the sum of the discounted funding costs over the period
- $F(t)$ is the amount in the cash account to fund the deal
- $D_r(t, T)$ is the discount factor observed at time t for maturity T , at the rate r
- $\bar{V}_t$ is the value of the deal including all costs, measured at time t .

The cash-flow method gives the total value $\bar{V}_t$ as the discounted expectation of all the cash-flows, i.e.

$$\bar{V}_t = E_t[\pi(t, T \wedge \tau) + \gamma(t, T \wedge \tau) + \varphi(t, T \wedge \tau) + \mathbf{1}_{\tau < T} D(t, \tau) g_\tau(\epsilon)] \quad (333)$$

A choice is made for the risk-free rate underlying $D(t, T)$ in a reference currency, say USD for MHI. This rate will be approximated by USDOIS, and all non-USD cash-flows due in the future are converted into USD at the future spot FX. With this in mind, we consider all flows as measured in the base currency, i.e. including the FX conversion as of payment date. The issue of cross-currency flows and discounting are addressed later in this document.

B.1.1 Margining cost

We consider a generic margining procedure with unspecified thresholds and MTAs. The comprises boundary cases such as fully collateralized or uncollateralized situations, as well as unilateral collateralization. While the analysis is doable with asymmetric collateral rates (i.e. different for each counterparty), this case is not of interest to us so we will simplify to the case where there is a single rate underlying the collateral account as seen by both counterparties when or if they post. This rate is denoted by c_t and is valid at margining time t for the current margining period.

The margining schedule is described as a discrete set of times $\{t_i\}_{i=1..n}$ where the collateral is updated. More precisely, over each period $[t_i, t_{i+1}]$ with accrual fraction where $\delta_i = \delta(t_i, t_{i+1})$, the following processes take place

- At t_i , C_{t_i} is posted
- At t_{i+1} , the amount $C(t_i)$ is returned with interest on the posted amount, that is $C(t_i) \times c_{t_i} \delta_i$
- At t_{i+1} , margining is performed according to the characteristics of the CSA and the new amount $C_{t_{i+1}}$ is posted.

On the last margining date t_n the current collateral is returned with its interest and the procedure stops with no further margining.

Taking the example of the period above, we will write the value of all these flows by bringing the returned collateral and interest to time t_i by discounting them from t_{i+1} to t_i with the risk-free rate (with a short margining period of time in mind, typically daily). The combined flow seen from t_i is thus

$$C_{t_i} - C_{t_i}(1 + c_{t_i} \delta_i) P_{t_i}(t_{i+1}) \quad (334)$$

where $P_{t_i}(t_{i+1})$ is the risk-free discount bond as of t_i for time t_{i+1} . Furthermore let us define the equivalent quantity for the collateral rate $P_{t_i}^c(t_{i+1})$, i.e. the curve curve

$$P_t^c(T) = \frac{1}{1 + c_t(T - t)}. \quad (335)$$

Then the margining flow at t_i is

$$C_{t_i} - C_{t_i} \frac{P_{t_i}(t_{i+1})}{P_{t_i}^c(t_{i+1})} \quad (336)$$

and the total margining cost is

$$\gamma(t, T \wedge \tau) = \sum_{i=1}^{n-1} \mathbf{1}_{t_i < (T \wedge \tau)} D(t, t_i) C_{t_i} \left(1 - \frac{P_{t_i}(t_{i+1})}{P_{t_i}^c(t_{i+1})}\right). \quad (337)$$

To see a little more intuitively what this term is, we may adopt a linearization approximation over these intervals in which case the margining cost becomes

$$\gamma(t, T \wedge \tau) \approx \sum_{i=1}^{n-1} \mathbf{1}_{t_i < (T \wedge \tau)} D(t, t_i) C_{t_i} (r_{t_i} - c_{t_i}) \delta_i \quad (338)$$

where we can see the spread between collateral and risk-free rate $r - c$. In particular the margining cost term $\gamma(\cdot)$ vanishes when the collateral rate is the risk-free rate.

B.1.2 Close-out payment

We have defined in a very generic manner the symbol Γ as the close-out amount. Now it is time to specify a bit further. [Brigo14a] writes this term in a very generic way handling the potential situation where the counterparties do not agree on what the close-out should be. Here we will focus our interest on the more practical situation where they do agree, such that the close-out amount is independent on the counterparty and is written as the amount Γ_s at time s .

The reason why we keep this amount unspecified for now is due to the fact that it is open to interpretation what it should be, with 2 main views: it could be the risk-free value of the trade at default, or its risky value including the valuation adjustments, with various possibilities depending on which adjustments are included.

We will see later what such choices imply for valuation. For now we want to calculate the total value of the flow assuming one has agreed on the type of close-out. We start by a few words on the timing of the margining. The calculation of the margining flows in the previous section starts with the first posting and ends with the last return including interest. These marginings only happen up until default, a fact that we captured with the default indicator functions in (337). This cost is thus calculated such that its last value when the trade ends is 0. However, in practice upon default there will be an amount on the collateral account, which will effectively be the result of the last margining. This flow is not captured in part of the margining cost so we will include it here. Let us denote by C_τ the amount on the collateral account right before the default happens.

The total flow splits into 2 components depending on who defaults first between us (B) and the counterparty (C). We will explain in more details the case where the counterparty defaults first, the other case being simply symmetric. When the counterparty defaults, the amount C_τ is on the collateral account. As discussed above, this amount has not been included in the margining cost, so we capture it here by saying that the counterparty posts (or just posted) C_τ . The default is settled by comparing the close-out value Γ_τ to what is already on the collateral account C_τ . If the close-out value is larger than the collateral amount, the counterparty owes us the difference, but we will only receive the recovery fraction of it. However, if the close-out value is smaller than the collateral amount, we owe the difference and pay it in full. As a consequence, the flow is

$$C_\tau + RR_C(\Gamma_\tau - C_\tau)^+ + (\Gamma_\tau - C_\tau)^- = \Gamma_\tau - LGD_C(\Gamma_\tau - C_\tau)^+ \quad (339)$$

where RR_C (resp. LGD_C) is the recovery rate (resp. LGD) on the counterparty. The situation is symmetric for our default, which yields the flow

$$C_\tau + (\Gamma_\tau - C_\tau)^+ + RR_B(\Gamma_\tau - C_\tau)^- = \Gamma_\tau - LGD_B(\Gamma_\tau - C_\tau)^-. \quad (340)$$

Indeed, if the settlement amount $\Gamma_\tau - C_\tau$ is positive, it means the counterparty owes it to us and has to pay in full because they are surviving. However, if it is negative, we owe it to the counterparty but only pay our recovery rate RR_B on it.

The total flow must take into account the fact that these flows happen with different probabilities depending on who defaults first, which we write as

$$\begin{aligned} g_\tau &= \mathbf{1}_{\tau=\tau_C < \tau_B} [\Gamma_\tau - LGD_C(\Gamma_\tau - C_\tau)^+] \\ &\quad + \mathbf{1}_{\tau=\tau_B < \tau_C} [\Gamma_\tau - LGD_B(\Gamma_\tau - C_\tau)^-] \end{aligned} \quad (341)$$

The default indicator $\mathbf{1}_{\tau=\tau_C < \tau_B}$ counting the probability that the first defaulting entity is the counterparty. In particular one can use the identity

$$\mathbf{1}_{\tau=\tau_C < \tau_B} + \mathbf{1}_{\tau=\tau_B < \tau_C} = \mathbf{1} \quad (342)$$

to further reduce the default flow to

$$g_\tau = \Gamma_\tau - \mathbf{1}_{\tau=\tau_C < \tau_B} LGD_C(\Gamma_\tau - C_\tau)^+ - \mathbf{1}_{\tau=\tau_B < \tau_C} LGD_B(\Gamma_\tau - C_\tau)^- \quad (343)$$

where we can recognize the CVA-like and DVA-like terms containing the positive and negative exposures on default. Not much more can be said without specifying what the close-out amount Γ_τ is. We will do this later in this section.

B.1.3 Funding cost

The analysis proceeds in a similar way to that of the margining cost. For now we take a generic funding amount F_{t_i} on a time grid of funding roll periods, and will specify later what this amount is. We assume that there is such an amount that needs to be funded by borrowing at the funding rate for borrowing if $F > 0$ or the funding rate for lending if $F < 0$. The reference [Brigo14a] we are following here handles the case where the 2 funding

rates are different, but for our specific MHI methodology we consider them identical and denote their common rate by f_{t_i} , with associated funding curve $P_{t_i}^f(\cdot)$. This curve will typically be specified as a spread curve above the risk-free rate (USD OIS), but here we just keep it as a total funding curve.

On this funding time grid, the funding amount F_{t_i} is received at t_i and returned at t_{i+1} with interest at the funding rate, and at t_{i+1} a new funding position $F_{t_{i+1}}$ is entered into. Again we aggregate the flows of 1 period at its beginning t_i with risk-free discounting of the t_{i+1} returns down to t_i , and find

$$\varphi(t, T \wedge \tau) = \sum_{i=1}^{n-1} \mathbf{1}_{t_i < (T \wedge \tau)} D(t, t_i) F_{t_i} \left(1 - \frac{P_{t_i}(t_{i+1})}{P_{t_i}^f(t_{i+1})}\right). \quad (344)$$

up until t_n where the last funding amount is returned and the position is closed.

The difference with the handling of the collateral lies in the expression of the funding amounts F_{t_i} . Quoting [Brigo14a],

When we talk about a derivative deal's funding, we essentially mean the cash position that is required as part of the hedging strategy, and with funding costs we refer to the costs of maintaining this cash position [...] In general, we may borrow/lend the cash needed to establish [the hedge] from/to our treasury, and we may then use the risk assets in [the hedge] for repo or stock lending/borrowing in the market. This means that we could include the funding costs and benefits coming from this use of the risky asset.

and further assuming that the Treasury recognizes this benefit/cost at the same funding rate as the cash. We will do similarly from now, such that the hedge amount is just aggregated to the cash amount following the same growth under the funding rate. The funding amount F_t is thus the whole of the cash amount necessary to fund the position. If rehypothecation is not allowed, the amount from the collateral cannot be used and the funding is the whole value of the derivative, see [Brigo14a] for more details. In this document we will assume work under the rehypothecation case such that

$$F_t = \bar{V}_t - C_t. \quad (345)$$

B.1.4 Continuous time formulas

Now let us put together all the terms described above in the pricing formula (333), introduce the XVAs and go to the limit where the various schedules are with infinitesimally small time steps. We start by writing explicitly the sum of discounted payoffs $\pi(t, T \wedge \tau)$ appearing in (333) as

$$\pi(t, T \wedge \tau) = \int_t^T D(t, s) \mathbf{1}_{s < \tau} \pi_s ds \quad (346)$$

where $\pi_s ds$ denotes the flows on the infinitesimal interval $[s, s + ds]$.

The default flow g_τ is paid on default, and to write this in continuous time we define the probability $\mathbf{1}_{\tau \in ds}$ to mean that default happens on the interval $[s, s + ds]$. The CVA and DVA will further need to specify which entity defaults first. We thus define $\mathbf{1}_{\tau = \tau_C \in ds < \tau_B}$ as the probability that default happens in $[s, s + ds]$ and that this default is the one of the counterparty while the investor has not defaulted yet by time $s + ds$, with symmetric notation for the DVA. We can then write the total value $\bar{V}_t$ in several components as

$$\bar{V}_t = V_t^0 + CVA + DVA + FVA + CollVA \quad (347)$$

$$V_t^0 = E_t \left[\int_t^T D_r(t, s) (\mathbf{1}_{\tau > s} \pi_s ds + \mathbf{1}_{\tau \in ds} \Gamma_s) \right] \quad (348)$$

$$CVA = -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau = \tau_C \in ds < \tau_B} LGD_C(\Gamma_s - C_s)^+ \right] \quad (349)$$

$$DVA = -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau = \tau_B \in ds < \tau_C} LGD_B(\Gamma_s - C_s)^- \right] \quad (350)$$

$$FVA = -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau > s} (\bar{V}_s - C_s) (f_s - r_s) ds \right] \quad (351)$$

$$CollVA = -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau > s} C_s (c_s - r_s) ds \right]. \quad (352)$$

where for reminder c_s and f_s are the collateral and funding rates at time s . One can see the familiar formulas for CVA, DVA and FVA appearing, but we have to stress that without specifying the close-out amount Γ_s these formulas do not provide a prescriptive way of performing this valuation. Indeed in their general form these equations imply a non-linear behaviour of valuation where the current value $\bar{V}_t$ depends on the future values $\bar{V}_s$ for $s > t$. [Brigo14a] mentions that these non-linear effects are not necessarily small.

Before leaving this section we will rewrite the formulas above in a slightly more convenient way using the default intensity framework as briefly summarized in appendix A.3.1. In particular we define the default intensities λ_C for the counterparty and λ_B for ourselves. We will also often use the first-to-default intensity $\lambda = \lambda_C + \lambda_B$. With these notations and the properties (322) and (323), we can rewrite the valuation formulas above as

$$\begin{aligned} \bar{V}_t &= V_t^0 + CVA + DVA + FVA + CollVA \\ V_t^0 &= E_t \left[\int_t^T D_{r+\lambda}(t, s) (\pi_s + \lambda_s \Gamma_s) ds \right] \\ CVA &= -E_t \left[\int_t^T D_{r+\lambda}(t, s) \lambda_C LGD_C(\Gamma_s - C_s)^+ \right] \\ DVA &= -E_t \left[\int_t^T D_{r+\lambda}(t, s) \lambda_B LGD_B(\Gamma_s - C_s)^- \right] \\ FVA &= -E_t \left[\int_t^T D_{r+\lambda}(t, s) (f_s - r_s) (\bar{V}_s - C_s) ds \right] \\ CollVA &= -E_t \left[\int_t^T D_{r+\lambda}(t, s) (c_s - r_s) C_s ds \right]. \end{aligned} \quad (353)$$

B.1.5 Risky close-out

Full risky close-out corresponds to the assumption that the close-out value is the full risky value of the deal, i.e. that $\Gamma_s = \bar{V}_s$.

Limiting Case: Perfect Collateralization

We are going to show that under risky close-out and perfect collateralization, the total price $\bar{V}_t$ in the formulas above reduces to the well-known fact that the derivatives total value is calculated by discounting the flows with the collateral rate.

In this situation the collateral amount C_s is always perfectly matched with the close-out amount $\Gamma_s = \bar{V}_s$. This implies that CVA, DVA and FVA vanish and only the base value V_t and CollVA remain. The valuation becomes

$$\bar{V}_t = E_t \left[\int_t^T D_{r+\lambda}(t, s) \pi_s ds \right] + E_t \left[\int_t^T D_{r+\lambda}(t, s) (r_s + \lambda_s - c_s) \bar{V}_s ds \right]. \quad (354)$$

As mentioned above this does not give a prescriptive way of calculating $\bar{V}_t$ as it depends on the future $\bar{V}_s, s > t$. However, we can prove that collateral discounting is the solution. Indeed, let us postulate the candidate solution

$$\bar{V}_s = E_s \left[\int_t^T D_c(t, s) \pi_s ds \right] \quad (355)$$

that is, let us plug this collateral-discounted value in the full valuation equation (354). The r.h.s. of this equation is precisely in the configuration of Proposition 1 with base rate $r + \lambda$ and correction rate c , and therefore we can conclude using (326) that

$$E_t \left[\int_t^T D_{r+\lambda}(t, s) \pi_s ds \right] + E_t \left[\int_t^T D_{r+\lambda}(t, s) (r_s + \lambda_s - c_s) \bar{V}_s ds \right] = E_t \left[\int_t^T D_c(t, s) \pi_s ds \right] \quad (356)$$

$$= \bar{V}_t \quad (357)$$

and therefore that the collateral discounted value (355) is indeed the solution.

In particular, the risk-free rate r has dropped from the valuation which is therefore independent of the assumption of risk-free rate.

Limiting Case: No Collateralization, No Default

We take the other extreme situation for which we also have a standard pricing formula, that is no collateralization at all, i.e. $C_s = 0$. We are going to show that pricing reduces to the traditionally used result that the flows should be discounted with the funding rate (under further assumption of no default).

When $C_s = 0$, CollVA vanishes and under risky close-out together with assumption of no-default, valuation becomes CVA and DVA vanish as well, leading to

$$\bar{V}_t = V_t^0 + FVA \quad (358)$$

$$V_t^0 = E_t \left[\int_t^T D_r(t, s) \pi_s ds \right] \quad (359)$$

$$FVA = -E_t \left[\int_t^T D_r(t, s) \bar{V}_s (f_s - r_s) ds \right]. \quad (360)$$

We make use again of Proposition 1 with base rate r and correction rate f , and conclude using (326) that

$$\bar{V}_t = E_t \left[\int_t^T D_f(t, s) \pi_s ds \right] \quad (361)$$

is the solution, i.e. that we should discount future cash-flows with the funding rate.

In particular, the risk-free rate r has dropped from the valuation which is therefore independent of the assumption of risk-free rate.

Generic Case

We have above at 2 extreme situations, perfect collateralization (with default) and total absence of collateralization (without default). Both situations have exact pricing solutions that can be written in terms of discounting of future cash-flows with a specific rate, either the collateral or the funding rate. In particular, in both situations the final valuation does not depend on the assumption of risk-free rate.

In a generic case where collateralization is not perfect but still present, we are in a mixed situation with influence of both the collateral rate and the funding rate, and with non-linear terms in valuation. While this does not appear to be obvious from the equations in the generic case, [Brigo14a] shows that independence of the risk-free rate holds. However, he claims so after changes of measure to a "trade-dependent" measure. The meaning of this is not very clear to us. Furthermore this conclusion is reached after an assumption on the PDE evolution that is not obvious to us. In summary, we are not totally convinced about these arguments, but cannot rule them out.

B.1.6 Risk-free close-out

Now we assume risk-free close-out, i.e. that $\Gamma_s = V_s$, and further assume that the amount to be funded is the left-over from the risk-free value, or in other words that the funding component is proportional to $V_s - C_s$ rather than $\bar{V}_s - C_s$. *Beware that this is only a heuristic shortcut designed to recover familiar formulas, and we think it deserves clarification.* Without this shortcut, the formulas remain consistent with the literature such as [Brigo14a] and [Elouerkhaoui16a], but it is not easy to see the connection with familiar formulas that lend themselves to practical application. The replication approach seems easier to apply, see section 1.2 for more details.

Before presenting the formulas we simplify the V_t^0 component slightly. Since the close-out amount is the risk-free value V_s , this component becomes

$$V_t^0 = E_t \left[\int_t^T D_{r+\lambda}(t, s) \pi_s ds \right] + E_t \left[\int_t^T D_{r+\lambda}(t, s) \lambda_s V_s ds \right] \quad (362)$$

We can now use Proposition 1 with base rate $r + \lambda$ and correction rate r , and conclude using (326) that V_t^0 is in fact the risk-free value

$$V_t^0 = E_t \left[\int_t^T D_r(t, s) \pi_s ds \right]. \quad (363)$$

The XVAs can then be written as

$$\begin{aligned} \bar{V}_t &= V_t + CVA + DVA + FVA + CollVA \\ V_t &= E_t \left[\int_t^T D_r(t, s) \pi_s ds \right] \\ CVA &= -E_t \left[\int_t^T D_{r+\lambda}(t, s) \lambda_C LGD_C (V_s - C_s)^+ ds \right] \\ DVA &= -E_t \left[\int_t^T D_{r+\lambda}(t, s) \lambda_B LGD_B (V_s - C_s)^- ds \right] \\ FVA &= -E_t \left[\int_t^T D_{r+\lambda}(t, s) (f_s - r_s) (V_s - C_s) ds \right] \\ CollVA &= -E_t \left[\int_t^T D_{r+\lambda}(t, s) (c_s - r_s) C_s ds \right] \end{aligned} \quad (364)$$

where the close-out amount is the risk-free present value i.e.

$$\epsilon_s = V_s = E_s \left[\int_s^T D_r(s, u) \pi_u du \right]. \quad (365)$$

This results in a true split of risk components as the XVAs do not depend on each other any more.

Limiting Case: Perfect Collateralization

We are going to show that under risk-free close-out and perfect collateralization, as opposed to the case of the risky close-out, the total price $\bar{V}_t$ reduces to the collateral discounted price only approximately, with a correction amount that is dependent on the choice of risk-free rate.

The collateral amount C_s is always perfectly matched with the close-out amount $\Gamma_s = V_s$. CVA, DVA and FVA vanish and only the base value V_t^0 and CollVA remain. Under default intensity modelling, the valuation becomes

$$V_t^0 + \text{CollVA} = E_t \left[\int_t^T D_r(t, s) \pi_s ds \right] + E_t \left[\int_t^T D_{r+\lambda}(t, s) (r_s - c_s) V_s ds \right]. \quad (366)$$

With further algebra using Proposition 1 in the appendix and equation (326) we can write the solution as

$$\bar{V}_t = E_t \left[\int_t^T D_c(t, s) \pi_s ds \right] + \delta_{r, \lambda, c}. \quad (367)$$

This time the collateral discounting price is not an exact solution. The correction term $\delta_{r, \lambda, c}$ is present and in particular does not appear to be independent of r .

Limiting Case: No Collateralization

When $C_s = 0$, CollVA vanishes and under risk-free close-out linearity allows us to calculate the other XVAs independently of each other. This leads to

$$\bar{V}_t = V_t^0 + CVA + DVA + FVA \quad (368)$$

$$V_t^0 = E_t \left[\int_t^T D_r(t, s) \pi_s ds \right] \quad (369)$$

$$CVA = -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau = \tau_C \in ds < \tau_B} LGD_C V_s^+ \right] \quad (370)$$

$$DVA = -E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau = \tau_B \in ds < \tau_C} LGD_B V_s^- \right] \quad (371)$$

$$FVA = E_t \left[\int_t^T D_r(t, s) \mathbf{1}_{\tau > s} V_s (r_s - f_s) ds \right]. \quad (372)$$

Under default intensity modelling we rewrite the base + FVA part as

$$V_t^0 + FVA = E_t \left[\int_t^T D_r(t, s) \pi_s ds \right] + E_t \left[\int_t^T D_{r+\lambda}(t, s) (r_s - f_s) V_s ds \right] \quad (373)$$

which takes the same form as the expression we just worked on in perfect collateralization. The same conclusion follows, i.e. that

$$V_t^0 + FVA = E_t \left[\int_t^T D_f(t, s) \pi_s ds \right] + \delta_{r, \lambda, f} \quad (374)$$

and in particular that the solution is not exactly equal to discounting with the funding rate. A correction term is present and is not obviously independent of r .

Generic case

Only looking at the standard limiting cases, it does not appear that the risk-free close-out allows for independence of pricing of the choice of risk-free rate.

B.1.7 Other close-outs

With the previous sections in mind, one could imagine other types of close-out that would at least have a chance of ensuring some form of risk-free rate independence. For instance, if a CSA exists, one could define the close-out as the collateral discounted value. In the perfect collateralization case, valuation would reduce to collateral discounting exactly and therefore be invariant of the choice of risk-free rate. It is reasonable to assume that dependence on the risk-free rate for non-perfect collateralization may be reduced in this manner, although that is not certain. Similarly one could imagine a funding rate close-out in case the CSA does not exist, and in this situation pricing would reduce to funding rate discounting under the no-default assumption. Such other close-outs are considered for instance in [Green16a]. We may be looking at them in the future but for now remain focused on the risk-free close-out for practical reasons.

B.2 More on the Replication

B.2.1 Zero-Cost

In this section we give some details about how to derive the valuation equations for the generic Zero-Cost strategy whose PDE is (50).

First we

We start by introducing in the PDE the dummy term $+r\hat{V} - r\hat{V}$. This is analogous to how we introduced PV in the linear case. Here, we can see the entirely dummy nature of this term, and therefore, that anything that is consistently derived (i.e. without r -dependent approximations or assumptions) below is invariant under the choice of r . After introducing this term and a little re-arranging algebra, we find the PDE

$$\begin{aligned} \frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} - (r + \tilde{\lambda}_B + \tilde{\lambda}_C)\hat{V} &= \tilde{\lambda}_C(V^{r_\epsilon} - X - I_C)^+ + \tilde{\lambda}_B(V^{r_\epsilon} - X + I_B)^- \\ &\quad + (r_X - r)X + r_{I_C}I_C - (r_{I_B} - r)I_B + (\gamma_K - \phi r)K \\ &\quad + (r_1 - r)(\hat{V} - X + I_B - \phi K)^+ + (r_2 - r)(\hat{V} - X + I_B - \phi K)^- \\ &\quad - \tilde{\lambda}_B LGD_1(\hat{V} - X + I_B - \phi K)^+ - \tilde{\lambda}_B LGD_2(\hat{V} - X + I_B - \phi K)^- \\ &\quad - (\tilde{\lambda}_B + \tilde{\lambda}_C)V^{r_\epsilon} \end{aligned} \quad (375)$$

where we can recognize, in order of appearance on the r.h.s, terms that broadly speaking will become CVA , DVA , $CollVA$, MVA , and RVA .

At least formally, one may apply Feynman-Kac now to obtain valuation equations. Let us discuss the case of the main payoff contribution first. As the boundary condition for $\hat{V}$ is equality with the payoff, Feynman-Kac leads to a PV-like term

$$\begin{aligned} \hat{V} &= V^{r+\tilde{\lambda}_B+\tilde{\lambda}_C} \\ &= V^r - E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C}(\tilde{\lambda}_B + \tilde{\lambda}_C)V^r \right]. \end{aligned} \quad (376)$$

We are now in a position to write the valuation equations, at least formally, as

$$\begin{aligned}
\hat{V} &= PV + CVA + DVA + CollVA + MVA + KVA \\
&\quad + FCA + FBA + DVA_2 + DVA_3 + RVA \\
PV &= E_t \left[\int_t^T D_r \pi \right] \\
CVA &= -E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} \tilde{\lambda}_C LGD_C (V^{r_\epsilon} - X - I_C)^+ \right] \\
DVA &= -E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} \tilde{\lambda}_B LGD_B (V^{r_\epsilon} - X + I_B)^- \right] \\
CollVA &= -E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} (r_X - r) X \right] \\
MVA &= -E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} (r_{I_C} I_C - (r_{I_B} - r) I_B) \right] \\
KVA &= -E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} (\gamma_K - \phi r) K \right] \\
FCA &= -E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} (r_1 - r) (\hat{V} - X + I_B - \phi K)^+ \right] \\
FBA &= -E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} (r_2 - r) (\hat{V} - X + I_B - \phi K)^- \right] \\
DVA_2 &= E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} \tilde{\lambda}_B LGD_{P_1} (\hat{V} - X + I_B - \phi K)^+ \right] \\
DVA_3 &= E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} \tilde{\lambda}_B LGD_{P_2} (\hat{V} - X + I_B - \phi K)^- \right] \\
RVA &= -E_t \left[\int_t^T D_{r+\tilde{\lambda}_B+\tilde{\lambda}_C} (\tilde{\lambda}_B + \tilde{\lambda}_C) (V^r - V^{r_\epsilon}) \right].
\end{aligned} \tag{377}$$

B.2.2 Multiple Reference Rates

In this section we derive the valuation equations in the case where different reference rates are used to define PV. We take the case of the hedged 2-bond funding with PDE in (50), before introducing the notional of "PV". For simplicity as this is only for illustration purposes, we take the initial margins and capital terms to be 0, symmetric funding, and define

$$\tilde{r}_1 = r_1 - LGD_{P_1} \lambda_B = \tilde{r}_F \tag{378}$$

$$\tilde{r}_2 = r_2 - LGD_{P_2} \lambda_B = \tilde{r}_F \tag{379}$$

where we have define the funding rate r_F . Then the PDE becomes

$$\begin{aligned}
\frac{\partial \hat{V}}{\partial t} + \mathcal{A} \hat{V} &= r_X X - \lambda_C (g_C - \hat{V}) - \lambda_B (g_B - \hat{V}) \\
&\quad + \tilde{r}_F (\hat{V} - X).
\end{aligned} \tag{380}$$

As usual we introduce the on-default payments $g_C = Q - LGD_C(Q - X)^+$ and $g_B = Q - LGD_B(Q - X)^-$ and find

$$\begin{aligned}
\frac{\partial \hat{V}}{\partial t} + \mathcal{A}\hat{V} &= \lambda_C LGD_C(Q - X)^+ + \lambda_B LGD_B(Q - X)^- \\
&\quad + \tilde{r}_F(\hat{V} - X) + r_X X \\
&\quad - (\lambda_C + \lambda_B)(Q - \hat{V})
\end{aligned} \tag{381}$$

where we can find the terms that will become, after introducing PV, the future CVA, DVA, FVA, ColVA and RVA.

Note that FVA (and ColVA) are not written as "spreads" yet, that is they are proportional to full rates, not to spreads against some rate.

Now let us introduce the notion of "PV". We define PV as the sum of the expectations of the discounted flows of each trade, where the discount is done at a specific discount rate for each trade. Let us say that there are N trades counted by the subscript $i = 1..N$, and let us say that each trade has its PV defined by discounting with the rate r_i . In our notations, this means that PV, which we denote by V , is

$$V = \sum_{i=1}^N V_i^{r_i} \tag{382}$$

where $V_i^{r_i}$ is the expectation of the future flows of trade i discounted at rate r_i . PV then satisfies the following PDE

$$\frac{\partial V}{\partial t} + \mathcal{A}V = \sum_{i=1}^N r_i V_i^{r_i}. \tag{383}$$

Now we introduce the split of the risky value $\hat{V}$ into PV (V) and XVAs (U), i.e.

$$\hat{V} = V + U \tag{384}$$

and find the PDE on U

$$\begin{aligned}
\frac{\partial U}{\partial t} + \mathcal{A}U - (\tilde{r}_F + \lambda_B + \lambda_C)U &= \lambda_C LGD_C(Q - X)^+ + \lambda_B LGD_B(Q - X)^- \\
&\quad + \tilde{r}_F(V - X) - \sum_{i=1}^N r_i V_i^{r_i} + r_X X \\
&\quad - (\lambda_C + \lambda_B)(Q - V).
\end{aligned} \tag{385}$$

in which we do not recognize easily the familiar FVA and ColVA in the second line.

Now we take a closer look at this second line. When all trades are discounted with the same rate in the calculation of PV, i.e. $r_i = r$ for all i , the second line takes the form

$$\tilde{r}_F(V - X) - \sum_{i=1}^N r_i V_i^{r_i} + r_X X = \tilde{r}_F(V - X) - rV + r_X X \tag{386}$$

$$= (\tilde{r}_F - r)(V - X) + (r_X - r)X \tag{387}$$

where we can easily identify the familiar FVA and ColVA as

$$FVA \sim (\tilde{r}_F - r)(V - X) \tag{388}$$

$$ColVA \sim (r_X - r)X \tag{389}$$

However, this decomposition does not appear when the discount rates are different on different trades, and in that case we are left with the unfamiliar combination

$$FVA + ColVA = \tilde{r}_F(V - X) - \sum_{i=1}^N r_i V_i^{r_i} + r_X X \quad (390)$$

$$FVA = ? \quad (391)$$

$$ColVA = ? \quad (392)$$

where we cannot say what is FVA and what is ColVA.

B.3 Defaultable Underlyings a la Fries

Now let us introduce the concept of PV in order to be able to split the total value $\hat{V}$ into PV and XVAs. As before PV is defined as the discounted expected value of 'risk-free' flows. Here by risk-free we mean that both the C/P and the underlying do not default, so PV, still denoted by V^r is the discounted value of the payoff under assumption that the underlying has not defaulted by the time we reach maturity. It is still solution to the PDE (46) but now expressed with the new operator (168). We again decompose as

$$\hat{V} = V^r + U \quad (393)$$

and find the PDE

$$\begin{aligned} \frac{\partial U}{\partial t} + \mathcal{A}U - \gamma U &= -\lambda_C g_C - \lambda_B g_B - \lambda_{S/C} g_{S/C} - \lambda_{S \wedge C} g_{S \wedge C} \\ &+ (r_2 - LGD_{P_2} \lambda_B - r)(V^s - X + I_B - \phi K)^+ \\ &+ (r_3 - LGD_{P_3} \lambda_B - r)(V^s - X + I_B - \phi K)^- \\ &+ (r_X - r)X - (r_{I_B} - r)I_B + r_{I_C} I_C + (\gamma_K - r\phi)K \\ &+ (r_1 - LGD_{P_1} \lambda_B)(V^r - V^s) + (\lambda_B + \lambda_C + \lambda_{S/C} + \lambda_{S \wedge C} - r)V^r + rV^s. \end{aligned} \quad (394)$$

where we have defined

$$\gamma = r_1 + R_{P_1} \lambda_B + \lambda_C + \lambda_{S/C} + \lambda_{S \wedge C}. \quad (395)$$

Now let us introduce the on-default payments (23) and (22). This last one, the default resolution when the C/P defaults, is meant here under the assumption that the underlying has not defaulted. Its expression is unchanged such that we can still use (22). We find

$$\begin{aligned} \frac{\partial U}{\partial t} + \mathcal{A}U - \gamma U &= \lambda_C LGD_C(Q - X - I_C)^+ + \lambda_B LGD_B(Q - X + I_B)^- \\ &- \lambda_{S/C} g_{S/C} - \lambda_{S \wedge C} g_{S \wedge C} \\ &+ (r_2 - LGD_{P_2} \lambda_B - r)(V^s - X + I_B - \phi K)^+ \\ &+ (r_3 - LGD_{P_3} \lambda_B - r)(V^s - X + I_B - \phi K)^- \\ &+ (r_X - r)X - (r_{I_B} - r)I_B + r_{I_C} I_C + (\gamma_K - r\phi)K \\ &+ (r_1 - LGD_{P_1} \lambda_B)(V^r - V^s) + (\lambda_B + \lambda_C + \lambda_{S/C} + \lambda_{S \wedge C} - r)V^r + rV^s \\ &- (\lambda_B + \lambda_C)Q. \end{aligned} \quad (396)$$

At least informally, we may now apply Feynman-Kac's theorem and find the following valuation equations

Component	Hedged	
XVA Discount	$r_1 + R_{P_1} \lambda_B + \lambda_C + \lambda_{S/C} + \lambda_{S \wedge C}$	
CVA	$-\lambda_C LGD_C (Q - X - I_C)^+$	
CVA(S/C)	$\lambda_{S/C} g_{S/C}$	
CVA($S \wedge C$)	$\lambda_{S \wedge C} g_{S \wedge C}$	
DVA	$-\lambda_B LGD_B (Q - X + I_B)^-$	
FCA	$-(r_F^b - r)(V^s - X + I_B - \phi K)^+$	
FBA	$-(r_F^l - r)(V^s - X + I_B - \phi K)^-$	
DVA2	$\lambda_B LGD_{P_1} (V^s - X + I_B - \phi K)^+$	
CVA2	$\lambda_B LGD_{P_2} (V^s - X + I_B - \phi K)^-$	
CoIVA	$-(r_X - r)X$	
MVA	$-\left[r_{I_C} I_C - (r_{I_B} - r)I_B\right]$	
KVA	$-(\gamma_K - \phi r)K$	
GVA	$-(r_1 - LGD_{P_1} \lambda_B)(V^r - V^s) - (\lambda_B + \lambda_C + \lambda_{S/C} + \lambda_{S \wedge C} - r)V^r - rV^s + (\lambda_B + \lambda_C)Q$	

C Cross-Currency Measures

We write down a step-by-step derivation of change of measure and discounting formulas for pricing under single and cross-currency collateralization. Throughout this section, collateralization is assumed to be perfect (i.e. 0 threshold, 0 MTA, instantaneous margining) and we assume that the only asset eligible for collateral is cash in a unique currency specified in the CSA. Imperfect collateralization is valued by the XVAs in section 1.

This section is strongly inspired by the works of V. Piterbarg in [Piterbarg12a] and T. Wong in [MHSCRMDTP51]. We start by gathering results from these articles, while expanding on some of the justifications and proofs, tailoring the expose for our specific purposes⁹. We then discuss the practical implications of these results in terms of effective discounting, addressing the commonly used simplifying assumption of *invariance of the forward* in relation to basis modelling.

C.1 Single Currency Collateral Measure

[Piterbarg12a] went back to first principles in order to derive pricing formulas under collateralization. It is through such arguments that one can prove the validity of the now well-known principle that cash-flows should be discounted according to the collateral rate.

This section serves as a warm-up to set up notations and methodologies, to be extended to the cross-currency case in the following section.

Let us assume that there are $N + 1$ assets with values V_t^i , $i = 1..N + 1$ at time t . First we derive the evolution of positions on these assets, and then we explain the change of measure argument leading to using the risk-neutral measure.

C.1.1 Asset evolution under funding

Here we use the "old" argument in terms of the existence of a risk-free rate to fund the position.

- At t , buy the asset V_t^i by borrowing V_t^i at rate r for a period dt . This position π_t^i requires no cash and has a total value of 0, as

$$\pi_t^i = V_t^i - V_t^i = 0 \quad (397)$$

- At $t + dt$, give back the cash amount plus interest $V_t^i \times (1 + rdt)$ and sell the asset at its new value V_{t+dt}^i , which results in the cash-flow

$$d\pi^i = V_{t+dt}^i - V_t^i \times (1 + rdt) = dV^i - rV^i dt \quad (398)$$

- Here the important point is that the change in cash-flows is not just the change in the asset, it also has to contain the change in the funding position.
- We can assume an evolution for the asset under the real world measure as

$$dV^i = \mu^i V^i dt + \Sigma^i dW^i \quad (399)$$

with drift μ^i in that measure and diffusion Σ^i . The change in the position is then

$$d\pi^i = (\mu^i - r)V^i dt + \Sigma^i dW^i. \quad (400)$$

⁹Of connecting this to XVA calculations and calculation of resetting notional quanto.

C.1.2 Asset evolution under collateralization

Now we redo this but assuming that each asset is collateralized at its own collateral rate c^i .

- At t , buy the asset V_t^i . As soon as this is done, the PV becomes positive for us with value V_t^i so that the counterparty has to post precisely that amount to us. So we do not need to borrow the money to enter the position. Effectively there is no cash-flow exchange, and again the position π_t^i requires no cash and has a total value of 0, as

$$\pi_t^i = V_t^i - V_t^i = 0 \quad (401)$$

- At $t + dt$, we must return the collateral plus interest, that is, $V_t^i \times (1 + c^i dt)$. Now the asset value has changed to V_{t+dt}^i , which is the amount the counterparty has to give to us for collateralization. This results in the cash-flow

$$d\pi^i = V_{t+dt}^i - V_t^i \times (1 + c^i dt) = dV^i - c^i V^i dt \quad (402)$$

- Effectively we have funded the trade with the collateral.
- In the same way, the risk-neutral dynamics of the position is

$$d\pi^i = (\mu^i - c^i) V^i dt + \Sigma^i dW^i. \quad (403)$$

C.1.3 Collateral measure

We have seen above that the former case of "risk-free" rate is leading to the same evolution of the position as for a set of collateralized asset with the risk-free rate replaced by the collateral rates. So we follow the reasoning with collateral rates now. The bottom line is that assuming there are more assets than sources of randomness, we can find a linear combination of assets, that is, a portfolio, that has no randomness. Non-arbitrage imposes that its evolution is 0 since it's entered into at 0 cost, and this creates a relation between the real-world drifts, the collateral rates and the diffusion, which, through Girsanov's theorem, lets us find an equivalent measure in which all assets evolve at their collateral rate (formerly risk-free rate).

We mostly repeat the arguments in [Piterbarg12a]. Assuming there are only N sources of diffusion, i.e. Brownian motions, that is, 1 less than the number of assets, we can write the dynamics of all the assets in vectorial form as

$$dV = \mu V dt + \Sigma dW \quad (404)$$

where all quantities are $N + 1$ vectors, but the Brownian motion dW is only N -dimensional and the diffusion matrix Σ is of rank N .

This implies that there exist a vector w in the null space of Σ such that

$$w^T \Sigma = 0. \quad (405)$$

We can use this vector to entered into a portfolio π of weighted assets $w^T V$ with collateralization for each, and this has evolution

$$d\pi = w^T (dV - c dt) = w^T (\mu - c) dt \quad (406)$$

where c is the collateral rate vector. This portfolio has no randomness due to the particular value of w^T in (405), and by no-arbitrage its evolution must be 0. This leads to

$$w^T (\mu - c) = 0, \quad (407)$$

or in other words that the vector $\mu - c$ is orthogonal to the weights w . Because of this, we can rewrite it as the matrix Σ applied to some vector λ i.e.

$$\mu - c = \Sigma\lambda, \quad (408)$$

and the evolutions of the assets become

$$dV = cVdt + \Sigma(dW + \lambda dt). \quad (409)$$

Thanks to Girsanov's theorem, we can find a measure where $dW + \lambda dt$ is a Brownian motion, and in that measure the evolution of all assets has the collateral rates as drifts, i.e.

$$dV = cVdt + \Sigma d\tilde{W}^Q. \quad (410)$$

Q denotes this particular measure, analogous to the risk-neutral measure in the old framework of the risk-free rate, called "collateral measure" in [MHSCRMTP51].

For the sake of completeness, we can then proceed to pricing in this measure. As all assets have their collateral rate as drift, the discounted assets

$$Y^i(T) = e^{-\int_t^T c^i(u)du} V^i(T) \quad (411)$$

have no drifts and are Martingales, which yields the pricing formula

$$Y^i(t) = E_t^Q[Y^i(T)] \quad (412)$$

$$V^i(t) = E_t^Q[e^{-\int_t^T c^i(u)du} V^i(T)] \quad (413)$$

where PVs are obtained by discounting payoffs with the collateral rate.

C.2 Cross-Currency Collateralization

In this section we extend the arguments of the previous section to the case of cross-currency collateralization. We derive a number of pricing formulas for bonds in both domestic and foreign currencies collateralized in either currency, and connect these quantities to the change of measure between domestic and foreign collateral measures.

From now on we assume that the domestic currency has only one possible collateral that is cash, and denote its rate by $c_d(t)$. In the same way, the foreign currency has only one possible collateral, cash, with rate $c_f(t)$. These rates are single currency rates and would typically be the OIS rates in each currency. Zero-coupon bonds, which we will call bonds in situations with no ambiguities, are denoted with a first letter for the currency and a second letter for the currency of the collateralization. For instance, $P_{d,f}(t, T)$ is the bond in domestic currency, collateralized in foreign currency, observed at time t for maturity $T > t$.

C.2.1 Dynamics in domestic measure

Domestic bond

A domestic bond collateralized in domestic currency is simply a particular case of asset as in the previous section, that pays 1 at maturity. We can thus apply the previous results directly and write the evolution of a collateralized position on it as

$$dP_{d,d} - c_d P_{d,d} dt \quad (414)$$

and pricing formula

$$P_{d,d}(t, T) = E_t^{Q^d}[e^{-\int_t^T c_d(u)du}] \quad (415)$$

where Q^d is the collateral measure in the domestic currency.

Foreign bond

Now let us take the case of the foreign bond in domestic collateral $P_{f,d}(t, T)$ and adapt the arguments of the previous section to the cross-currency case.

- We (A) buy $P_{f,d}(t, T)$ at time t as an amount in foreign currency, from our counterparty (B).
- As the bond is collateralized in domestic currency, B must post to us its value in domestic currency which is $P_{f,d}(t, T) \times X(t)$ where $X(t)$ is the spot FX as a number of domestic currency $\mathcal{D}$ for 1 foreign currency $\mathcal{F}$
- The total value of this position at t is zero
- At $t + dt$, we return the collateral to B with interest, that is, we pay

$$P_{f,d}(t, T)X(t)(1 + c_d(t)dt) \quad (416)$$

- Now both the value of FX and of the bond have changed and B must give us the new collateral amount

$$P_{f,d}(t + dt, T)X(t + dt) \quad (417)$$

- The total flow at $t + dt$ is therefore

$$P_{f,d}(t + dt, T)X(t + dt) - P_{f,d}(t, T)X(t)(1 + c_d(t)dt) = d[P_{f,d}X] - c_d[P_{f,d}X]dt. \quad (418)$$

This is just equivalent to looking at the asset $P_{f,d}X$ collateralized in domestic currency. In particular, the pricing formula is

$$P_{f,d}(t, T)X(t) = E_t^{Q^d} [e^{-\int_t^T c_d(u)du} X(T)]. \quad (419)$$

FX

We consider now an FX repo transaction collateralized in foreign currency, with r_{df} as repo rate.

- At t , we borrow $X(t)$ in domestic currency from a counterparty and give $1\mathcal{F}$ as collateral in the foreign currency. This has value 0.
- At $t + dt$, we receive the principal back plus the repo interest, i.e. we receive

$$X(t)(1 + r_{df}dt) \quad (420)$$

- We return the collateral $1\mathcal{F}$, which in domestic currency now has value $X(t + dt)$.
- The total flow (its opposite in fact) at $t + dt$ is therefore

$$X(t + dt) - X(t)(1 + r_{df}dt) = dX - r_{df}Xdt \quad (421)$$

and the pricing formula is

$$X(t) = E_t^{Q^d} [e^{-\int_t^T r_{df}(u)du} X(T)]. \quad (422)$$

Note that in principle we would think of having interest earned on the collateral here. But this would simply add/subtract c_f to r_{df} , which, at this point, is not specified. The crucial point is that argued by Piterbarg in [Piterbarg12a] that r_{df} is an additional degree of freedom, with at least a component that is independent from

c_d and c_f . As a consequence, we can simply absorb the c_f interest into the definition of r_{df} and consider that the transaction is fair with r_{df} defined in this manner and no interest on the collateral.

Hybrid Model Dynamics

As a consequence of the above, in the domestic collateral measure the dynamics of the hybrid model is

$$\begin{pmatrix} \frac{dX}{X} \\ \frac{dP_{d,d}}{P_{d,d}} \\ \frac{dP_{f,d}X}{P_{f,d}X} \end{pmatrix} = \begin{pmatrix} r_{df} \\ c_d \\ c_d \end{pmatrix} dt + \Sigma dW^d \quad (423)$$

where Σ is the diffusion matrix and W^d is a multi-dimensional Brownian motion in the domestic collateral measure denoted by the subscript d .

C.2.2 Dynamics in foreign measure

Foreign bond

A foreign bond collateralized in foreign currency is simply a particular case of single currency asset as in the previous section, that pays 1 at maturity. We can thus apply the previous results directly and write the evolution of a collateralized position on it as

$$dP_{f,f} - c_f P_{f,f} dt \quad (424)$$

and pricing formula

$$P_{f,f}(t, T) = E_t^{Q^f} [e^{-\int_t^T c_f(u) du}] \quad (425)$$

where Q^f is the collateral measure in the foreign currency.

Domestic bond

Now let us take the case of the domestic bond in foreign collateral $P_{d,f}(t, T)$.

- We (A) buy $P_{d,f}(t, T)$ at time t as an amount in domestic currency, from our counterparty (B).
- As the bond is collateralized in foreign currency, B must post to us its value in foreign currency which is $\frac{P_{d,f}(t, T)}{X(t)}$
- The total value of this position at t is zero
- At $t + dt$, we return the collateral to B with interest, that is, we pay

$$\frac{P_{d,f}(t, T)}{X(t)} (1 + c_f(t) dt) \quad (426)$$

- Now both the value of FX and of the bond have changed and B must give us the new collateral amount

$$\frac{P_{d,f}(t + dt, T)}{X(t + dt)} \quad (427)$$

- The total flow at $t + dt$ is therefore

$$\frac{P_{d,f}(t + dt, T)}{X(t + dt)} - \frac{P_{d,f}(t, T)}{X(t)} (1 + c_f(t) dt) = d\left[\frac{P_{d,f}}{X}\right] - c_f\left[\frac{P_{d,f}}{X}\right] dt. \quad (428)$$

In particular, the pricing formula is

$$\frac{P_{d,f}(t, T)}{X(t)} = E_t^{Q^f} [e^{-\int_t^T c_f(u) du} \frac{1}{X(T)}]. \quad (429)$$

FX

We consider now an FX repo transaction collateralized in domestic currency.

- At t , we borrow $1/X(t)$ in foreign currency from a counterparty and give $1\mathcal{D}$ as collateral in the domestic currency. This has value 0. Looking at this transaction in the opposite way, the opposite side is effectively borrowing $1\mathcal{D}$ and is giving $1/X(t)$ as collateral. This transaction should be fair as in the previous set-up, which means that...
- At $t + dt$, they receive the "principal" back plus the repo interest, i.e. they receive

$$1 \times (1 + r_{df} dt) \quad (430)$$

- They return the "collateral" $1/X(t)$
- In the foreign currency, the total flow at $t + dt$ is therefore

$$\frac{1 + r_{df} dt}{X(t + dt)} - \frac{1}{X(t)} \approx d[1/X] + r_{df}[1/X]dt \quad (431)$$

and the pricing formula is

$$1/X(t) = E_t^{Q^f} [e^{\int_t^T r_{df}(u) du} 1/X(T)]. \quad (432)$$

Hybrid Model Dynamics

As a consequence of the above, in the foreign collateral measure the dynamics of the hybrid model is

$$\begin{pmatrix} \frac{d(1/X)}{(1/X)} \\ \frac{dP_{d,f}/X}{P_{d,f}/X} \\ \frac{dP_{f,f}}{P_{f,f}} \end{pmatrix} = \begin{pmatrix} -r_{df} \\ c_f \\ c_f \end{pmatrix} dt + \tilde{\Sigma} dW^f \quad (433)$$

where $\tilde{\Sigma}$ is the diffusion matrix and W^f is a multi-dimensional Brownian motion in the foreign collateral measure denoted by the subscript f .

C.2.3 Cross-currency change of measure

In this section we are mainly looking for formulas that enable us to price in the domestic or foreign measures, that is, we are looking for transformation formulas back and forth between the 2 measures such as

$$E_t^d[R] = E_t^f[H \times R] \quad (434)$$

$$E_t^f[R] = E_t^d[K \times R] \quad (435)$$

where we use interchangeably the notations $E_t^{Q^f} = E_t^f$ and its domestic analogue. R is a generic quantity such as a payoff and H, K are the terms we are looking for that let us change measures.

To do so we look at the pricing formulas obtained in the previous sections for the basic components that are the bonds and FX spot. We are looking for a normalized Martingale to play the role of the likelihood process satisfying the properties (298) and (299).

We can find it by looking at the pricing formula for the FX rate X in the domestic measure (422). We rewrite this equation as

$$\frac{X(t)}{e^{\int_0^t r_{df}}} = E_t^d \left[\frac{X(T)}{e^{\int_0^T r_{df}}} \right] \quad (436)$$

and dividing this by $X_0 = X(0)$ we find that

$$M(t) = \frac{X(t)}{e^{\int_0^t r_{df}} X_0} \quad (437)$$

is a normalized Martingale in the domestic measure. As such it can be used to change measure from domestic to another measure.

Now we are going to show that this other measure $M(\cdot)$ brings us to is in fact the foreign measure. Let us denote this measure by the question mark symbol '?'. The change of measure from domestic to '?' reads

$$E_t^?[R] = E_t^d \left[\frac{M(T)}{M(t)} R \right]. \quad (438)$$

Now we take inspiration from (432) and consider the quantity R

$$R = \frac{e^{\int_t^T r_{df}}}{X(T)}, \quad (439)$$

and then apply the candidate change of measure formula (438). We find

$$\begin{aligned} E_t^?[R] &= E_t^d \left[\frac{M(T)}{M(t)} R \right] \\ &= E_t^d \left[\frac{X(T) e^{\int_0^t r_{df}}}{e^{\int_0^T r_{df}} X(t)} \frac{e^{\int_t^T r_{df}}}{X(T)} \right] \\ &= E_t^d \left[\frac{1}{X(t)} \right] \\ &= \frac{1}{X(t)}, \end{aligned} \quad (440)$$

which is precisely the answer we find in the foreign measure by virtue of (432). This is how we identify the measure '?' with the foreign measure.

To summarize, the change of measure from domestic to foreign is performed via

$$E_t^f[R] = \frac{1}{X(t)} E_t^d [e^{-\int_t^T r_{df}} X(T) R] \quad (441)$$

and from foreign to domestic via

$$E_t^d[R] = X(t) E_t^f \left[\frac{e^{\int_t^T r_{df}}}{X(T)} R \right]. \quad (442)$$

C.2.4 Cross-currency basis

Let us define the quantity q_{df} as

$$q_{df} = c_f - c_d + r_{df}. \quad (443)$$

We will see later in this document that this quantity is what plays the role of what we usually call "the cross-currency basis".

C.3 Applications

In most practical applications in this document we will take the example of the pair USD/JPY considering JPY as the domestic currency and USD as foreign.

C.3.1 OIS curves

Typically the OIS curves in each country give us the discount bonds $P_{d,d}(t, T)$ from JPY OIS and $P_{f,f}(t, T)$ from USD FedFunds.

C.3.2 FX forwards

Let us start with the pricing of FX forwards collateralized in the foreign currency. We will usually take the example of the pair USD/JPY considering JPY as the domestic currency. Let us take a trade that exchanges at maturity T 1\$ against K ¥ and collateralized it in USD. We payoff at maturity is

$$Payoff = 1 - \frac{K}{X_T} \quad (444)$$

and can be viewed as a single currency trade collateralized in cash of its own currency USD. As such, the pricing formula is

$$\begin{aligned} PV_{USD} &= E_t^f [e^{-\int_t^T c_f} (1 - \frac{K}{X_T})] \\ &= P_{f,f}(t, T) - K \frac{P_{d,f}(t, T)}{X_t} \end{aligned} \quad (445)$$

which we get through the use of the pricing formulas in the foreign measure (295). Let us denote by $X_f(t, T)$ the forward rate for maturity T collateralized in USD, i.e. the strike K that makes the above PV vanish. We find

$$X_f(t, T) = X_t \frac{P_{f,f}(t, T)}{P_{d,f}(t, T)}. \quad (446)$$

Since the bonds $P_{f,f}(t, T)$ are known from the single currency world (i.e. USD FedFunds), knowing the quotes $X_f(t, T)$ allows us to obtain the initial curve $P_{df}(t, T)$ of the domestic bonds collateralized in foreign currency. This is under the assumption that the quoted FX forwards are collateralized in foreign currency.

On the other hand we can repeat this calculation for the case of domestic collateralization. In domestic currency ¥ the payoff is

$$Payoff = X_T - K \quad (447)$$

and can be viewed as a single currency trade collateralized in cash of its own currency JPY. As such, the pricing formula is

$$\begin{aligned} PV_{JPY} &= E_t^d[e^{-\int_t^T c_d}(X_T - K)] \\ &= P_{f,d}(t, T)X_t - KP_{d,d}(t, T) \end{aligned} \quad (448)$$

obtained using the pricing formulas (294). We define the FX forward collateralized in domestic currency as $X_d(t, T)$ and obtain it as the strike that nullifies the PV above, that is

$$X_d(t, T) = X_t \frac{P_{f,d}(t, T)}{P_{d,d}(t, T)}. \quad (449)$$

If quotes exist for $X_d(t, T)$, they allow us to calculate the initial curve $P_{fd}(t, T)$ since $P_{d,d}(t, T)$ is known (i.e. JPY OIS).

C.3.3 FX forward invariance

Our understanding of the market is that only $X_f(t, T)$ is quoted, such that only $P_{d,f}(t, T)$ can be obtained without further assumptions. It seems common to add the assumption that $X_d(t, T) = X_f(t, T)$, which is called "FX Forward Invariance" or "Automatic CSA" in Murex. This lets us back out $P_{fd}(t, T)$ by the fact that

$$\frac{P_{f,f}(t, T)}{P_{d,f}(t, T)} = \frac{P_{f,d}(t, T)}{P_{d,d}(t, T)}. \quad (450)$$

Note that this is the consequence of *an assumption on non-existent market data, not a theoretical pricing identity*. We will see below that this assumption is in fact connected to *the cross-currency basis being deterministic*.

To do so we look for expressions of the "cross" bonds P_{df} and P_{fd} in terms of the "straight" bonds P_{dd} and P_{ff} . We use the change of measure formulas leading to the chain of equations

$$P_{f,d}(t, T) = \frac{1}{X_t} E_t^d[e^{-\int_t^T c_d} X_T] \quad (451)$$

$$= E_t^f[e^{-\int_t^T [c_d - r_{df}]}] \quad (452)$$

$$= P_{f,f}(t, T) E_t^{Tf}[e^{-\int_t^T [c_d - r_{df} - c_f]}] \quad (453)$$

$$= P_{f,f}(t, T) E_t^{Tf}[e^{\int_t^T q_{df}}] \quad (454)$$

where we have used, in chronological order, the pricing formula in domestic currency (294), the change of measure from domestic to foreign (441), the change of measure from foreign collateral to foreign T-forward (303)

We define the quantities

$$H_{f,d}(t, T) = E_t^{Tf}[e^{\int_t^T q_{df}}] \quad (455)$$

$$H_{d,f}(t, T) = E_t^{Td}[e^{-\int_t^T q_{df}}] \quad (456)$$

which leads to

$$P_{f,d}(t, T) = P_{f,f}(t, T) \times H_{f,d}(t, T) \quad (457)$$

$$P_{d,f}(t, T) = P_{d,d}(t, T) \times H_{d,f}(t, T). \quad (458)$$

This simplifies the relation between domestic and foreign collateralized FX forwards in that

$$X_f(t, T) = \frac{\tilde{X}(t, T)}{H_{d,f}(t, T)} \quad (459)$$

$$X_d(t, T) = \tilde{X}(t, T) \times H_{f,d}(t, T) \quad (460)$$

$$\tilde{X}(t, T) = X_t \frac{P_{f,f}(t, T)}{P_{d,d}(t, T)}. \quad (461)$$

The FX forward invariance is then equivalent to assuming

$$H_{fd}(t, T) = \frac{1}{H_{df}(t, T)} \quad (462)$$

which holds when the cross-currency basis q_{df} is deterministic.¹⁰

C.3.4 Libor flows

Domestic collateral

We start with the simple case of domestic Libor flow collateralized in domestic currency. We look at libor flow as any other collateralized instrument, only keeping track of their frequency with the subscript $i = 1M, 3M, 6M, \dots$

Single currency domestic swaps collateralized in domestic currency require the ability to value the quantity

$$E_t^d [e^{-\int_t^{T_P} c_d} L_d^i(T_F, T_P)] \quad (463)$$

where T_F, T_P are the fixing/paying times of the flow. We can therefore bootstrap these quantities from the value of the swaps on L_d^i and define the initial yield curve with discount bonds $P_{d,d}^i(t, T)$ by

$$E_t^d [e^{-\int_t^{T_P} c_d} L_d^i(T_F, T_P)] = P_{d,d}(t, T_P) \left[\frac{P_{d,d}^i(t, T_F)}{P_{d,d}^i(t, T_P)} - 1 \right] \frac{1}{\delta(T_F, T_P)} \quad (464)$$

which gives us the curve $P_{d,d}^i(t, T)$.

Foreign collateral

This is the situation of constant notional cross-currency swaps between JPY and USD, whose quotes are typically assumed under USD collateralization. We need to value the JPY Libor collateralized in USD. We price this by converting the single currency JPY flow into USD at its payment time with the FX spot rate at the time, then consider this as a USD instrument to be price naturally under USD collateralization. In other words, the present value of this JPY Libor flow collateralized in USD is

$$PV_{USD} = E_t^f [e^{-\int_t^{T_P} c_f} \frac{L_d^i(T_F, T_P)}{X_{T_P}}]. \quad (465)$$

We do not know how to calculate this directly but we know how to calculate the JPY Libor flows in the domestic currency so we can attempt to change measure, which gives us

$$PV_{USD} = \frac{1}{X_t} E_t^d [e^{-\int_t^{T_P} c_d + q_{df}} L_d^i(T_F, T_P)]. \quad (466)$$

If now we assume the cross-currency basis to be deterministic, this can be transformed into

¹⁰This is not an equivalence though, so we may imagine models with stochastic cross-currency basis but that satisfy the FX forward invariance. How to achieve that is a different question.

$$PV_{USD} = \frac{e^{-\int_t^{T_P} q_{df}}}{X_t} E_t^d [e^{-\int_t^{T_P} c_d} L_d^i(T_F, T_P)] \quad (467)$$

$$= \frac{1}{X_t} P_{d,d}(t, T_P) e^{-\int_t^{T_P} q_{df}} \left[\frac{P_{d,d}^i(t, T_F)}{P_{d,d}^i(t, T_P)} - 1 \right] \frac{1}{\delta(T_F, T_P)} \quad (468)$$

$$= \frac{1}{X_t} P_{d,f}(t, T_P) \left[\frac{P_{d,d}^i(t, T_F)}{P_{d,d}^i(t, T_P)} - 1 \right] \frac{1}{\delta(T_F, T_P)}. \quad (469)$$

In other words, we reach the intuitive formula where we calculate the forward Libor using its own single-currency based tenor curve and discount it with the curve for domestic flows collateralized in foreign currency $P_{df}(t, T)$. Note that while this is the result we have been expecting, this only holds under the assumption of deterministic cross-currency basis.

If the cross-currency basis is not deterministic, we should not be able to calculate this flow in this manner. This hypothesis can then be tested if we have quotes for FX forwards up to maturities where the cross-currency swaps are quoted. Indeed, we could derive the curve $P_{df}(t, T)$ from the FX forward quotes, the curve $P_{dd}^i(t, T)$ from the single currency L_d^i swaps, and price the cross-currency swaps with the formula above. The distance to par would then give us an estimate of the cross-currency basis volatility implied by the market.

C.3.5 MtM XCcy quanto

Fujii's view

In the previous section we showed how to value a domestic Libor flow collateralized in foreign currency, which is the situation that happens when pricing constant notional cross-currency swaps. When valuing resetting notional swaps (MtM), we must value a slightly different flow. Indeed, one side of the swap (USD) has resetting notionals whose value is not known in advance but determined at fixing time by the spot FX applied to the other (constant) notional (JPY).

In more details, the resetting leg (USD) is made of a series of N periodical loans as follows

- At fixing time T_F , the spot FX is observed (X_{T_F}) and defines the USD notional for the coming period as N_d/X_{T_F} where N_d is the constant JPY notional. The lending party pays this USD amount at T_F .
- At paying time T_P , the borrowing party must repay the notional of the loan plus the interest on it, that interest being the foreign Libor.

The pricing formula for this loan, collateralized in USD, is thus

$$PV_{USD} = -E_t^f \left[e^{-\int_t^{T_F} c_f} \frac{1}{X_{T_F}} \right] + E_t^f \left[e^{-\int_t^{T_P} c_f} \frac{1}{X_{T_F}} (1 + L_f^i(T_F, T_P) \delta_{FP}) \right] \quad (470)$$

where $\delta_{FP} = \delta(T_F, T_P)$ is the Libor accrual.

Let us now introduce the tenor basis between L_f^i and the OIS rate c_f . We will see that if this basis is 0, the above quantity vanishes. Let us denote it by $B(T_F, T_P)$ and define it as

$$L_f^i(T_F, T_P) = L_f(T_F, T_P) + B(T_F, T_P) \quad (471)$$

where $L_f(T_F, T_P)$ is the spot rate from the foreign collateral curve, i.e.

$$L_f(T_F, T_P) = \left(\frac{1}{e^{-\int_{T_F}^{T_P} c_f}} - 1 \right) \frac{1}{\delta_{FP}}. \quad (472)$$

Then a little algebra shows that the PV reduces to

$$PV_{USD} = E_t^f \left[e^{-\int_t^{T_P} c_f} \frac{1}{X_{T_F}} B(T_F, T_P) \delta_{FP} \right]. \quad (473)$$

In all generality $B(T_F, T_P)$ would be stochastic, but let us assume the more practical case where it is deterministic and thus drops out of the expectation. We still have to evaluate

$$PV_{USD} = B(T_F, T_P) \delta_{FP} E_t^f \left[e^{-\int_t^{T_P} c_f} \frac{1}{X_{T_F}} \right] \quad (474)$$

which cannot be priced in a model-independent way with time 0 FX and bonds due to the timing mismatch between the discounting (T_P) and the FX fixing (T_F). A model has to be assumed for FX and the rates, and in particular this quantity should depend on their correlation.

However, there are 2 reasons to believe this term's model-dependent value should be small:

- It is not proportional to the notional or to rates but to the tenor basis only
- The expectation is "mostly" model-independent. Indeed the mismatch is between T_F and T_P as

$$E_t^f \left[e^{-\int_t^{T_F} c_f} \frac{1}{X_{T_F}} \right] = \frac{P_{d,f}(t, T_F)}{X_t}. \quad (475)$$

As a consequence we can rewrite

$$PV_{USD} = B(T_F, T_P) \delta_{FP} E_t^f \left[e^{-\int_t^{T_F} c_f} \frac{1}{X_{T_F}} e^{-\int_{T_F}^{T_P} c_f} \right] \quad (476)$$

$$\approx B(T_F, T_P) \delta_{FP} \left(\frac{P_{d,f}(t, T_F)}{X_t} - E_t^f \left[e^{-\int_t^{T_F} c_f} \frac{1}{X_{T_F}} c_f \delta_{FP} \right] \right) \quad (477)$$

where we can see that the model-dependent term (in the expectation) is of even smaller order as it is proportional to rate times basis.

A different view

We have seen above that modelling the basis as suggested by Fujii in (471) and then assuming the basis thus defined to be deterministic leads to a residual term in (474) that cannot be priced simply with curves. Here we are going to show that a slightly different choice in basis modelling can lead to the full disappearance of the quanto.

First let us go back to the resetting swaplet as described in (470). The first component is the notional fixing and payment at the fixing time T_F , and this one does not pose a problem, i.e. is valued with no quanto. The problems come from the other term, which we write as

$$PV_{USD}^+ = E_t^f \left[e^{-\int_t^{T_P} c_f} \frac{1}{X_{T_F}} (1 + L_f^i(T_F, T_P) \delta_{FP}) \right]. \quad (478)$$

We are now going to use the Tower rule to transform this expression a bit as

$$\begin{aligned} PV_{USD}^+ &= E_t^f \left[e^{-\int_t^{T_P} c_f} \frac{1 + L_f^i(T_F, T_P) \delta_{FP}}{X_{T_F}} | \mathcal{F}_t \right] \\ &= E_t^f \left[E_t^f \left[e^{-\int_t^{T_P} c_f} \frac{1 + L_f^i(T_F, T_P) \delta_{FP}}{X_{T_F}} | \mathcal{F}_{T_F} \right] | \mathcal{F}_t \right] \\ &= E_t^f \left[e^{-\int_t^{T_F} c_f} \frac{1 + L_f^i(T_F, T_P) \delta_{FP}}{X_{T_F}} E_t^f \left[e^{-\int_{T_F}^{T_P} c_f} | \mathcal{F}_{T_F} \right] | \mathcal{F}_t \right] \\ &= E_t^f \left[e^{-\int_t^{T_F} c_f} \frac{1 + L_f^i(T_F, T_P) \delta_{FP}}{X_{T_F}} P_f(T_F, T_P) | \mathcal{F}_t \right], \end{aligned} \quad (479)$$

where we used the fact that everything is fixed at T_F except the piece of numeraire from T_F to T_P , which conditional on information at T_F is the discount bond. Now we can use the expression of the Libor rate

$$L_f^i(T_F, T_P) = \left(\frac{1}{P_f^i(T_F, T_P)} - 1 \right) \frac{1}{\delta_{FP}} \quad (480)$$

to reduce the expression to

$$PV_{USD}^+ = E^f \left[e^{-\int_t^{T_F} c_f} \frac{P_f(T_F, T_P)}{P_f^i(T_F, T_P)} \frac{1}{X_{T_F}} | \mathcal{F}_t \right]. \quad (481)$$

At this point we choose to make a different assumption from Fujii's. Instead of decomposition the Libor at tenor i as the Libor of the OIS rate plus basis, we do this decomposition multiplicatively on the discount bond. In fact this assumption is more practical, as this is precisely what is done when modelling the yield curves under Hull-White with a deterministic basis ladder. Indeed, the most common way of modelling multiple curves in Hull-White is simply to assume only one random factor and make all short-rates at different tenors equal up to a deterministic basis spread. More precisely, under this typical modelling framework, we can write that

$$P_f^i(T_F, T_P) = P_f(T_F, T_P) Q^i(T_F, T_P) \quad (482)$$

where $Q^i(T_F, T_P)$ is the discount factor for the forward basis and is deterministic, i.e. estimated purely from the initial yield curves. This assumption leads to the disappearance of the randomness in the ratio

$$\frac{P_f^i(T_F, T_P)}{P_f(T_F, T_P)} \quad (483)$$

which is precisely the term we need to evaluate here. The expression of interest thus becomes

$$\begin{aligned} PV_{USD}^+ &= \frac{1}{Q^i(T_F, T_P)} E^f \left[e^{-\int_t^{T_F} c_f} \frac{1}{X_{T_F}} | \mathcal{F}_t \right] \\ &= \frac{P_{d,f}(t, T_F)}{X_t Q^i(T_F, T_P)} \end{aligned} \quad (484)$$

i.e. is equal to its value estimated only with initial curves. In other words, provided we do the basis modelling choice (482), which is very common, there does not seem to be any quanto.

C.3.6 Inversion of the FX dynamics

We would like to see what happens to the FX dynamics under such changes as FX into $1/FX$ and/or changing measure from domestic to foreign or vice-versa. The objective is to clarify the technicalities behind FX modelling and change of base currency in the XVAs. We will see that it is easiest to model FX under the domestic measure and $1/FX$ under the foreign measure, when FX is the number of domestic currency for 1 unit of foreign currency.

We start by writing the SDE of the FX process X_t in the domestic measure, where X_t is the number of domestic currency corresponding to 1 unit of foreign currency. This SDE is, according to (423) and (443)

$$\frac{dX}{X} = (c_d - c_f + q_{df})dt + \sigma(t, X)dW^d \quad (485)$$

where we assume a local volatility diffusion $\sigma(t, X)$. Now we want to obtain the dynamics of the inverse FX, which we denote by $\hat{X} = 1/X$.

We apply Ito's lemma with the function $f(X) = 1/X$ and obtain

$$\frac{d\hat{X}}{\hat{X}} = (c_f - c_d - q_{df} + \sigma^2(t, X))dt - \sigma(t, X)dW^d. \quad (486)$$

There are several remarks to make at this point:

- the dynamics is still written under the domestic measure
- the drift is almost the opposite of that of X but not quite because of the presence of an Ito correction. This might sound counter-intuitive as we would expect the inverse FX to have the opposite drift. This is a formulation of Siegle's paradox, which is only an apparent paradox due to the fact that we are still in the domestic measure.
- the volatility remains to be evaluated on the original X , that is, on $1/\hat{X}$.

We would expect the SDE of the inverse FX to take the same form as that of FX because the notions of domestic and foreign are purely conventional. The above seems to break this idea but this is because we have not achieved full inversion yet. We started from the SDE of FX in the domestic measure, and therefore a full inversion corresponds to the SDE of the inverse FX in the foreign measure. We still have to change measure to the foreign currency, which we do now using the cross-currency change of measure formula (321). The drift of $\hat{X}$ under the foreign measure becomes

$$\mu_f = \mu_d + \langle d \ln \hat{X}, d \ln X \rangle \quad (487)$$

$$= \mu_d - \sigma^2(t, X) \quad (488)$$

$$= c_f - c_d - q_{df}. \quad (489)$$

This finally leads to the fully inverted dynamics which does have the opposite drift in the foreign measure,

$$\frac{d\hat{X}}{\hat{X}} = (c_f - c_d - q_{df})dt - \sigma(t, X)dW^d. \quad (490)$$

Note again that the original local volatility is to be evaluated for $X = 1/\hat{X}$. We will say a few more words on this in section C.3.7.

C.3.7 Inversion of the FX calibration

In practice for the XVAs we will choose a base currency used for discounting and for measurement of the FVA spread. In financial engineering language, this translates into the fact that we will be writing the dynamics of the assets under the domestic measure, where the "domestic" is the base currency. This has the following implications:

- all assets that are not in the domestic currency will see their drifts modified with a quanto term coming from the change of measure from their native currency to the domestic currency. In particular, all non-base interest rates receive a quanto. The quanto term is typically the covariance between the asset and the FX rate.
- FX's SDE takes its traditional form (drift with rates only) in this domestic currency only if FX is defined as the number of domestic currency for 1 unit of foreign currency.

The above may lead to some (mild) ambiguity when FX quantities are typically quoted for the inverse FX, i.e. the number of "foreign" currency for 1 unit of domestic currency. Let us take the example of USD/JPY, typically quoted in that order, i.e. as a number of JPY for 1 USD. If the base currency for the XVA calculation is set to USD, which is the most common case in MHI, the FX rate that is naturally modelled is JPY/USD rather than USD/JPY.

This has 2 consequences:

1. the correlations of other assets with FX, often measured against USD/JPY, must be reversed to correctly represent the correlations with the modelled quantity JPY/USD (i.e. we have to change their signs)
2. the model volatility of the inverse FX is the same as that of the original FX but must be estimated on the inverse point.

This second point may lead to more difficulty when using a stochastic volatility model for FX. Indeed, the volatility process would pick a quanto term during the change of measure, which would be an additional quantity to simulate. It is possible but there is an alternative.

Let us summarize the possibilities in this particular case where inversion is needed.

1. We calibrate our FX model as if we were modelling USD/JPY, with typically quoted USD/JPY spot and option prices. We then model JPY/USD by inverting the correlations and estimating the model volatility of the inverse FX $\hat{\sigma}(t, \hat{X}) = \sigma(t, X)$. If this volatility process is stochastic, it must receive a quanto correction which will be non-zero as soon as the volatility process has non-zero correlation with the spot.
2. We calibrate our FX model by modelling JPY/USD during the calibration process, all expressed in the USD measure. In this manner we obtain directly the local volatility of our modelled quantity $\hat{X}$ and we do not need to include a quanto in the SDE of the volatility process if it is stochastic.

The first strategy has the advantage that the calibration process can be done entirely with already working implementations, with no need to transform or re-interpret spots and option prices. However, it requires extra care in the simulation process when using local and/or stochastic volatility processes.

The second strategy has the advantage that the simulation can be written right away in its most natural form, i.e. without quantos. However, it requires inversion of the FX spot, forwards, and option price surface before calibration. In the remaining of this section we show how this inversion can be performed. It is in fact a very intuitive result that we simply need to invert the strikes. Here we demonstrate it more rigorously.

Let us continue with the example of USD/JPY. This pair is typically handled in that order, i.e. with spots and forwards expressed as JPY amounts. Following the above analysis, this means that this pair is most conveniently calibrated and simulated with JPY as domestic currency. Let us see what this means for FX options. An FX call option is seen as the option to receive N_{USD} as a fixed USD amount in exchange for paying N_{JPY} as a fixed JPY amount, in the future at maturity T . We assume this market to be collateralized in USD.

The payoff expressed in JPY is

$$Payoff(JPY) = \text{Max}(N_{USD}X_T - N_{JPY}, 0) = N_{USD} \times \text{Max}(X_T - K, 0) \quad (491)$$

where X_T is the FX rate at maturity in the convenient order i.e. the number of JPY for 1 USD, and we have defined the "strike" $K = \frac{N_{JPY}}{N_{USD}}$. The present value of this trade in JPY is obtained with the usual Black-Scholes discounted according to the USD collateral

$$PV(JPY) = N_{USD}P_{JPY}^{USD}(T) \left[F(T)\mathcal{N}(d_+) - K\mathcal{N}(d_-) \right] \quad (492)$$

$$F(T) = X_0 \frac{P_{USD}^{OIS}(T)}{P_{JPY}^{USD}(T)} \quad (493)$$

$$d_{\pm} = \frac{\ln F/K}{\theta(T, K)\sqrt{T}} \pm \frac{1}{2}\theta(T, K)\sqrt{T}. \quad (494)$$

which defines the implied volatility surface $\theta(T, K)$. We can then calibrate the model volatility function $\sigma(t, X, \omega)$ of the USD/JPY process in the JPY measure (for some stochastic volatility process ω).

Our interest is instead to invert this process and consider this option with USD as domestic rate, i.e. consider it as a USD payoff. We would then write the same payoff in USD as

$$Payoff(USD) = \text{Max}(N_{USD} - \frac{N_{JPY}}{\hat{X}_T}, 0) = N_{JPY} \times \text{Max}(\hat{K} - \hat{X}_T, 0) \quad (495)$$

where the hat quantities are the inverse of the original ones. We can apply the standard Black-Scholes argument for this USD payoff (now a put option) collateralized in USD, and arrive at the price

$$PV(USD) = N_{JPY} P_{USD}^{OIS}(T) \left[\hat{K} \mathcal{N}(-\hat{d}_-) - \hat{F}(T) \mathcal{N}(-\hat{d}_+) \right] \quad (496)$$

$$\hat{F}(T) = \hat{X}_0 \frac{P_{JPY}^{USD}(T)}{P_{USD}^{OIS}(T)} \quad (497)$$

$$\hat{d}_{\pm} = \frac{\ln \hat{F}/\hat{K}}{\hat{\theta}(T, \hat{K})\sqrt{T}} \pm \frac{1}{2} \hat{\theta}(T, \hat{K})\sqrt{T} \quad (498)$$

which defines the implied volatility surface $\hat{\theta}(T, \hat{K})$, out of which we can calibrate the model volatility $\hat{\sigma}(t, \hat{X}, \hat{\omega})$ for the JPY/USD process in the USD measure.

All we need to do is find the relation between this implied volatility $\hat{\theta}(T, \hat{K})$ that we need for calibrating the inverse process and the one we obtain from the typically quoted market $\theta(T, K)$.

Note that this option expressed in USD is the same option and so the 2 prices must agree once expressed in the same currency today. This implies that

$$PV(USD) = \frac{PV(JPY)}{X_0} \quad (499)$$

$$= N_{USD} P_{JPY}^{USD}(T) \hat{X}_0 \left[\frac{1}{\hat{F}(T)} \mathcal{N}(d_+) - \frac{1}{\hat{K}} \mathcal{N}(d_-) \right] \quad (500)$$

$$= \frac{N_{USD} P_{JPY}^{USD}(T) \hat{X}_0}{\hat{F}(T) \hat{K}} \left[\hat{K} \mathcal{N}(d_+) - \hat{F}(T) \mathcal{N}(d_-) \right] \quad (501)$$

$$= N_{JPY} P_{USD}^{OIS}(T) \left[\hat{K} \mathcal{N}(d_+) - \hat{F}(T) \mathcal{N}(d_-) \right]. \quad (502)$$

A simple look at the d and $\hat{d}$ functions shows that this is equal to the USD PV in (496) provided the following relation holds

$$\hat{\theta}(T, \hat{K}) = \theta(T, K). \quad (503)$$

This means that we can calibrate the inverse process JPY/USD in the USD measure just like the USD/JPY process in the JPY measure while taking as implied volatility inputs for JPY/USD the implied volatilities of USD/JPY at the inverse strikes.

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